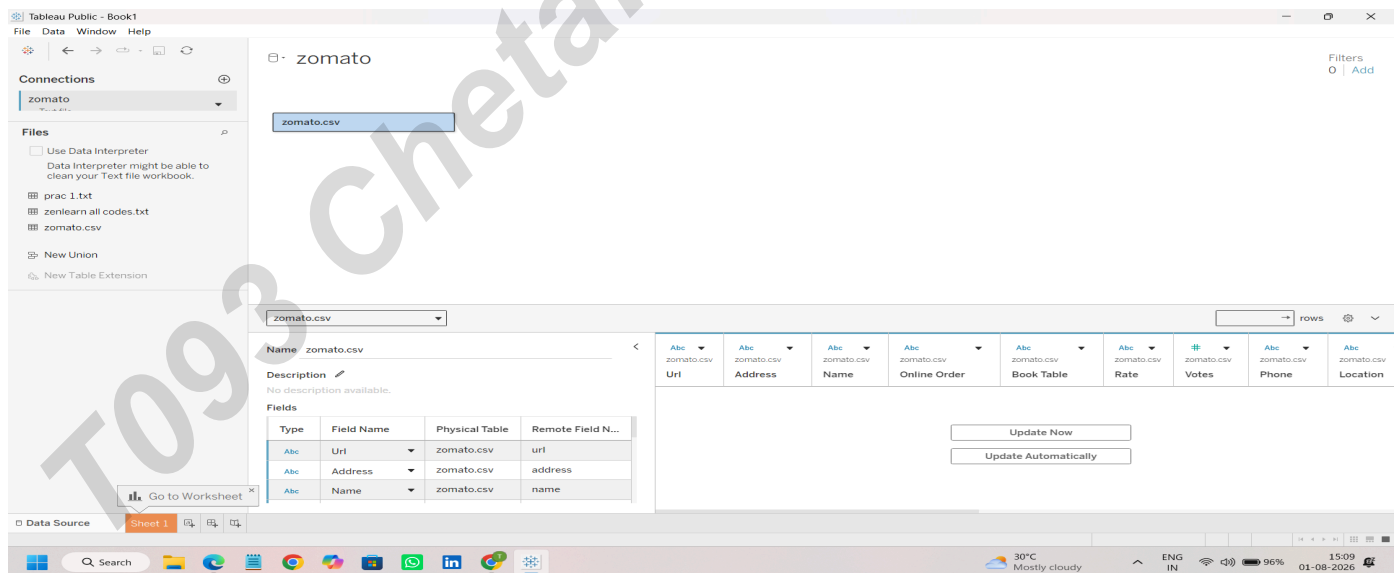
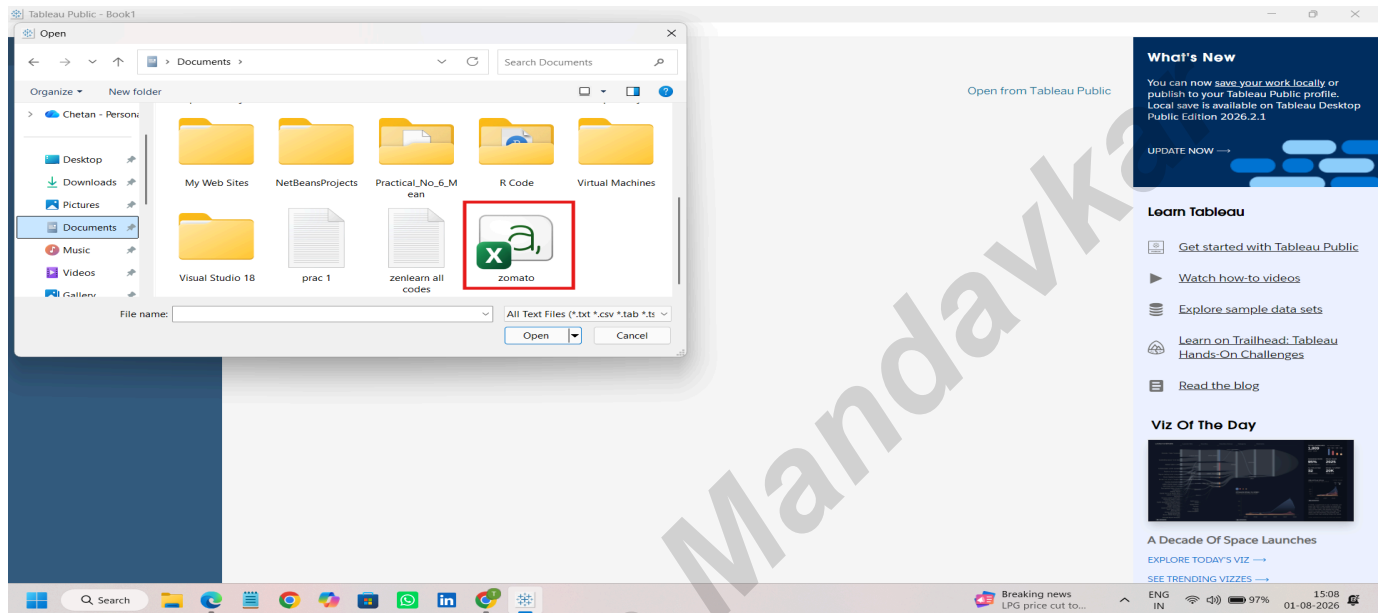


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Aim:- Analyzing Data using Table Calculations

A] Apply quick table calculations (running total, percent of total)

Step 1:- Open Tableau And Connect The Dataset



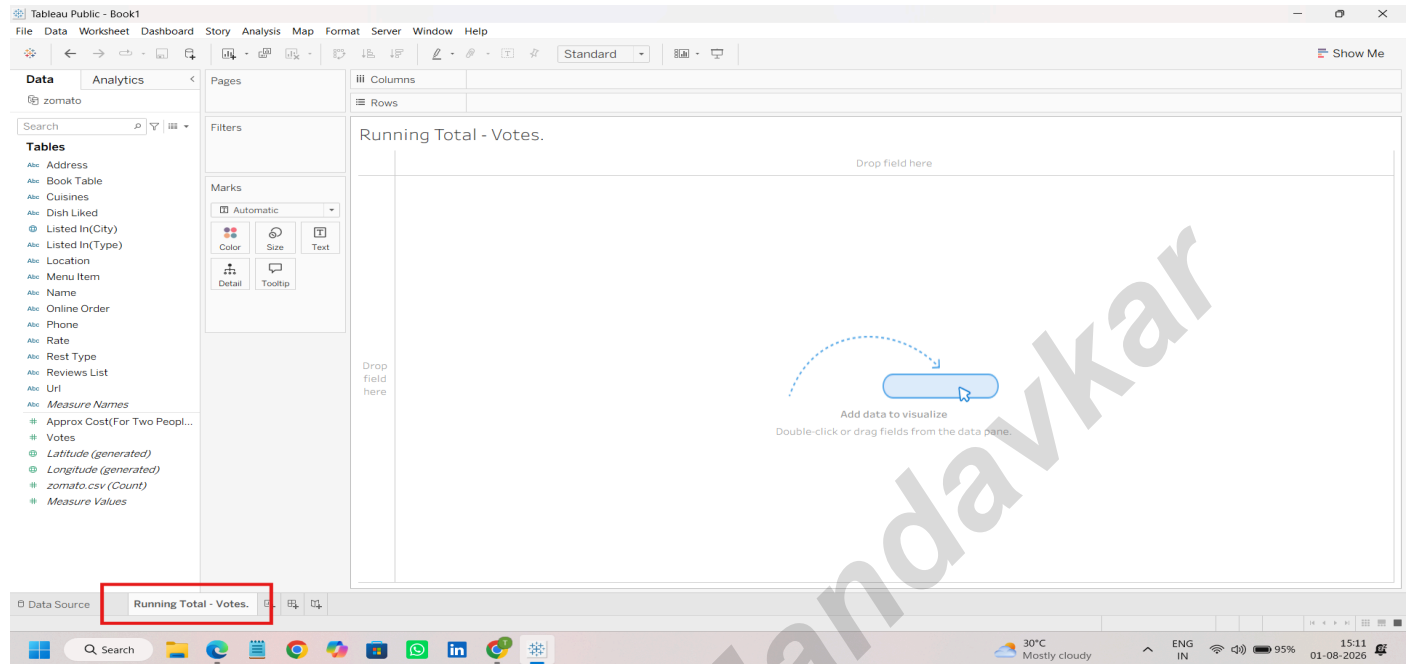
Name:- Chetan Mandavkar

Roll No. T093

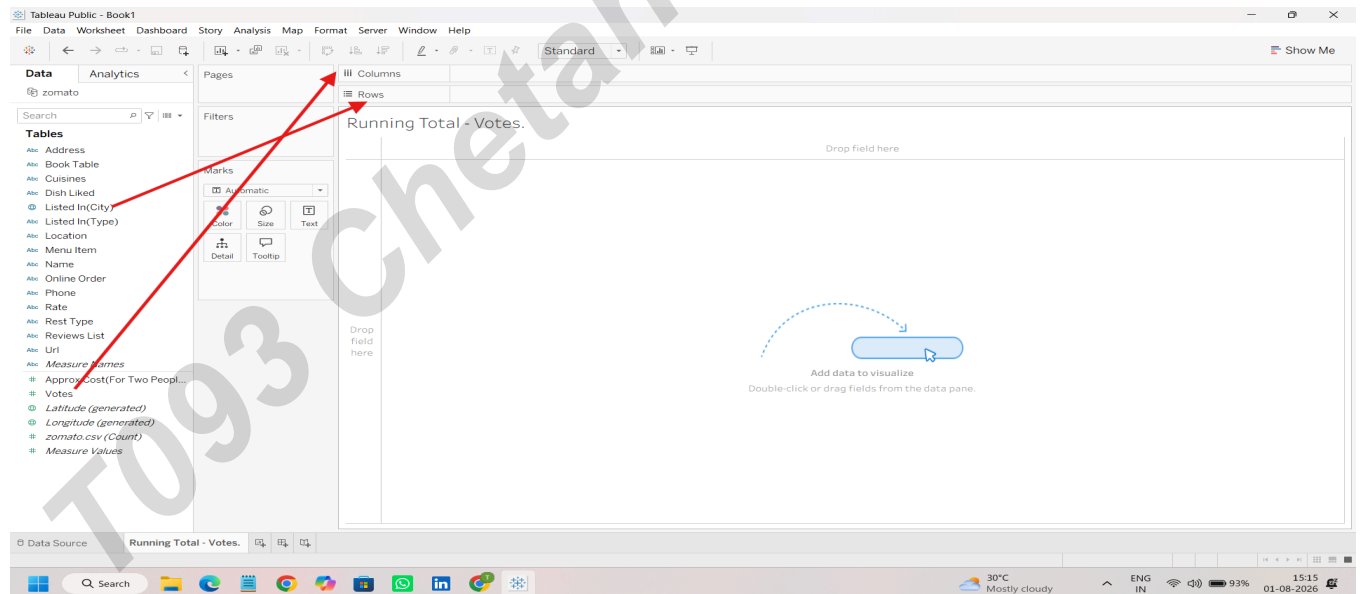
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Step 2:- Open a new worksheet and rename it: Running Total - Votes.



Step 3:- From the Data Pane, drag listed_in(city) to the Rows shelf. Drag votes to the Columns shelf

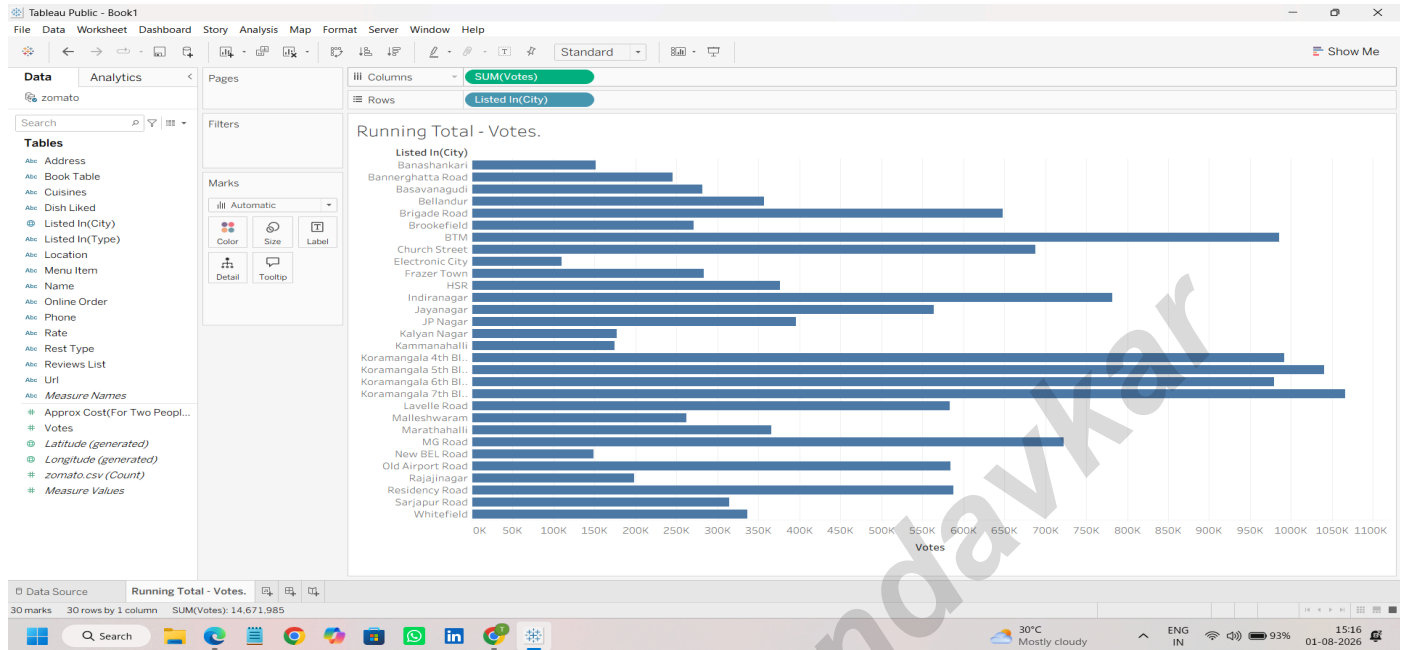


Name:- Chetan Mandavkar

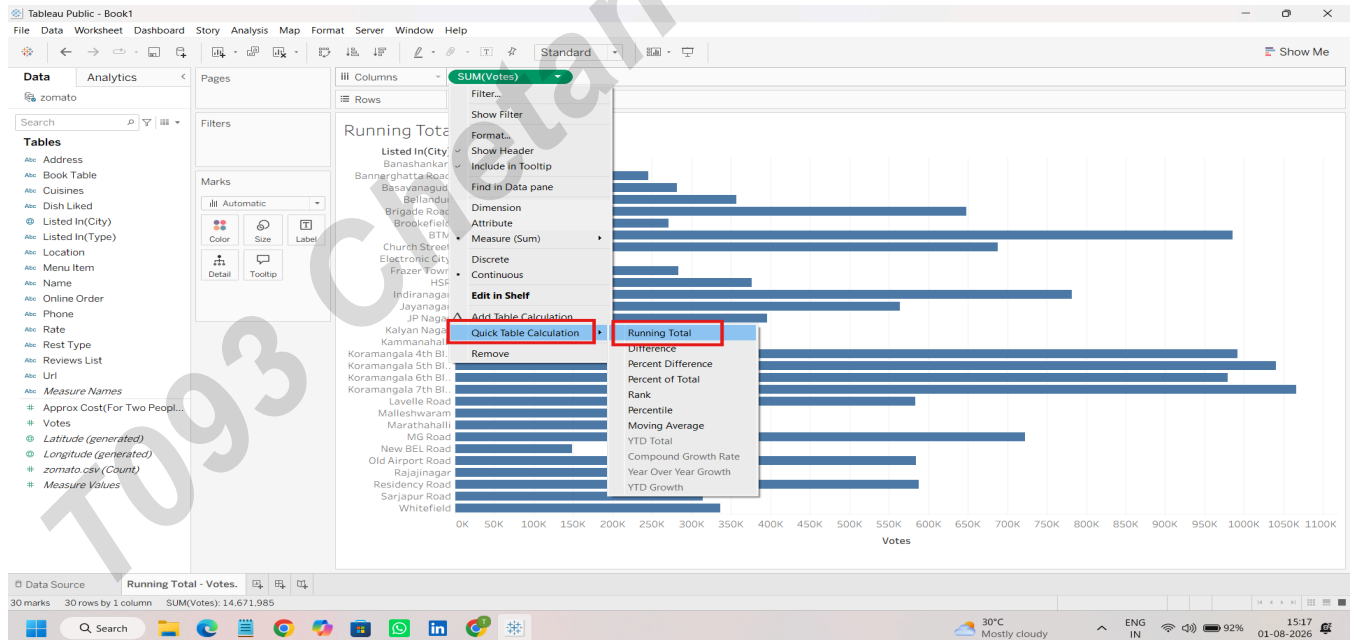
Roll No. T093

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Step 4:- Convert to Quick Table Calculation: Right-click the SUM(votes) pill on the Columns shelf. Hover over Quick Table Calculation and select Running Total.

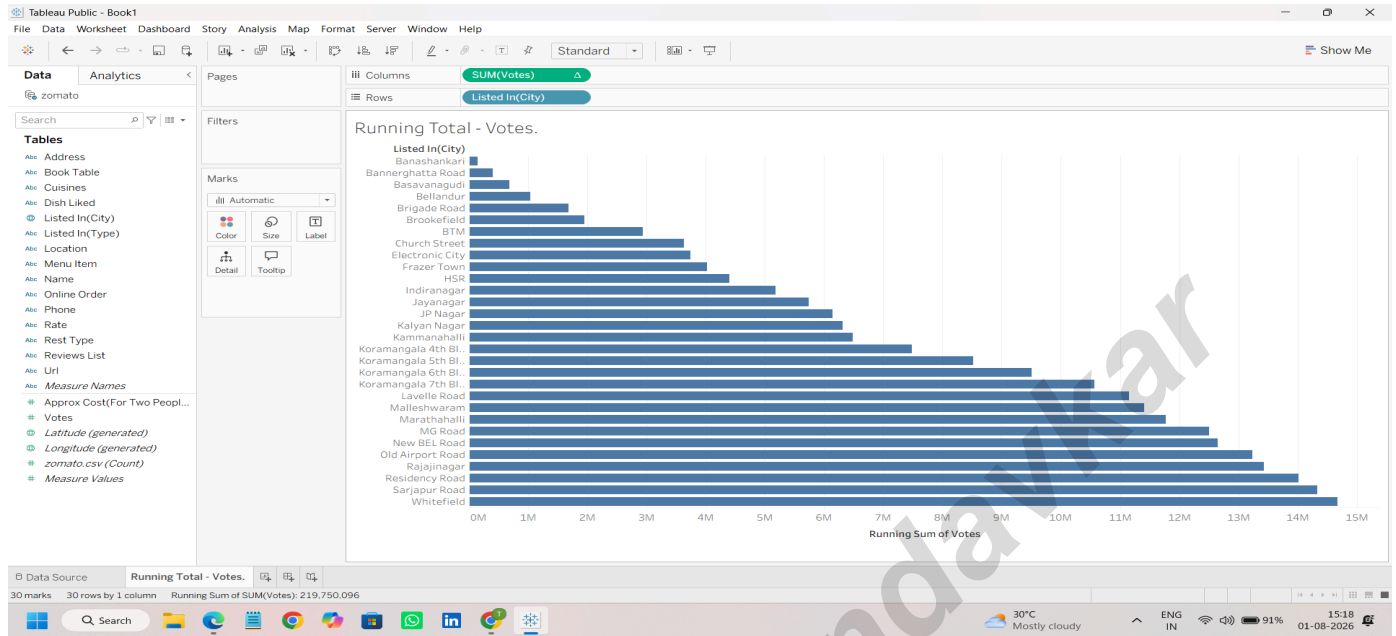


Name:- Chetan Mandavkar

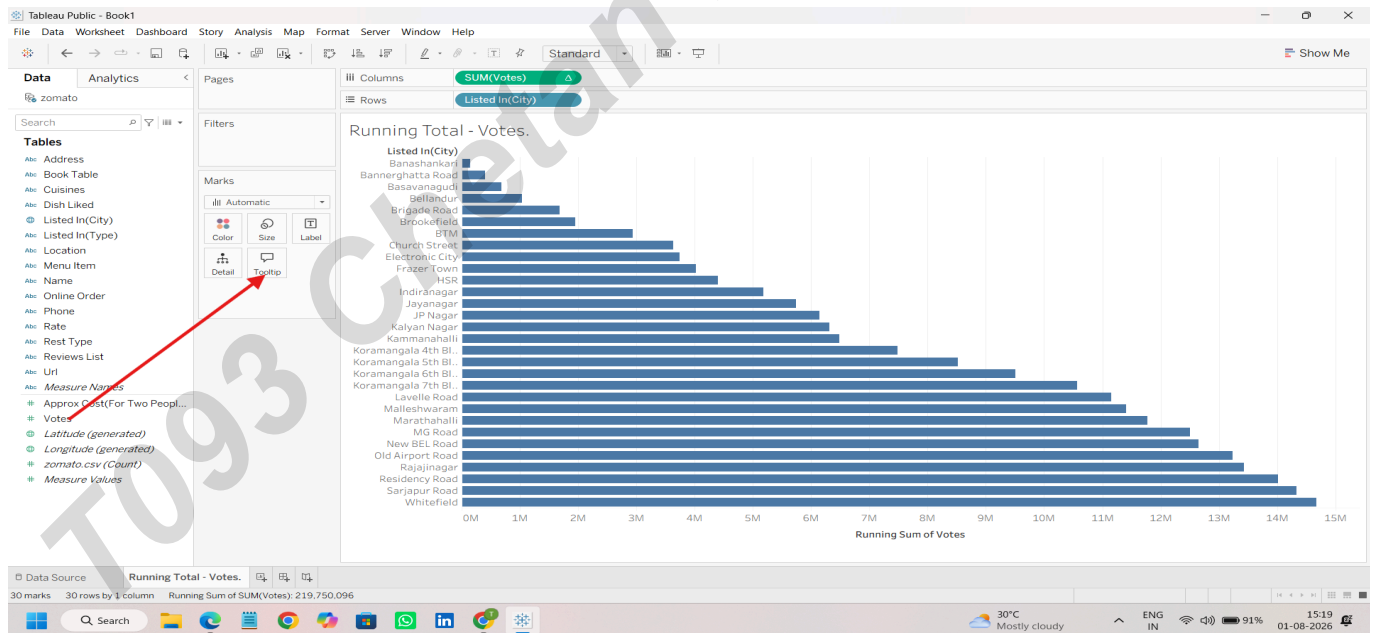
Roll No. T093

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Step 5:- Add Visual Context (Secondary Measure): Drag votes from the Data Pane again and drop it directly onto the Tooltip mark in the Marks Card.

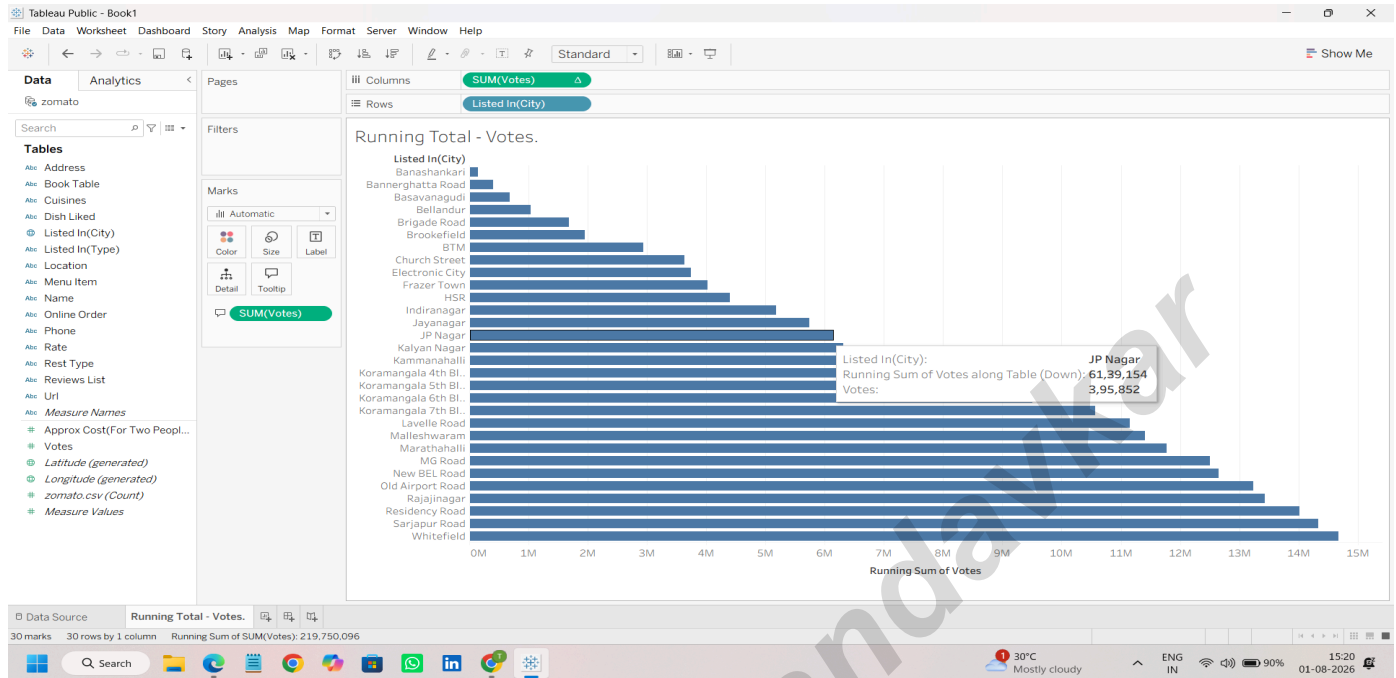


Name:- Chetan Mandavkar

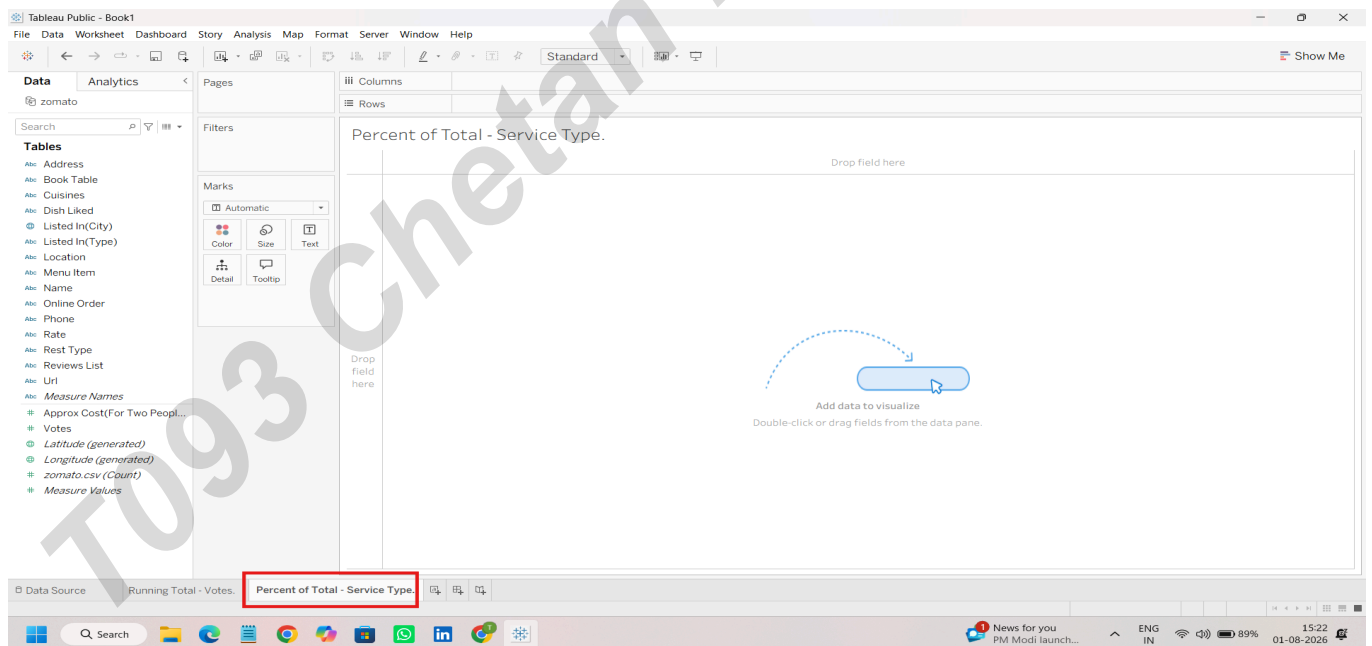
Roll No. T093

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Step 6:- Open a new worksheet and rename it: Percent of Total - Service Type.



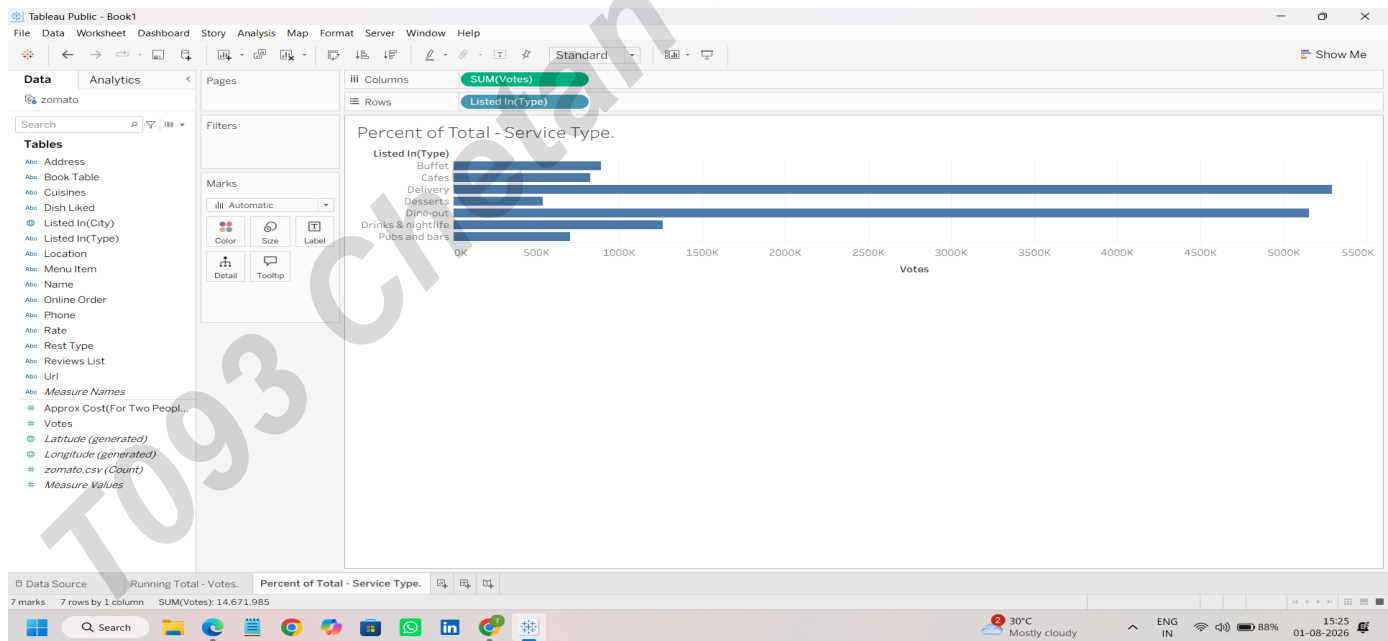
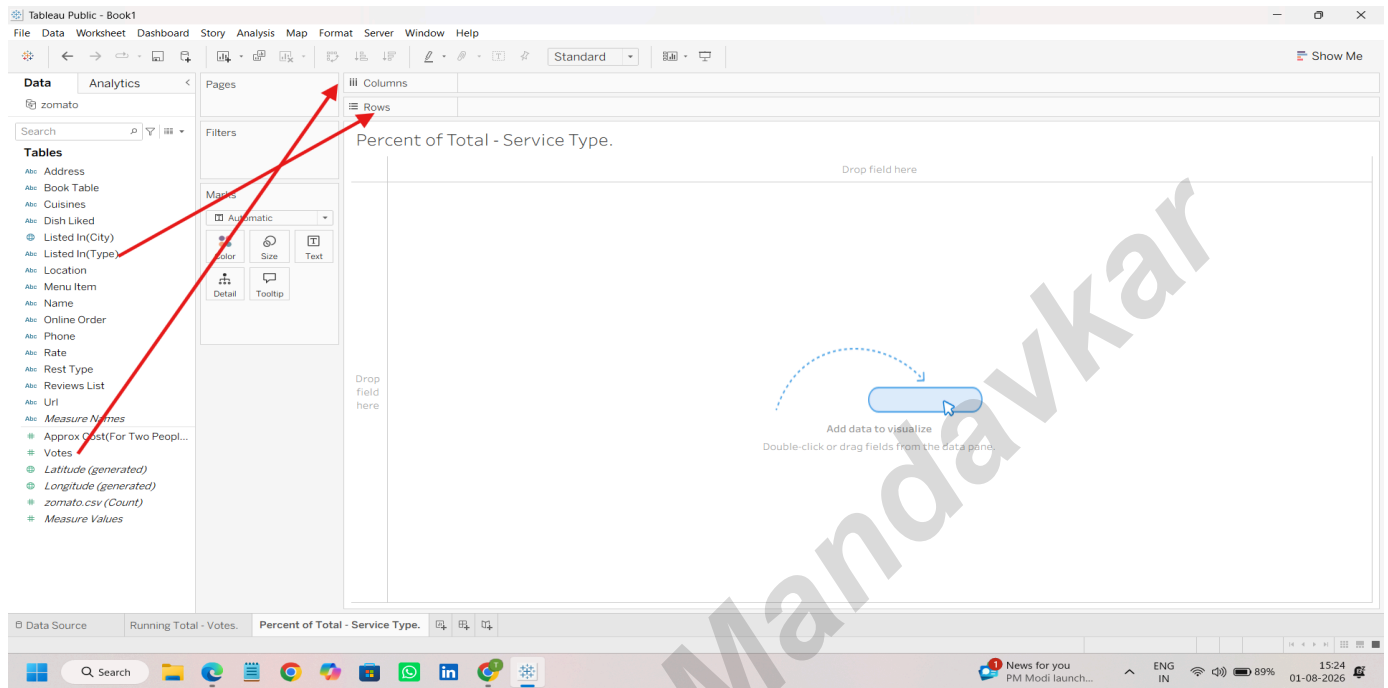
Name:- Chetan Mandavkar

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Step 7:- Drag listed_in(type) to the Rows shelf. Drag votes to the Columns shelf.



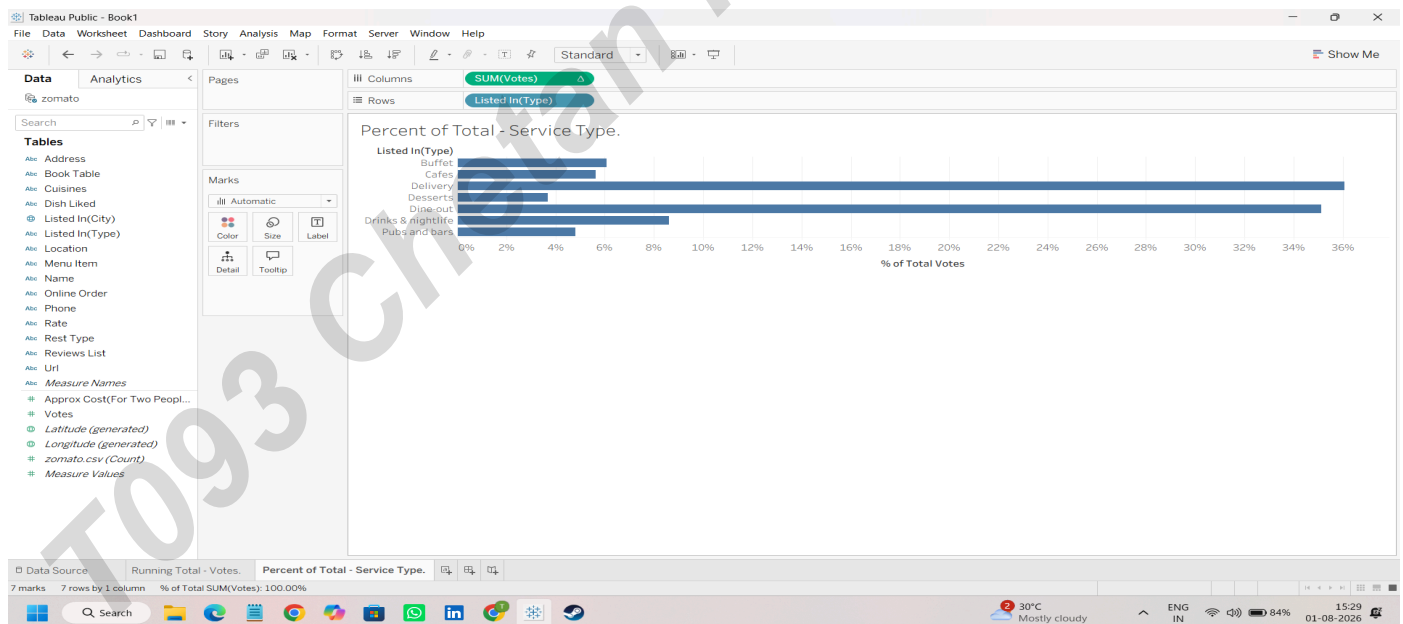
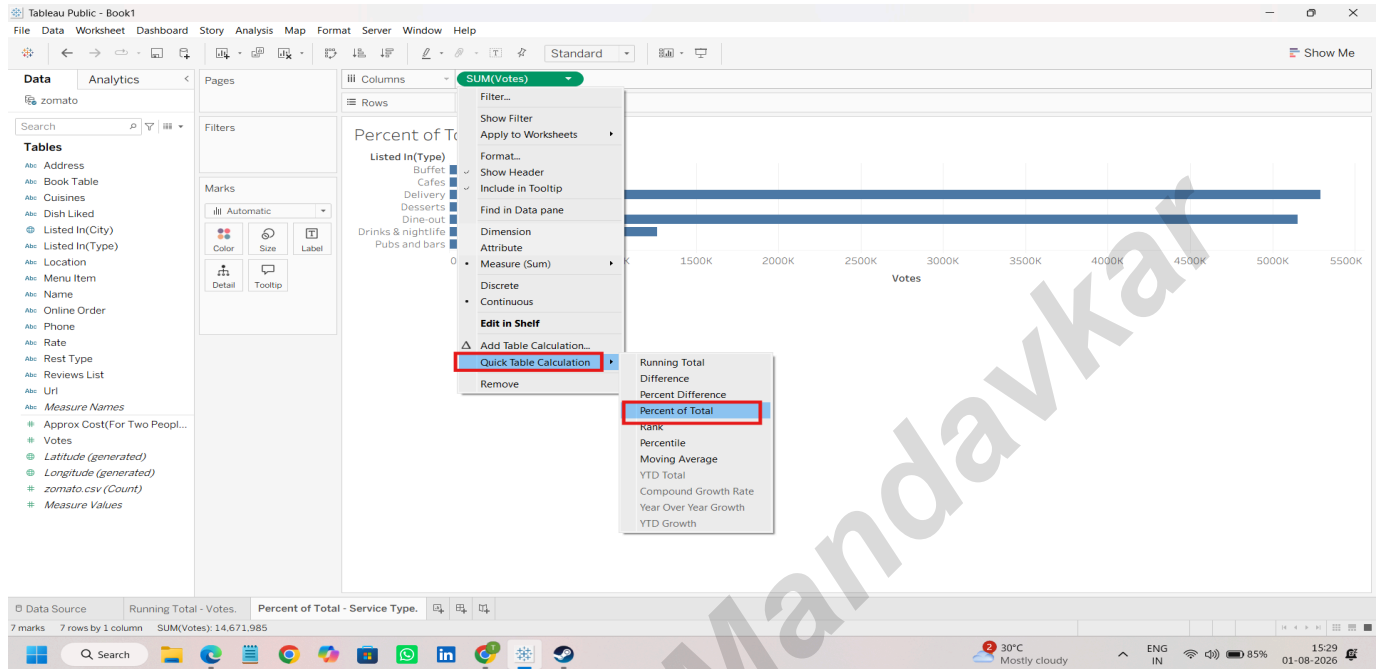
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Step 8:- Right-click the SUM(votes) pill on the Columns shelf. Hover over Quick Table Calculation and select Percent of Total.



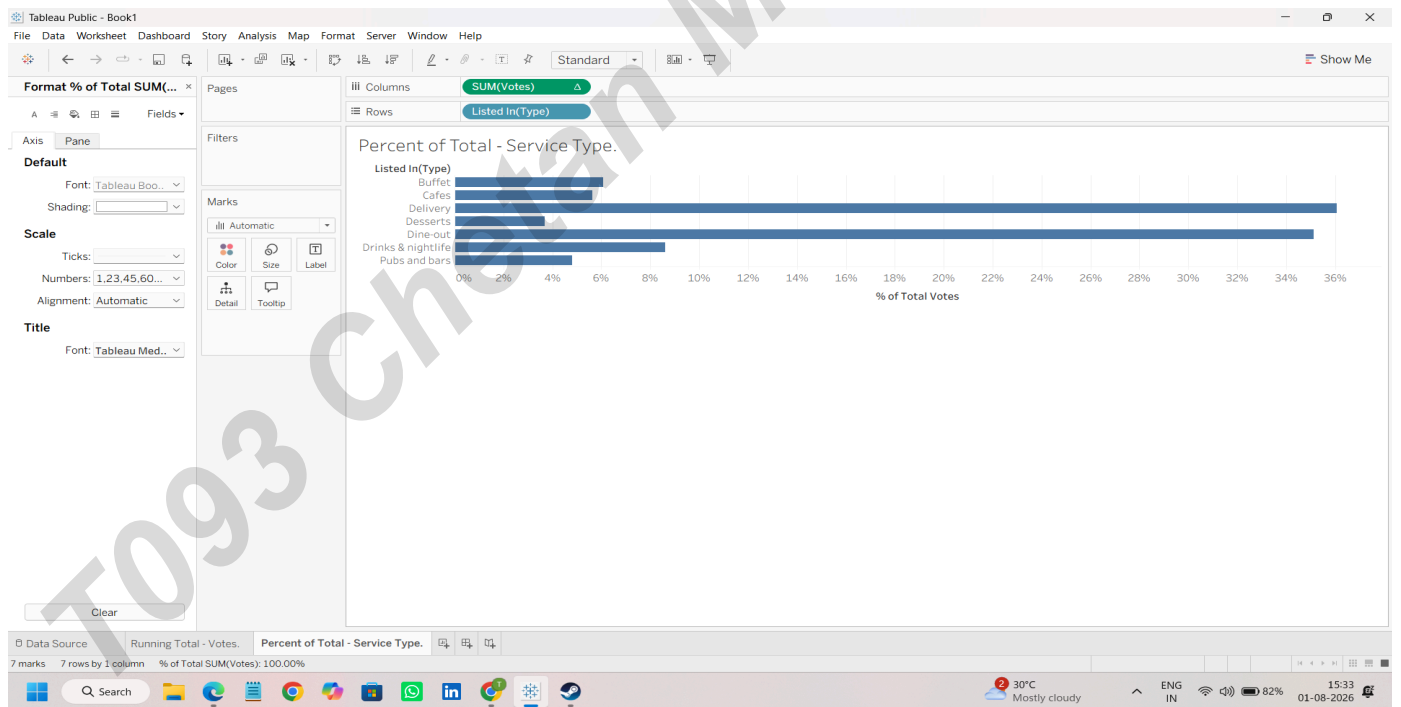
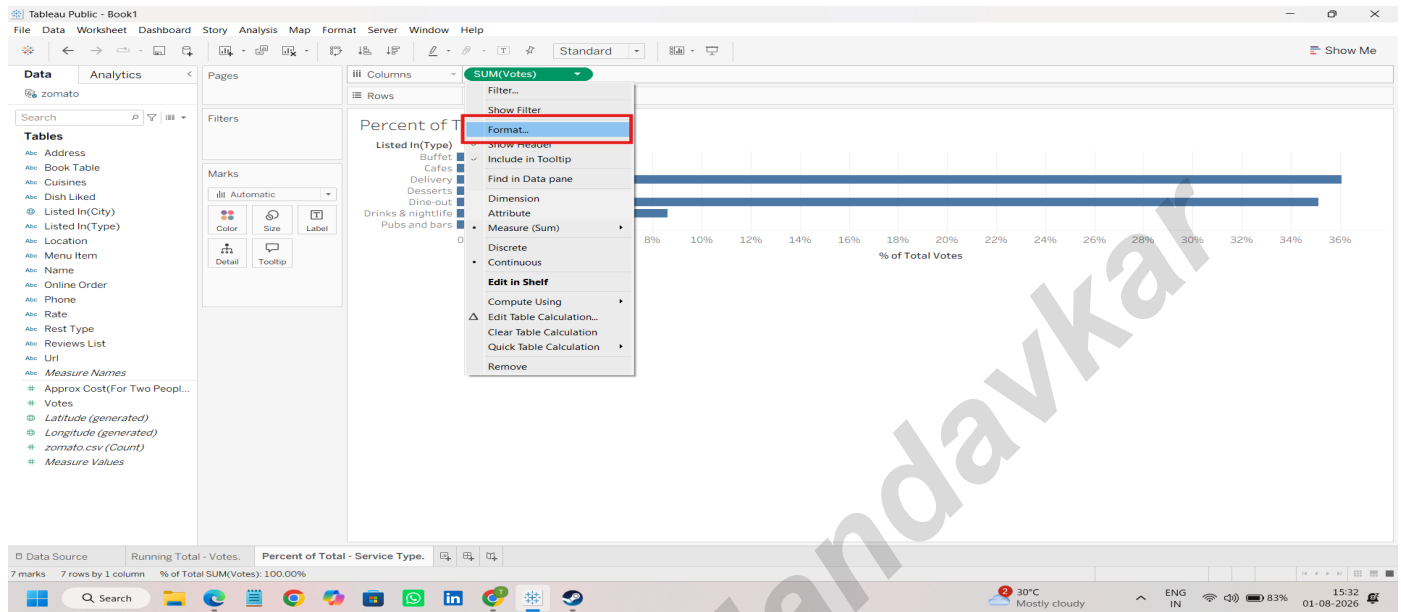
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Step 9:- Format as Percentage: Right-click the SUM(Votes) pill again and select Format...



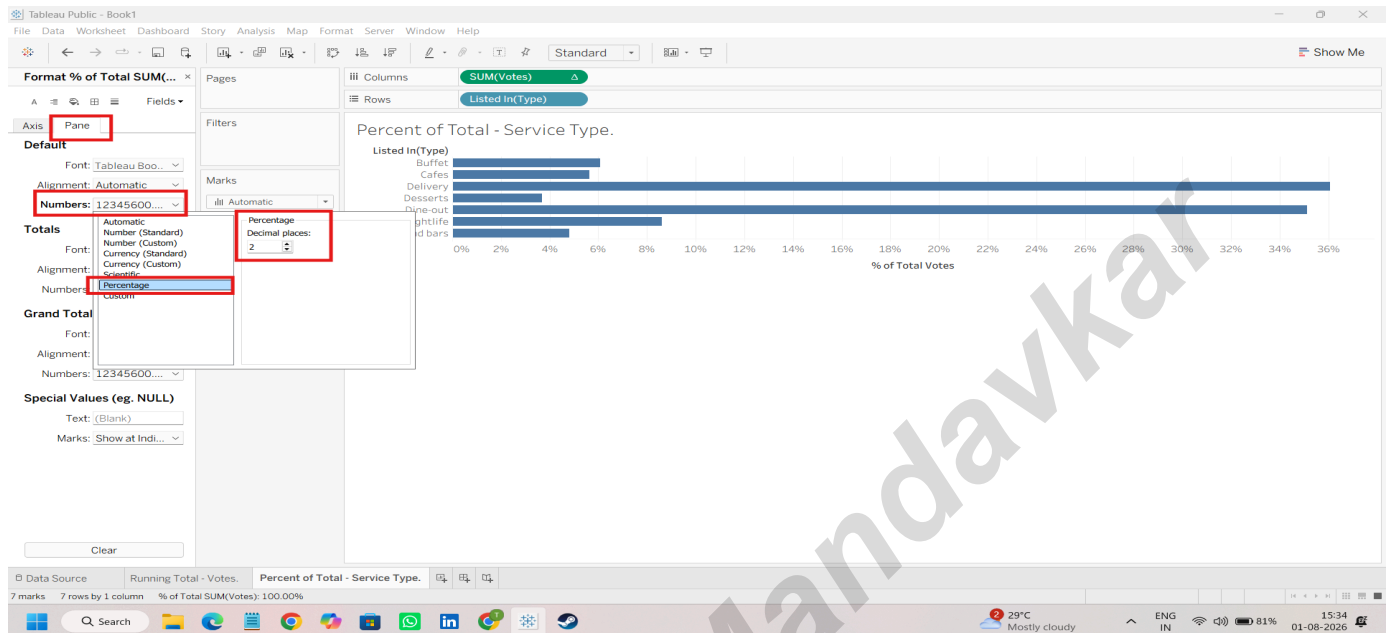
Name:- Chetan Mandavkar

Roll No. T093

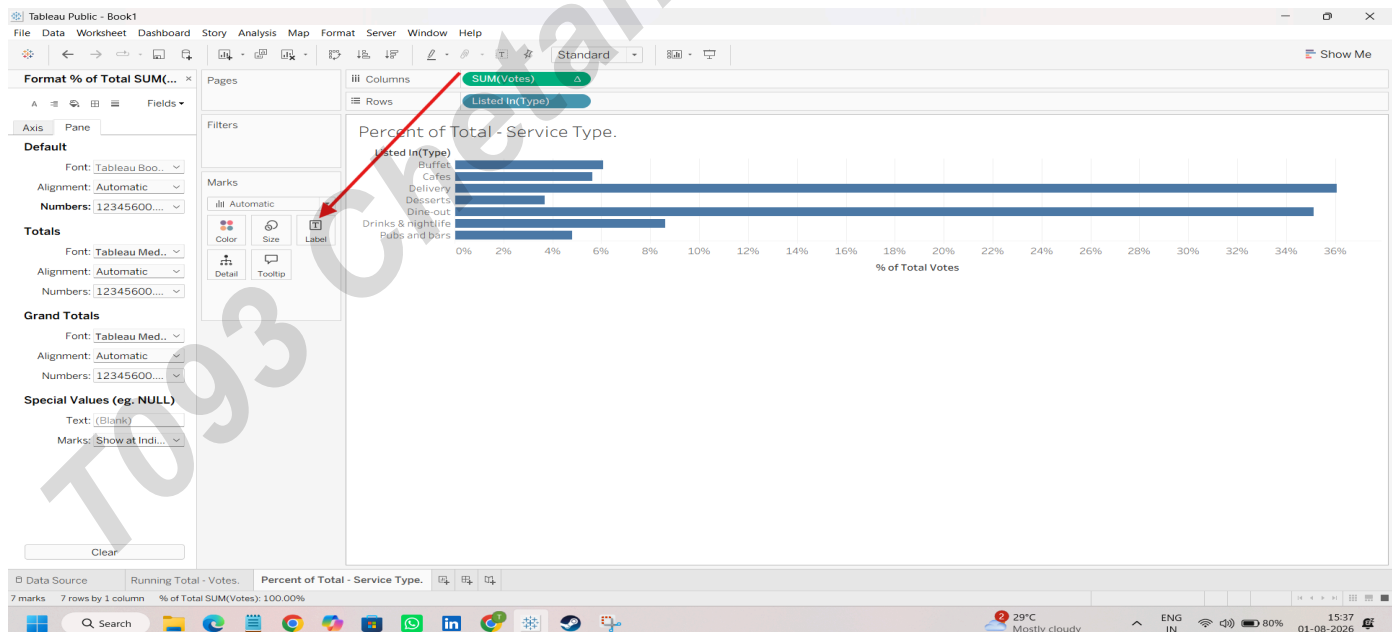
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Under the Pane tab on the left formatting panel, click Numbers $\rightarrow$ Percentage (set to 1 or 2 decimal places).



Step 10:- Drag SUM(votes) (with the table calculation icon Δ) onto the Label mark so the exact percentages display on top of each bar.

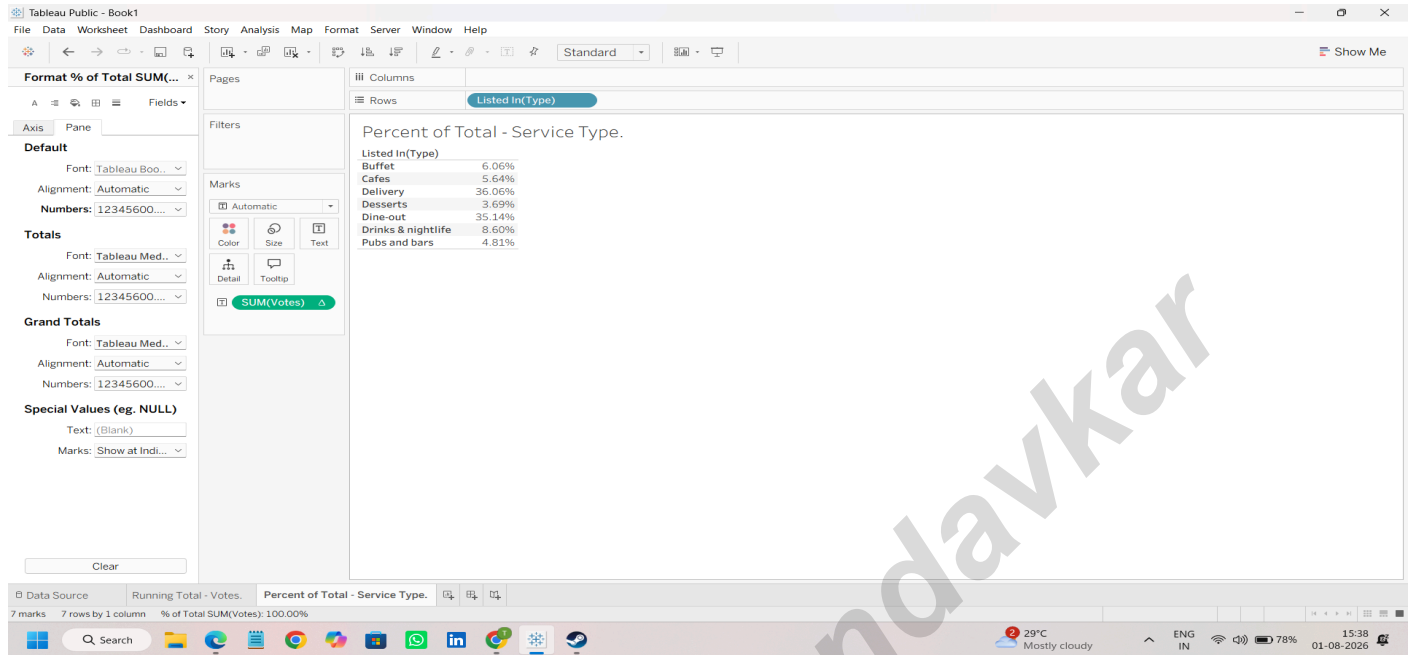


Name:- Chetan Mandavkar

Roll No. T093

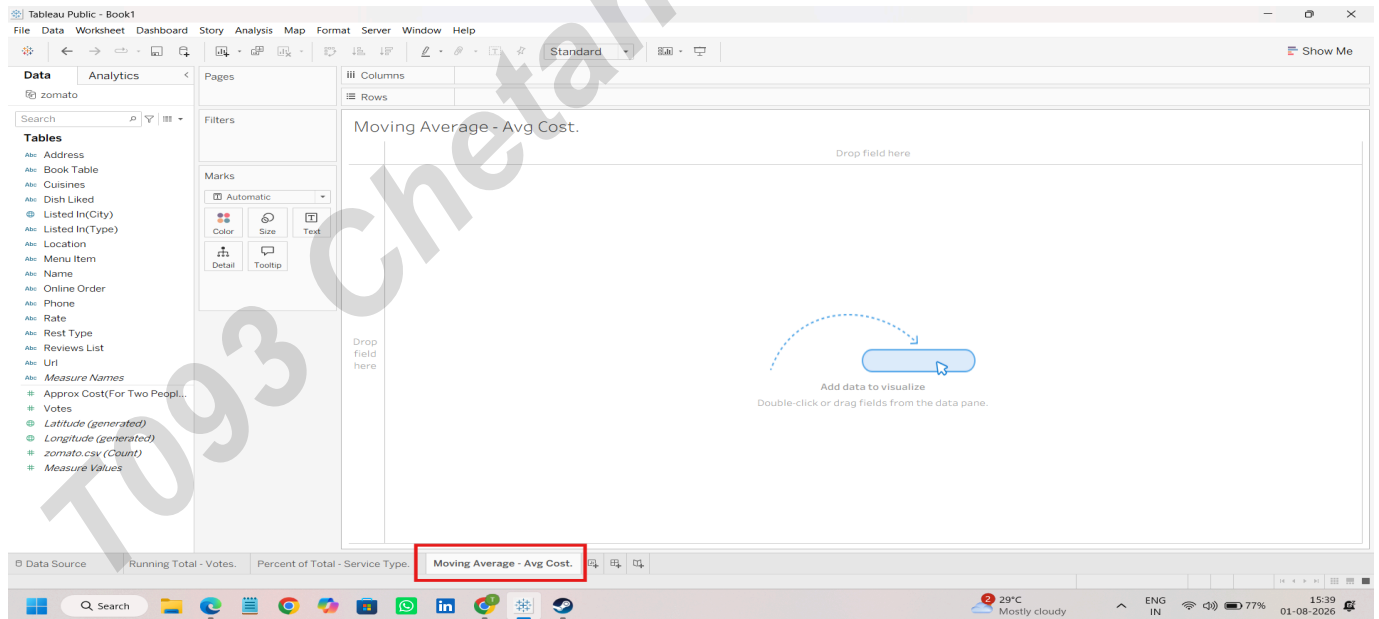
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B] Create moving averages and cumulative metrics

Step 1:- Open a new worksheet and rename it: Moving Average - Avg Cost.



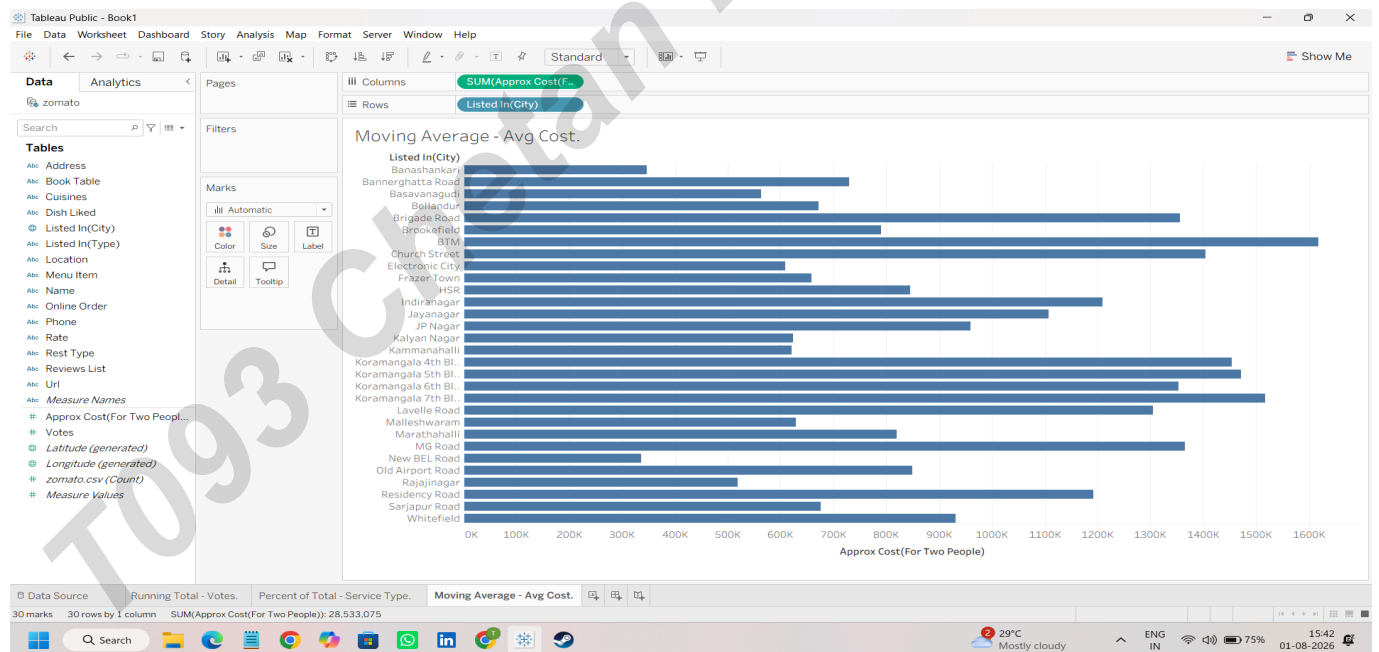
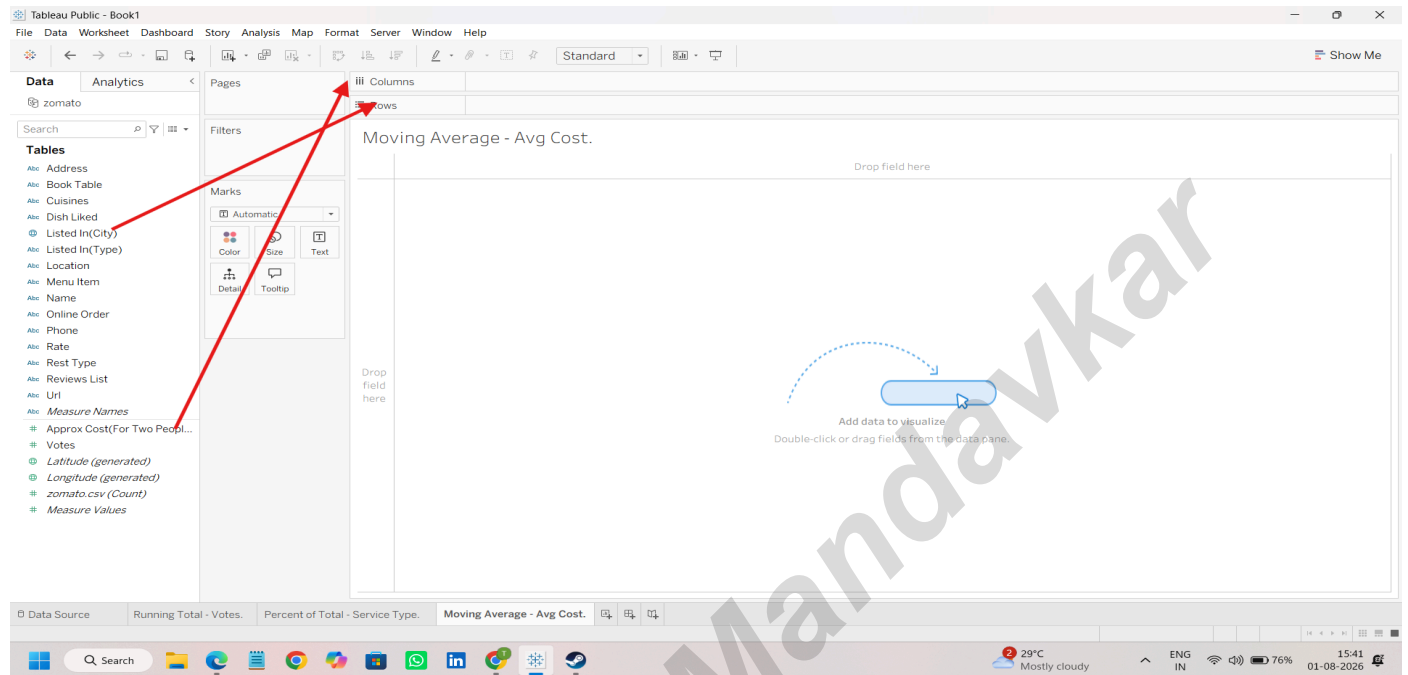
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Step 2:- Drag listed_in(city) to the Rows shelf. Drag approx_cost(for two people) to the Columns shelf.



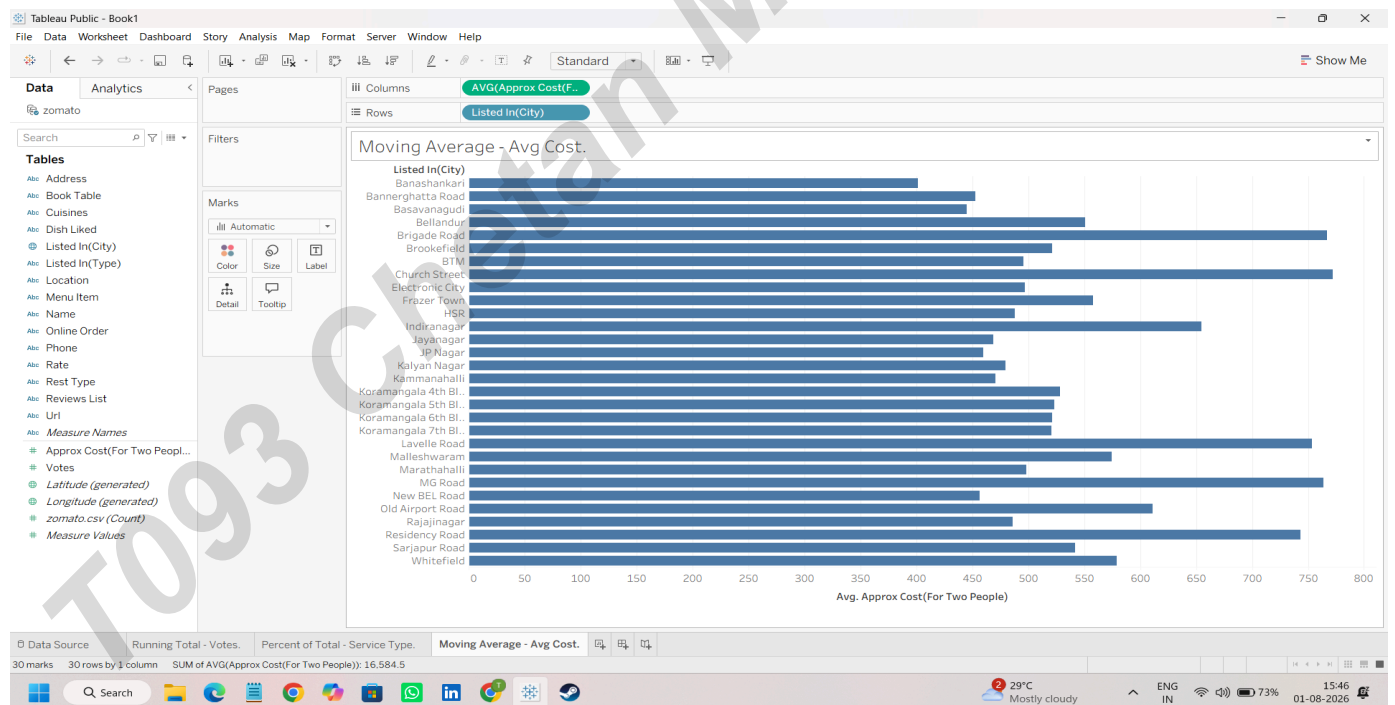
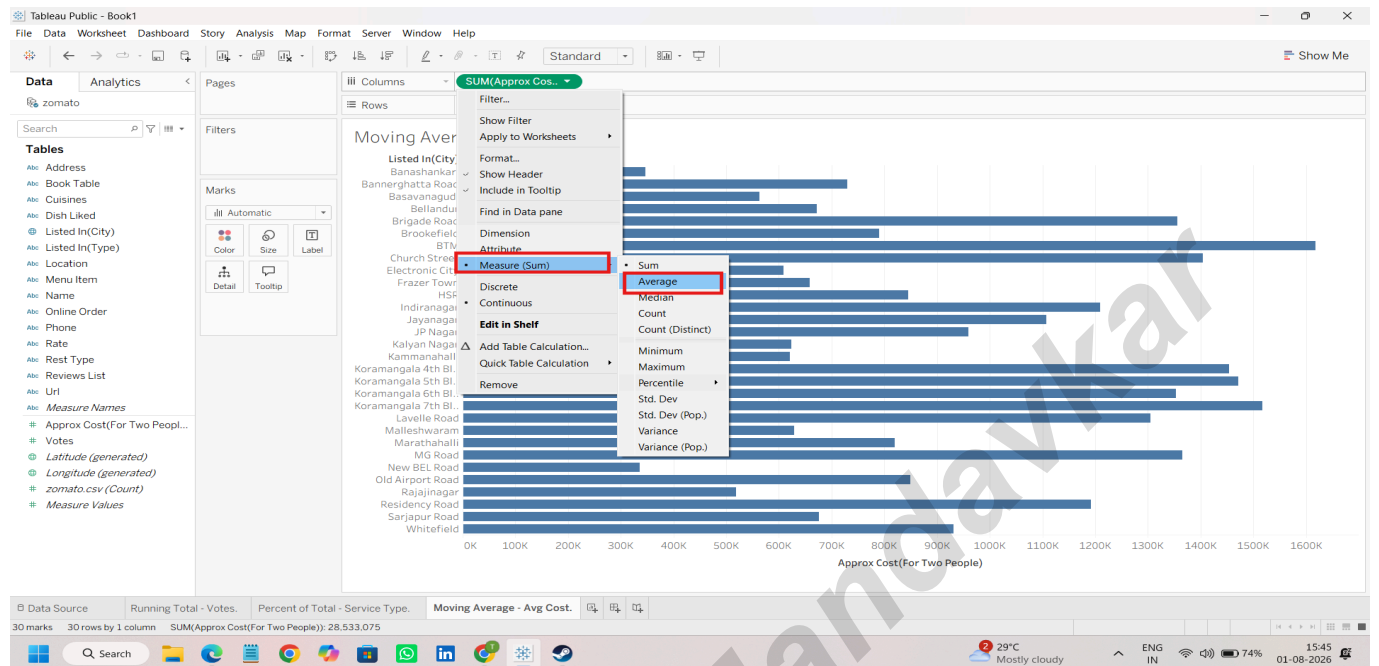
Name:- Chetan Mandavkar

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Step 3:- click on it Hover at Measure and click Average.



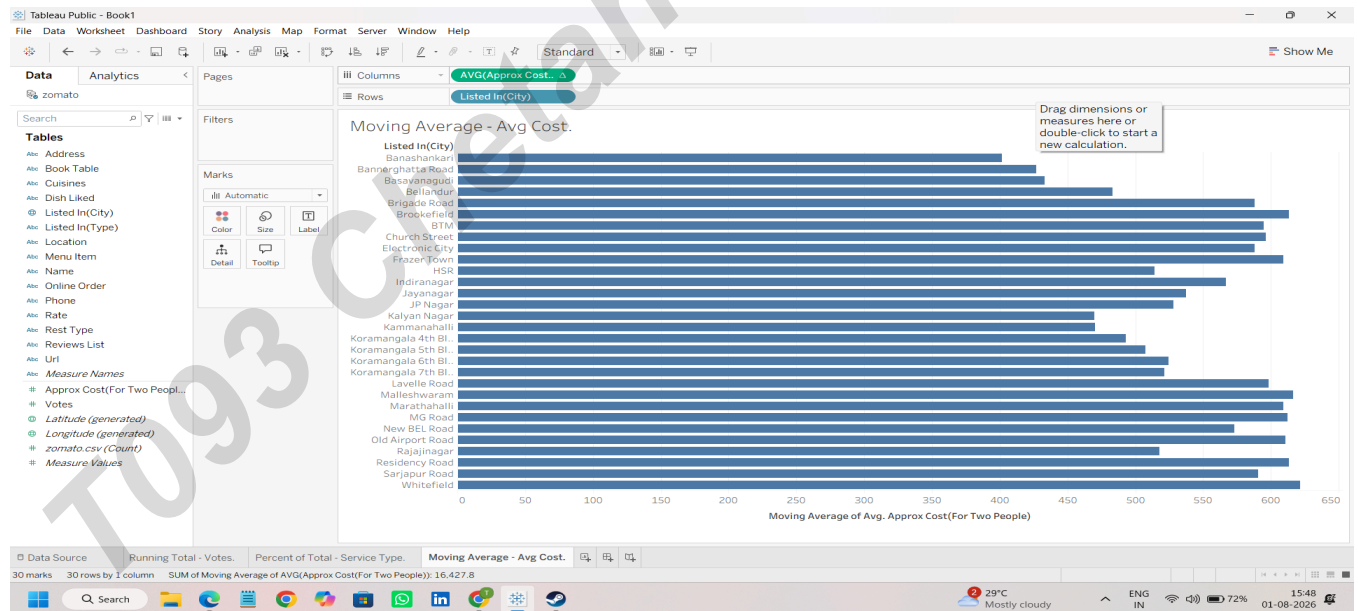
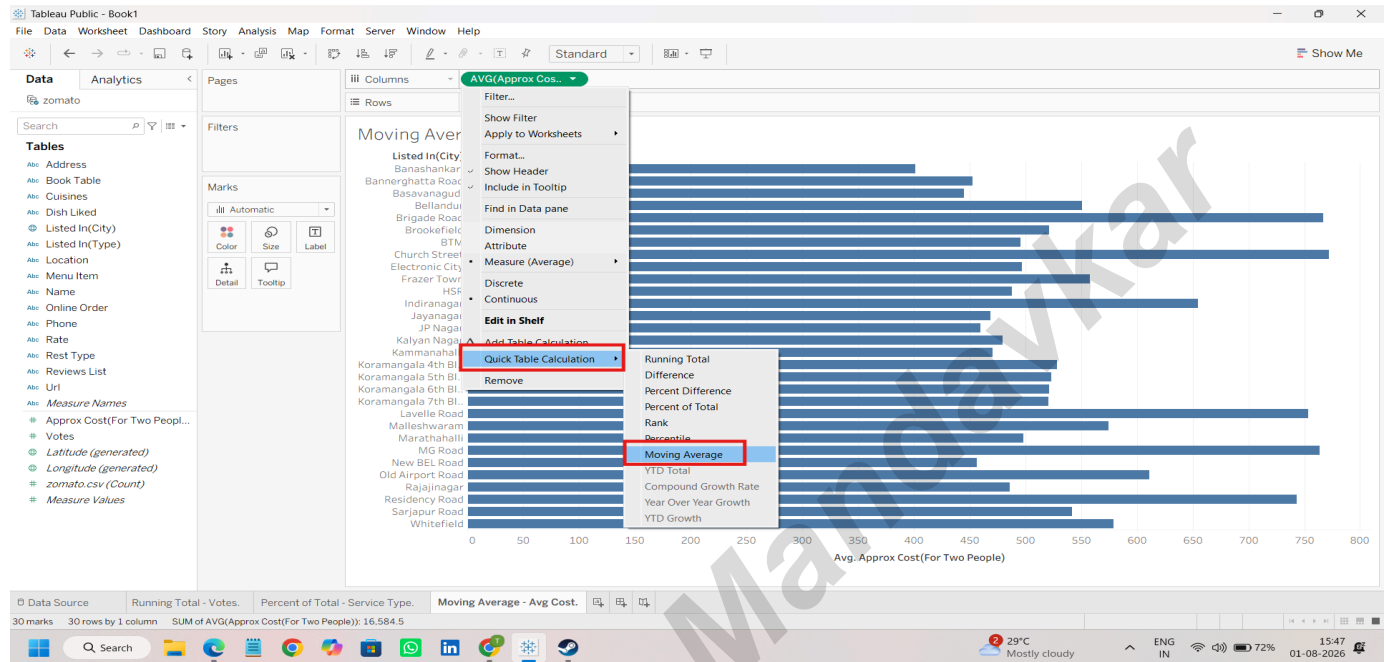
Name:- Chetan Mandavkar

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Step 4:- Convert to a Moving Average Quick Table Calculation:
Right-click the AVG(approx_cost...) pill on the Columns shelf. Hover over Quick Table Calculation and select Moving Average.



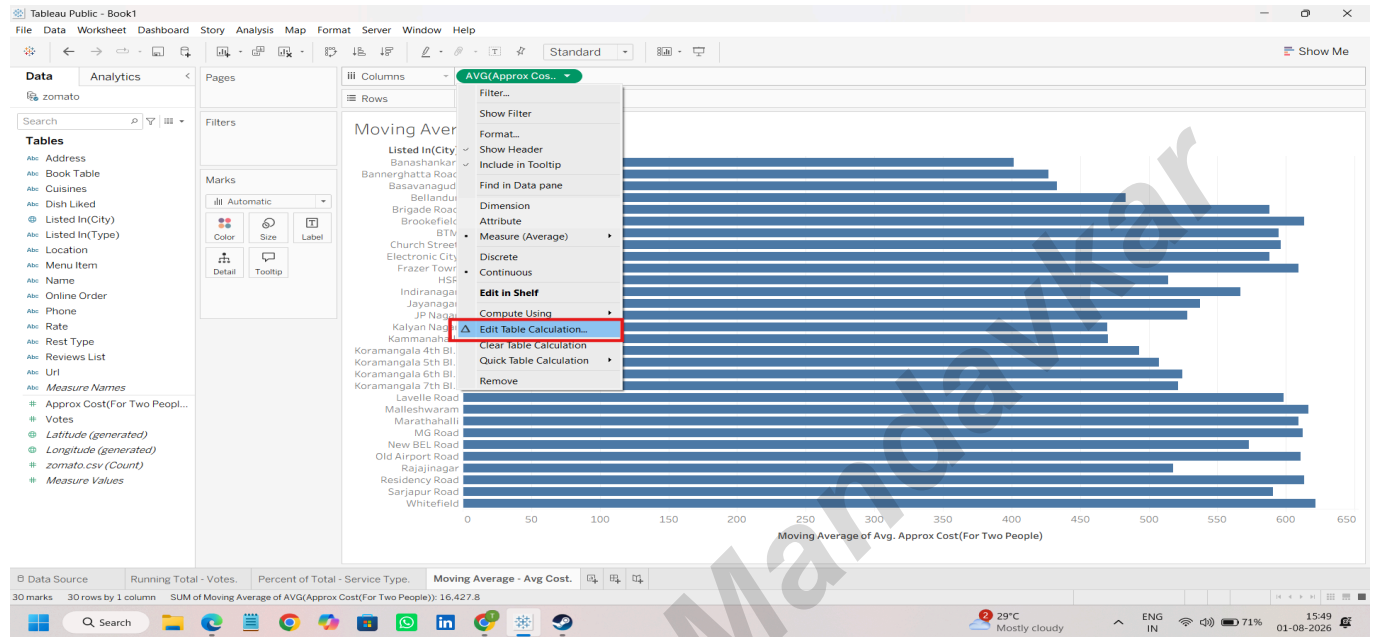
Name:- Chetan Mandavkar

Roll No. T093

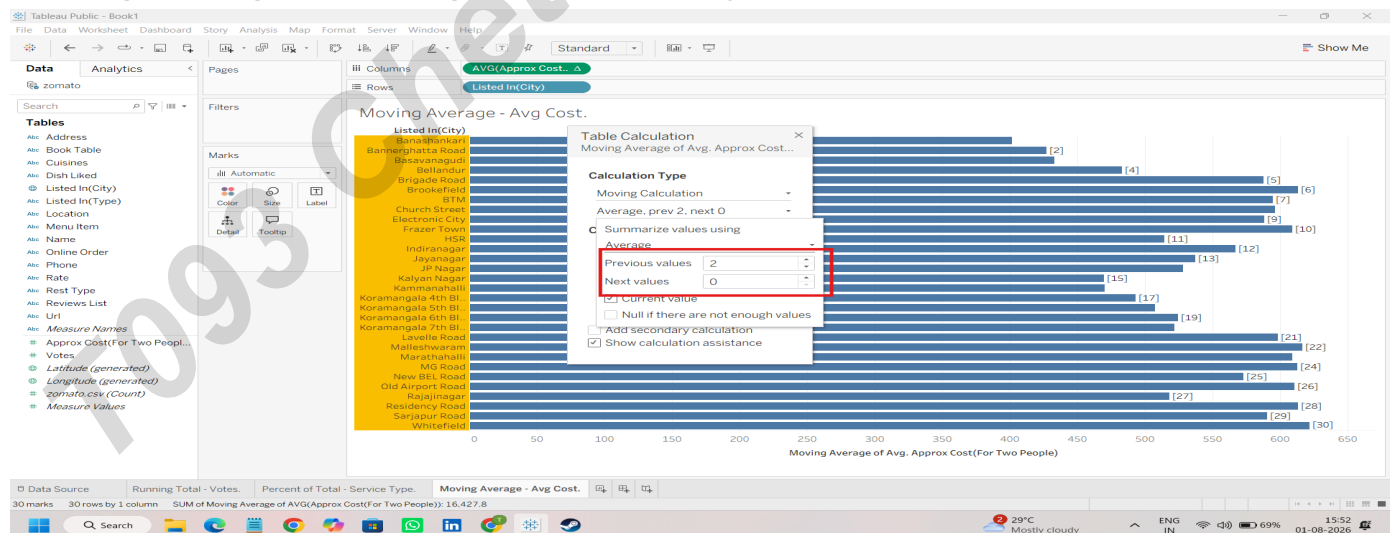
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Step 5:- Customize the Moving Window: Right-click the modified AVG(approx_cost...) pill again (it now has a Δ symbol). Select Edit Table Calculation...



In the configuration dialog, set Summarize values using: Average. Set Previous values: 2 and Next values: 0 (This calculates a 3-point moving average including the current city and 2 previous cities).



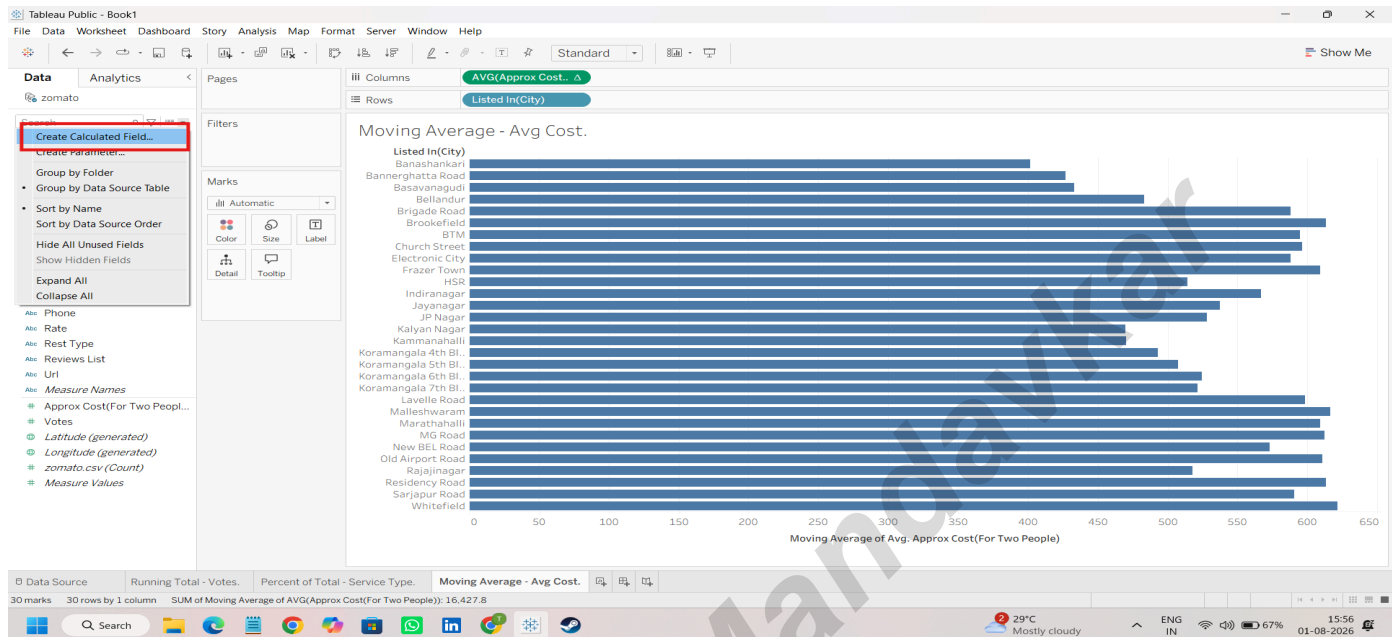
Name:- Chetan Mandavkar

Roll No. T093

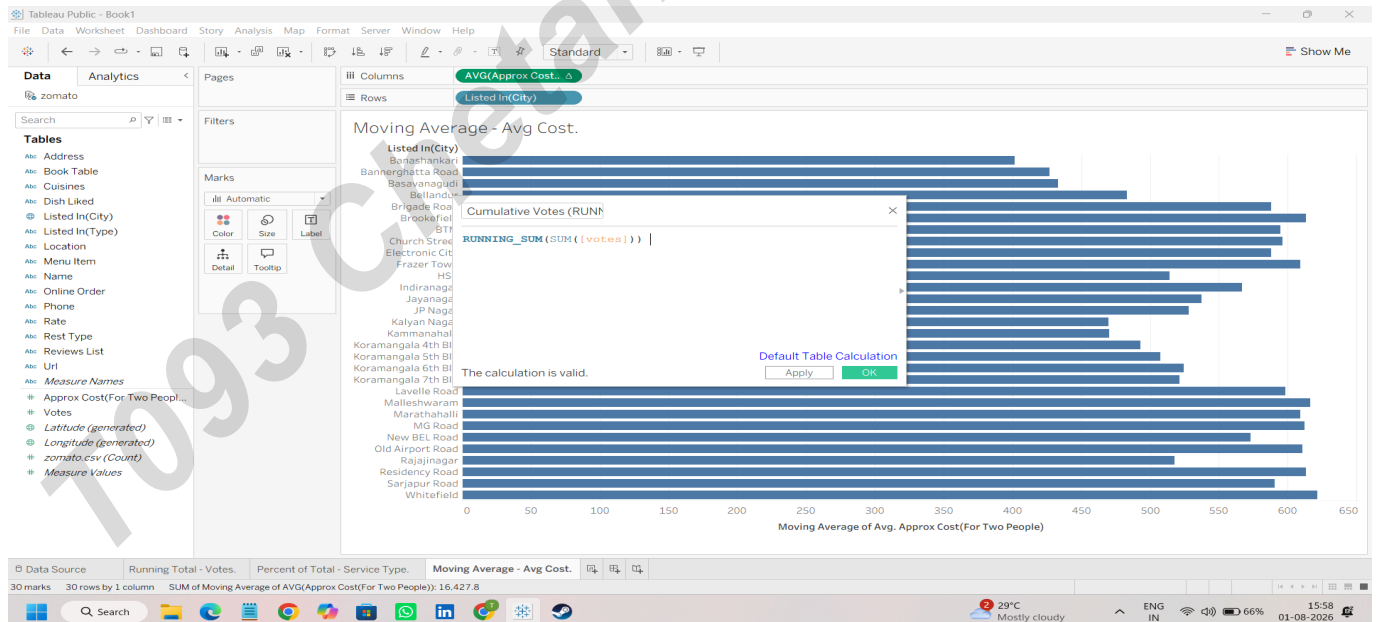
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Step 6:- Right-click anywhere on a specific field or dimension in the Data Pane Create Calculated Field...



Name the field: Cumulative Votes (RUNNING_SUM). Enter the following Table Calculation formula: RUNNING_SUM(SUM([votes]))



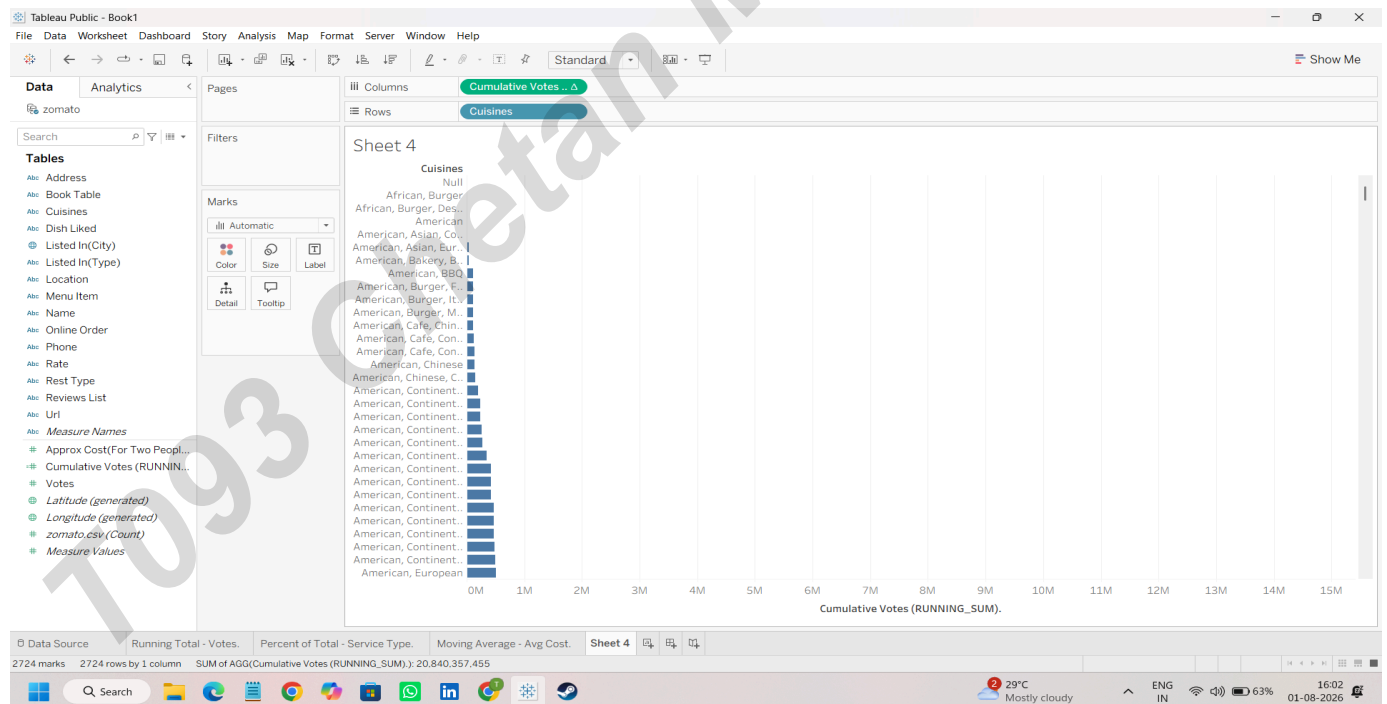
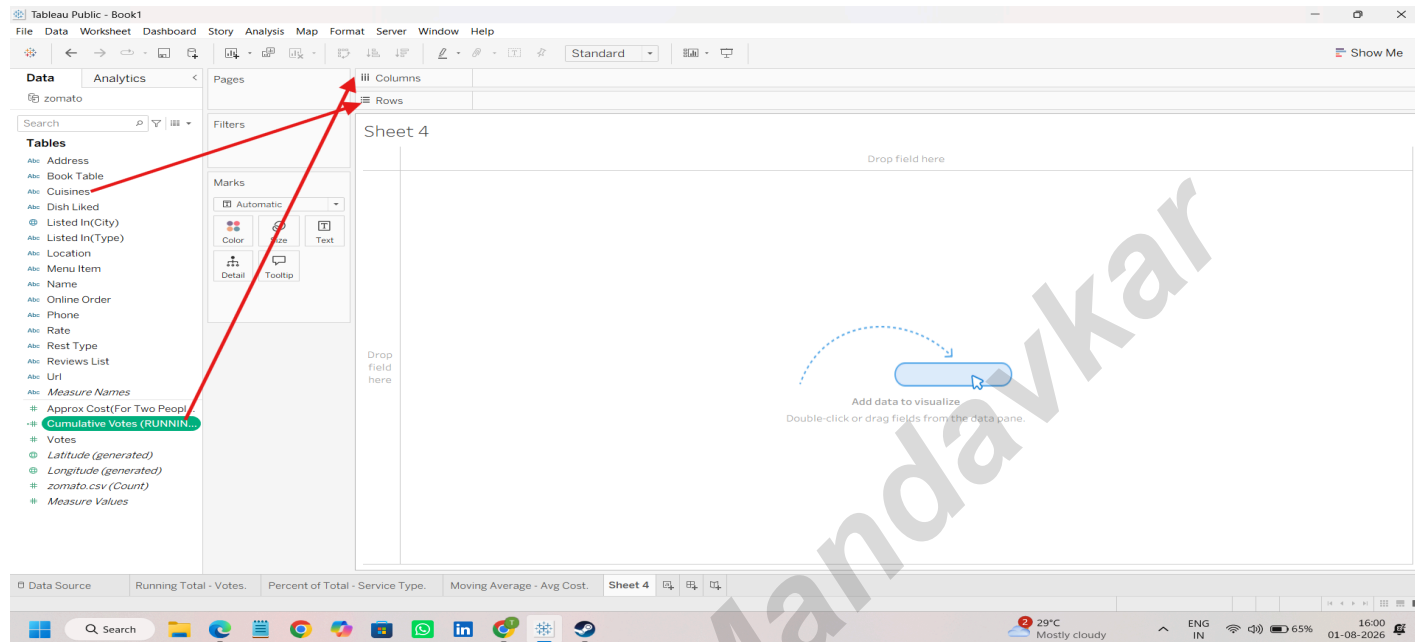
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Step 7:- Open a new worksheet. Drag cuisines to Rows. Drag your newly created Cumulative Votes (RUNNING_SUM) field onto Columns



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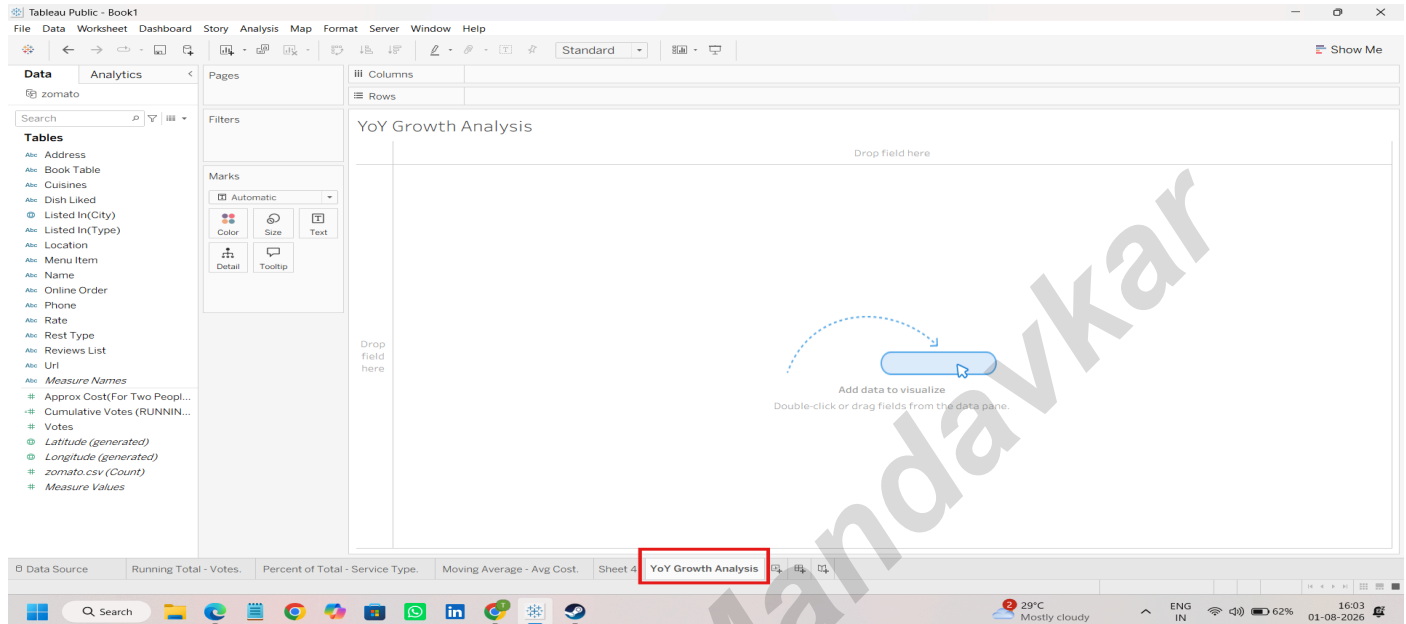
Roll No. T093

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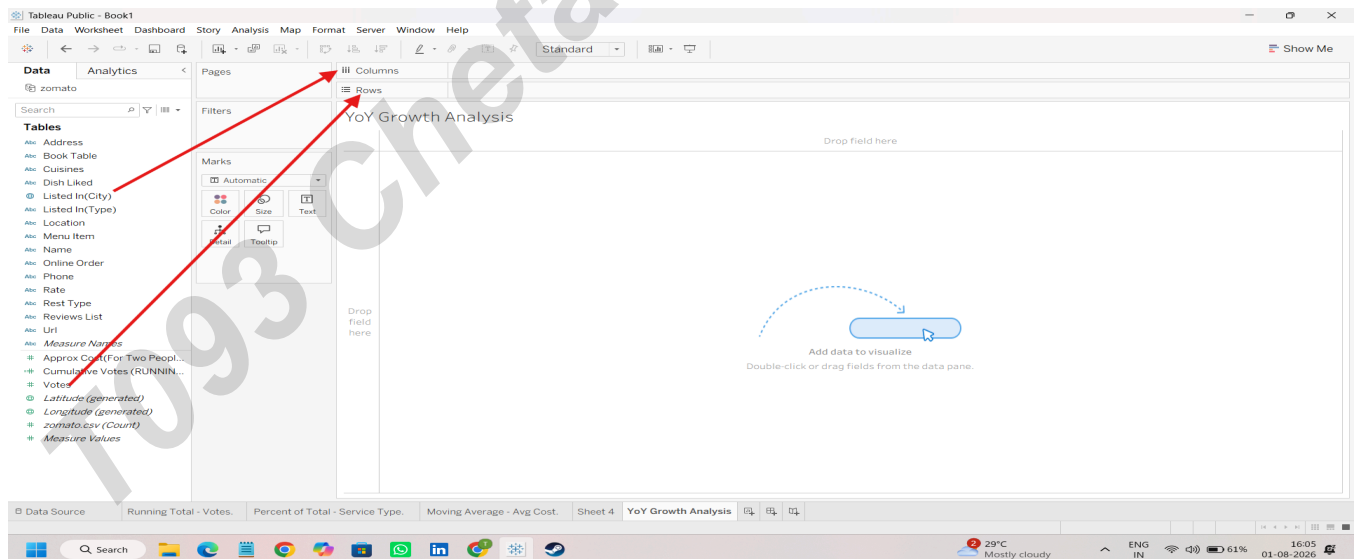
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C] Perform year-over-year growth analysis

Step 1:- Open a new worksheet and rename it: YoY Growth Analysis.



Step 2:- Place your time dimension (or ordinal grouped column like listed_in(city)) on the Columns shelf. Drag votes (or approx_cost(for two people)) onto the Rows shelf as SUM(votes).

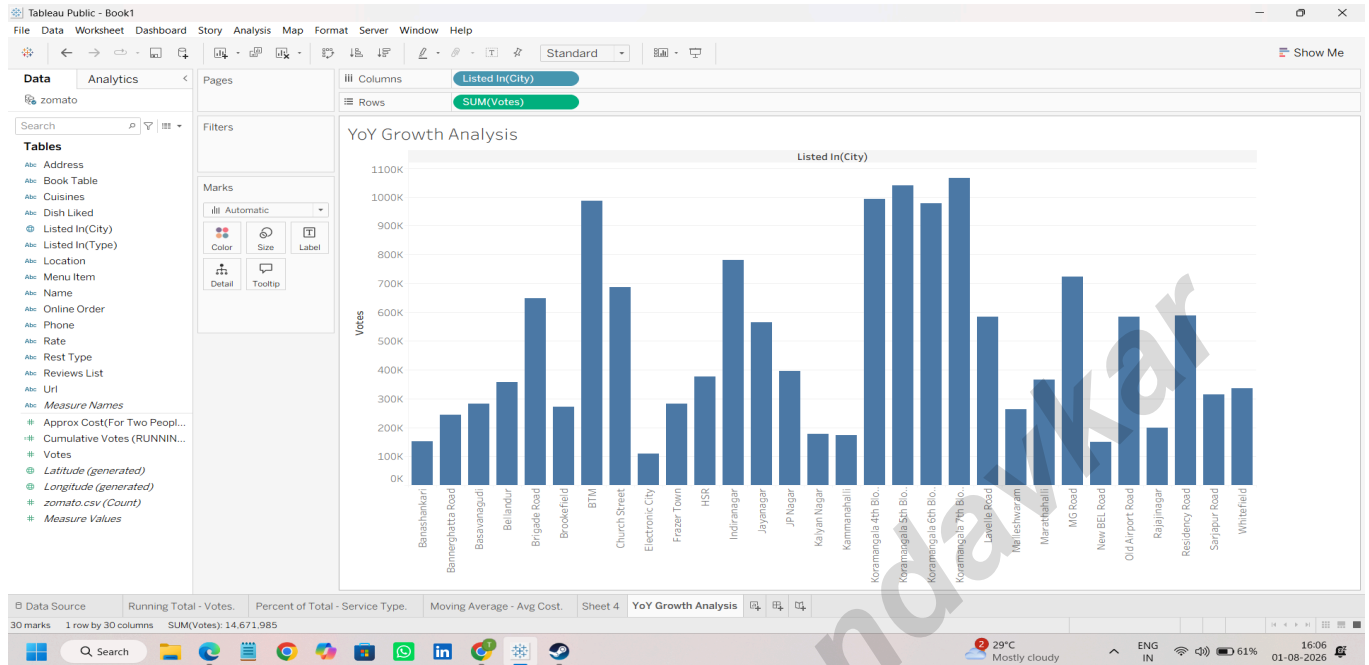


Name:- Chetan Mandavkar

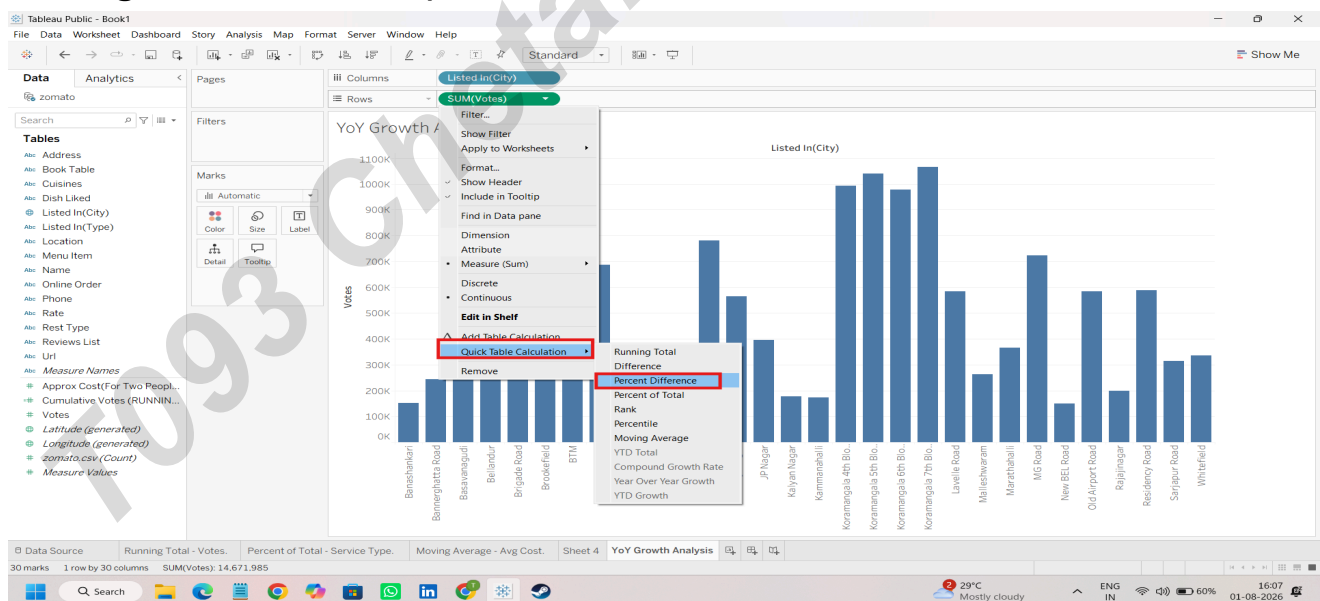
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Step 3:- Apply the Year-over-Year Growth Quick Table Calculation:
Right-click the SUM(votes) pill on the Rows shelf. Hover over Quick Table Calculation select Percent Difference (or Year Over Year Growth if using a true Date field).

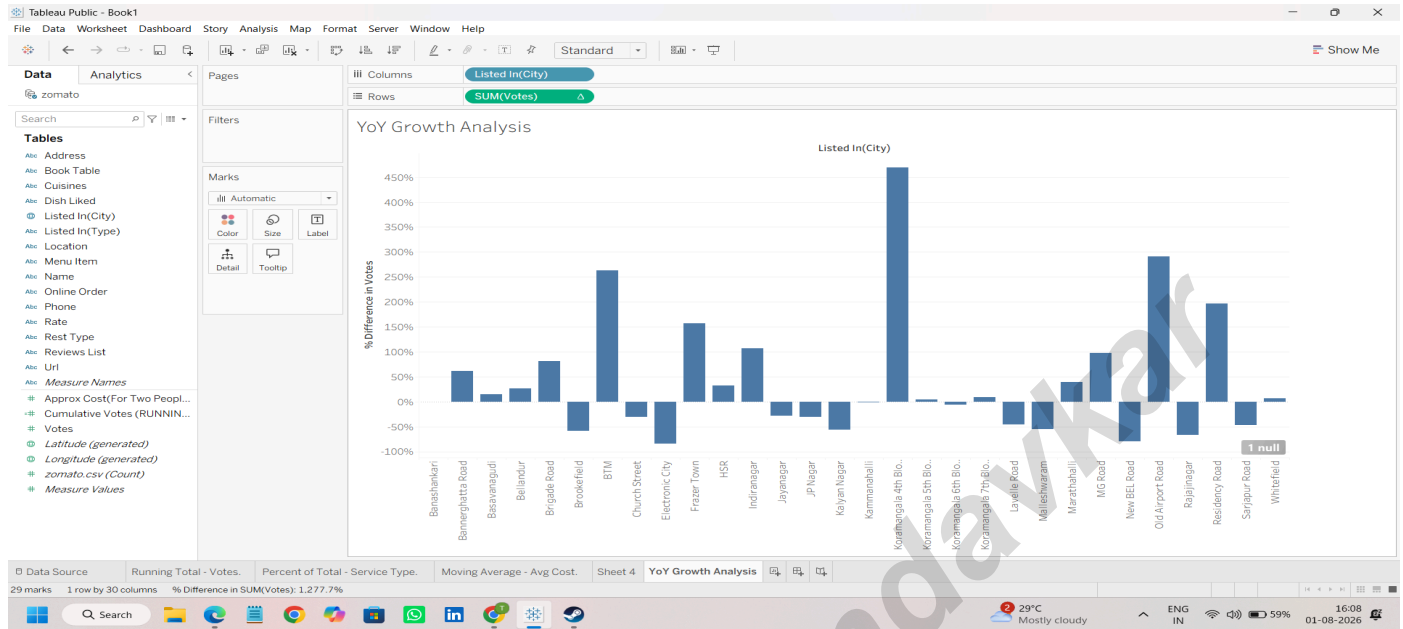


Name:- Chetan Mandavkar

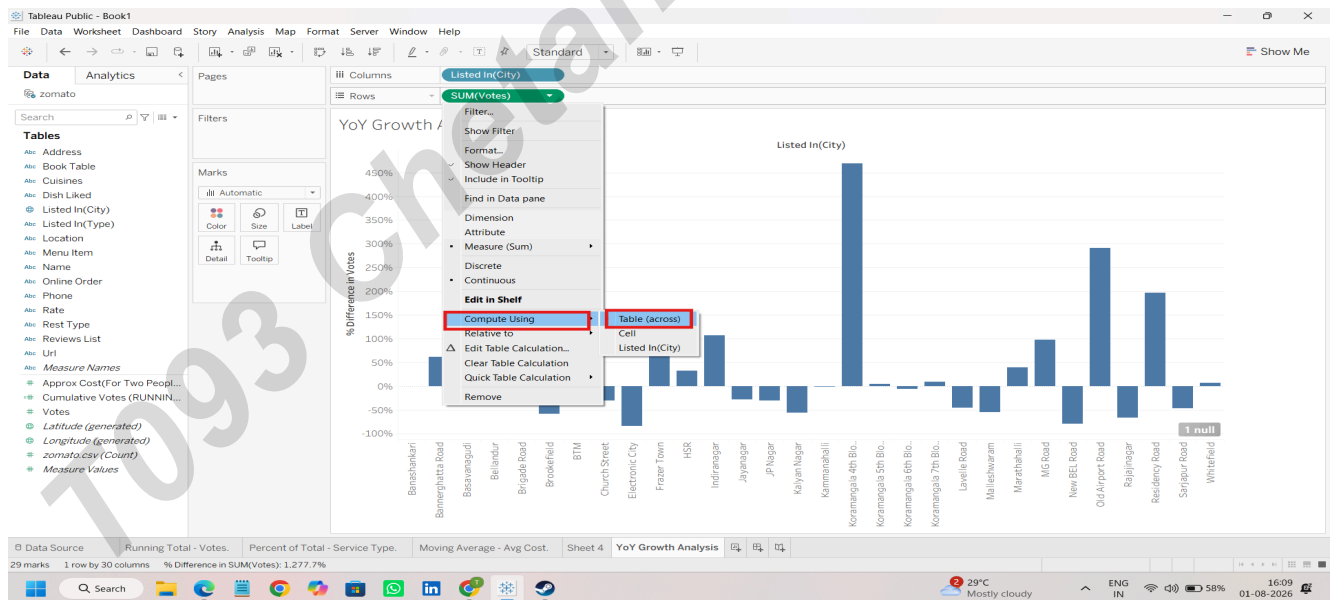
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Step 4:- Configure the Growth Calculation Direction: Right-click the modified SUM(votes) pill (with the Δ icon). Hover over Compute Using select Table (across) {In some cases it may have by selected by default}.



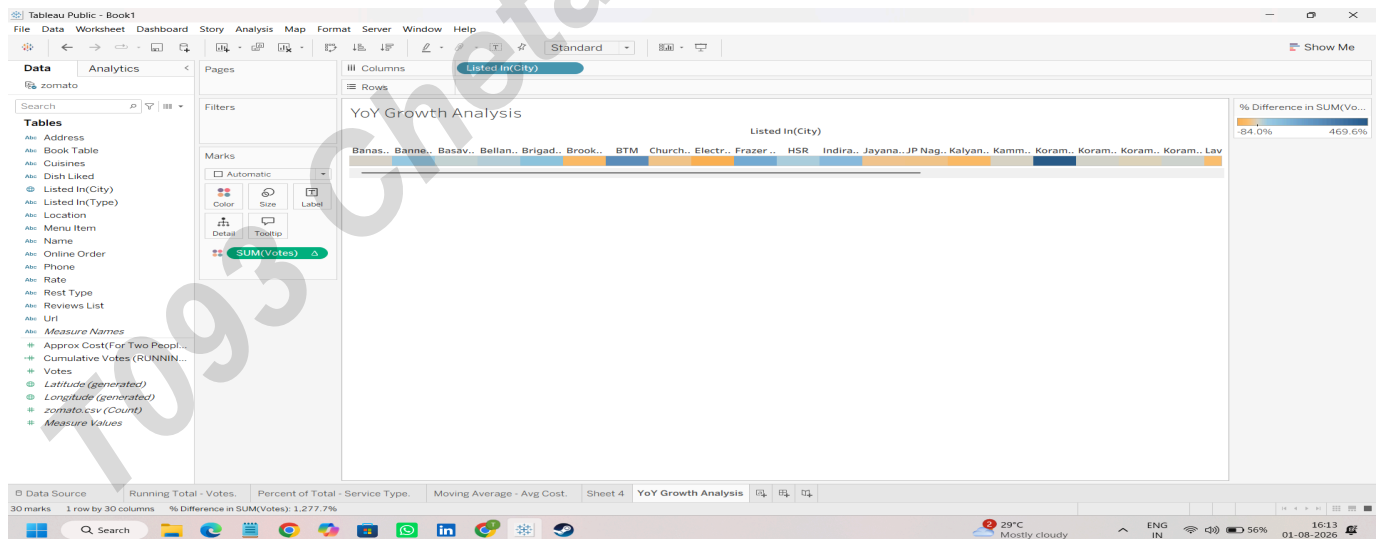
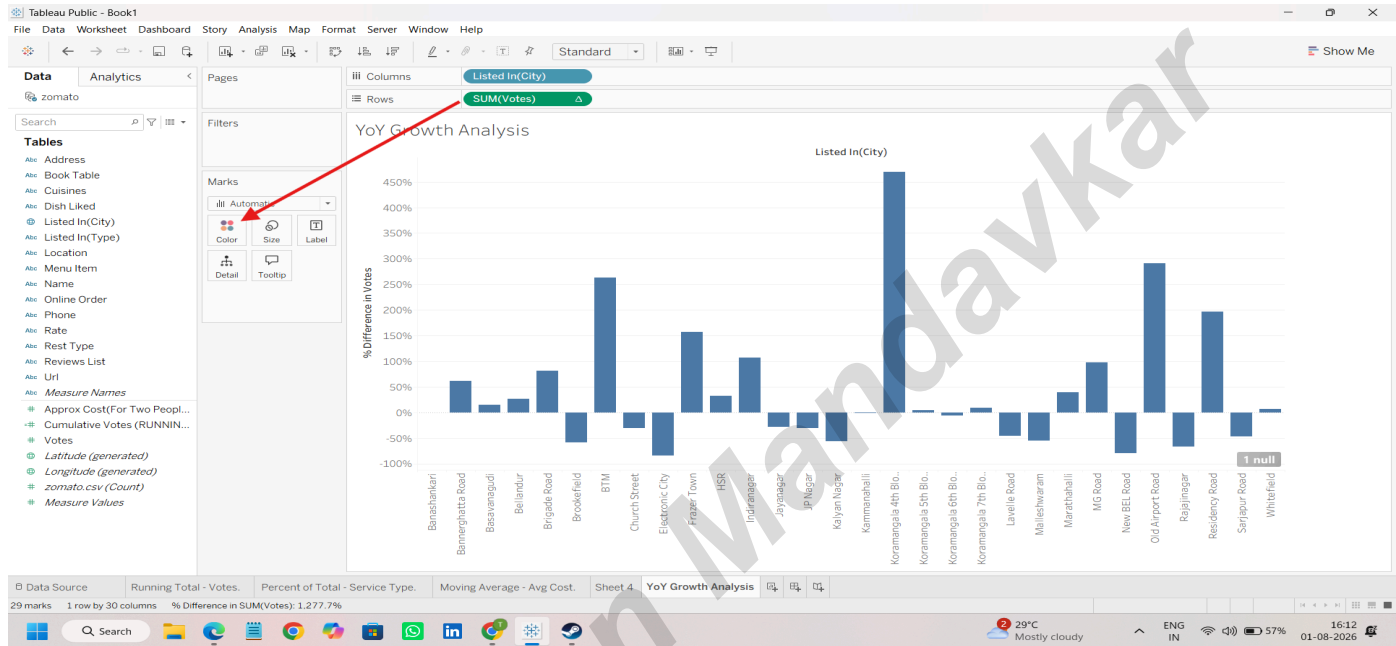
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Step 5:- Add Visual Color Indicators: Hold down Ctrl, click and drag the SUM(votes) pill from the Rows shelf, and drop it directly onto the Color mark on the Marks card. Positive growth will appear in one color (e.g., Green/Blue) while negative performance dips will turn Red/Orange!



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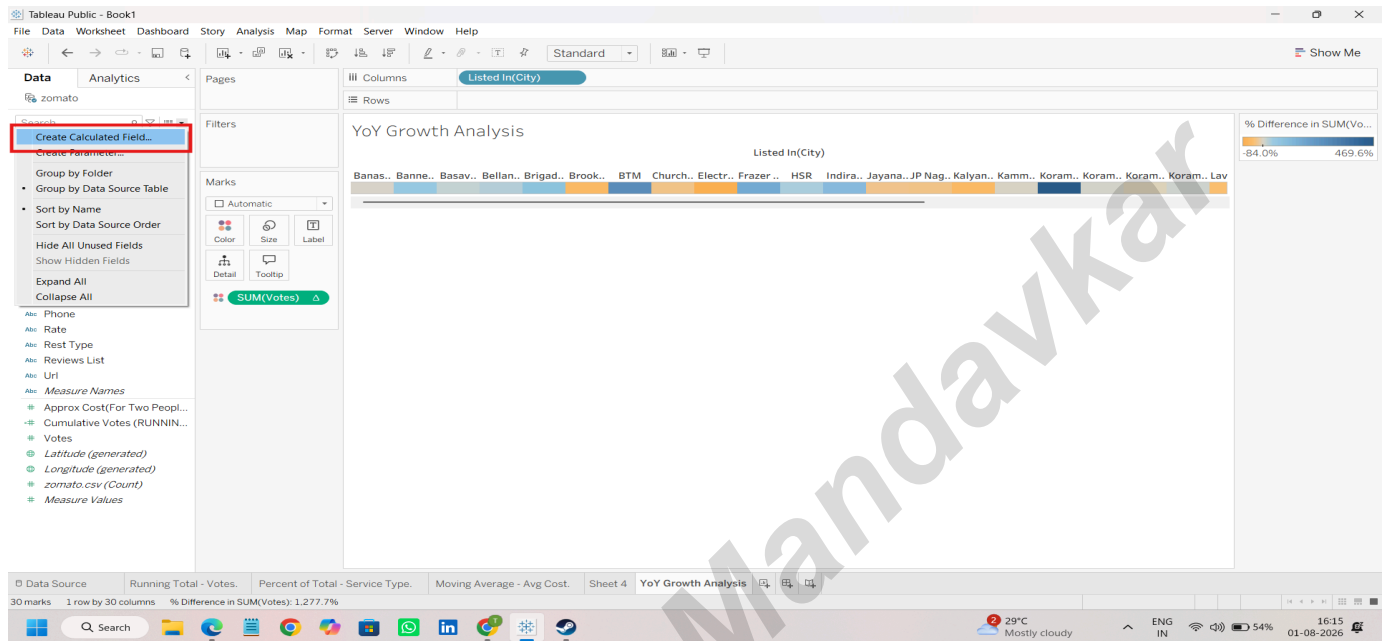
Roll No. T093

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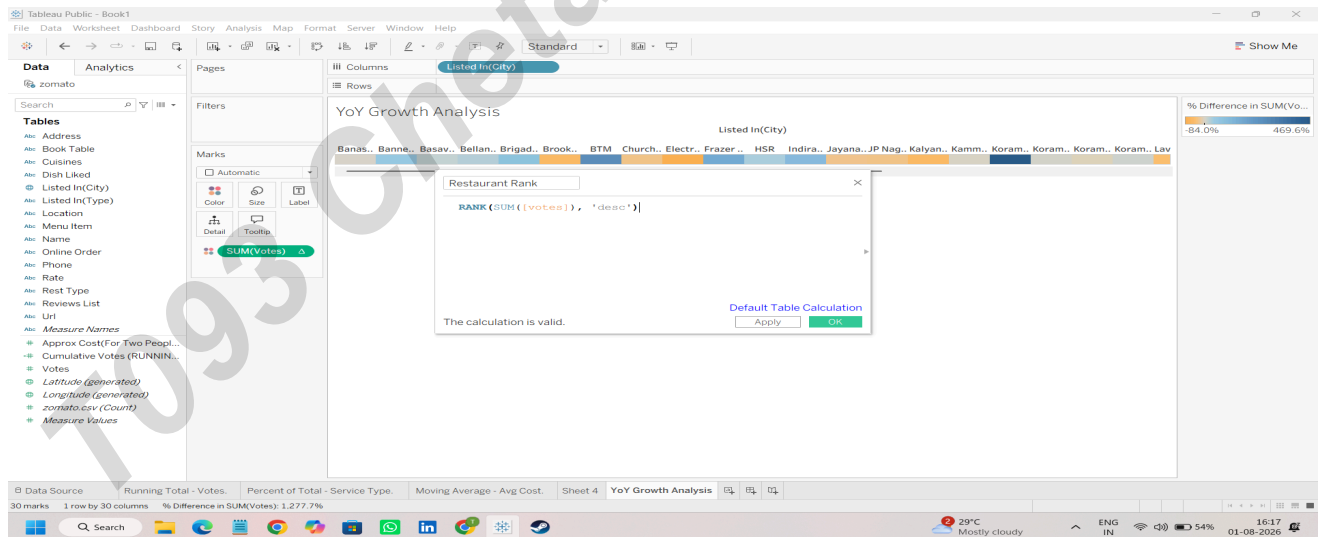
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D] Use ranking functions (Top N, Bottom N)

Step 1:- Click the small drop-down arrow at the top right of the Data Pane and select Create Calculated Field...



Step 2:- Name the field: Restaurant Rank. Input the following ranking calculation: RANK(SUM([votes]), 'desc')



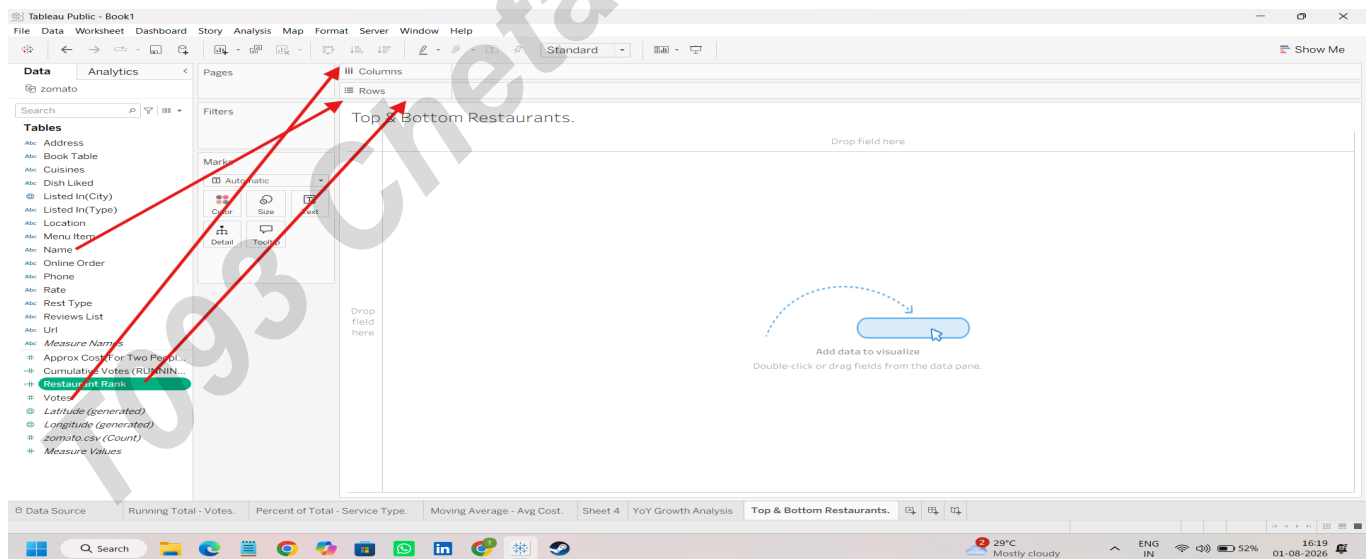
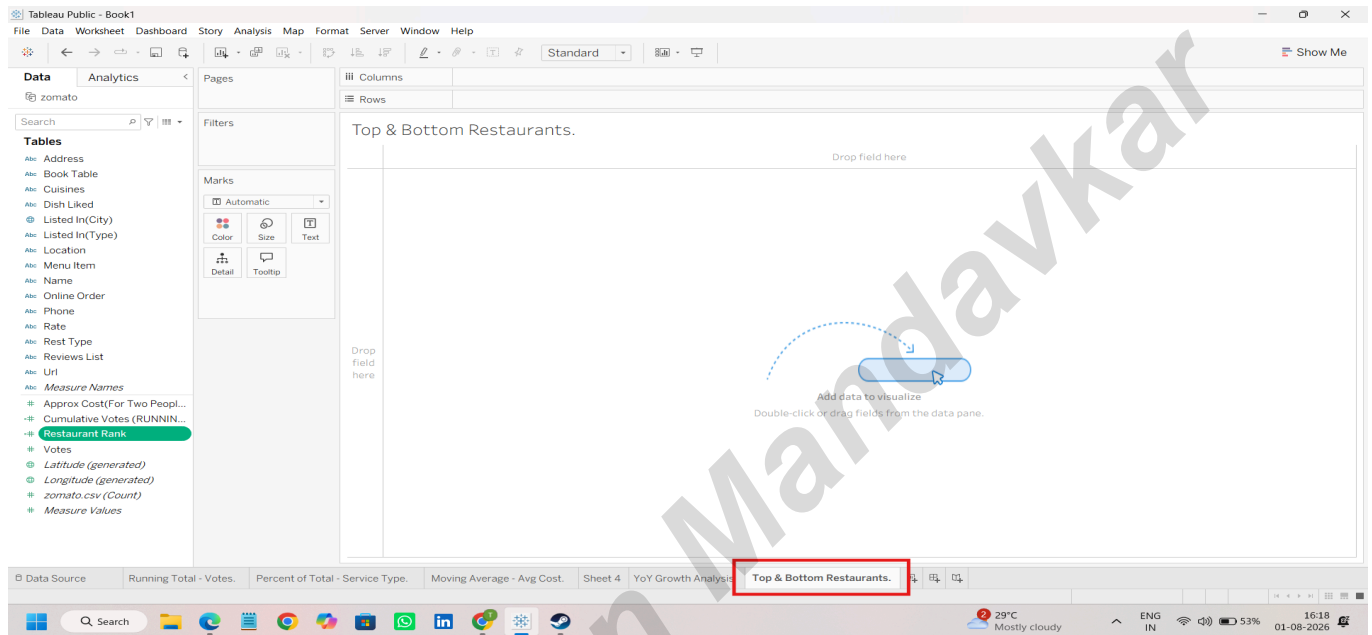
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Step 3:- Open a new worksheet and rename it: Top & Bottom Restaurants. Drag name (Restaurant Name) to the Rows shelf. Drag votes to the Columns shelf. Drag your newly created Restaurant Rank calculated field onto the Rows shelf, placing it to the left of name.

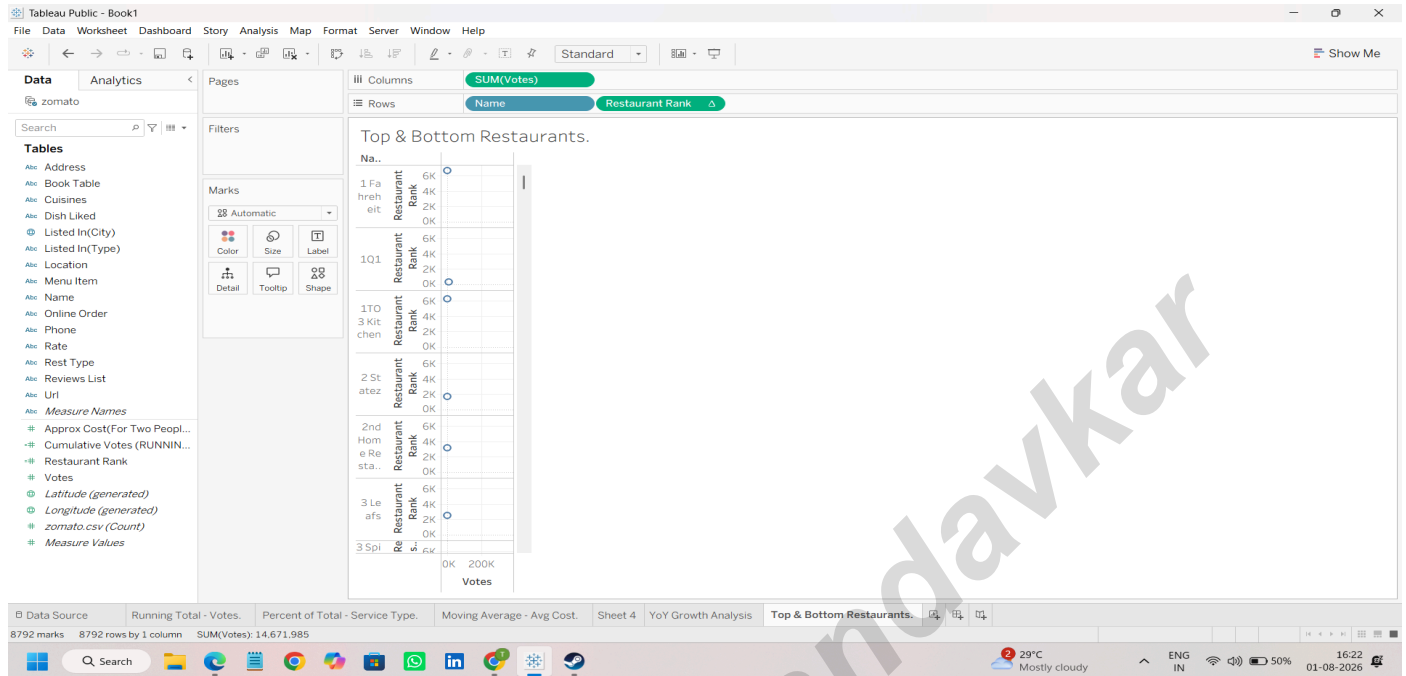


Name:- Chetan Mandavkar

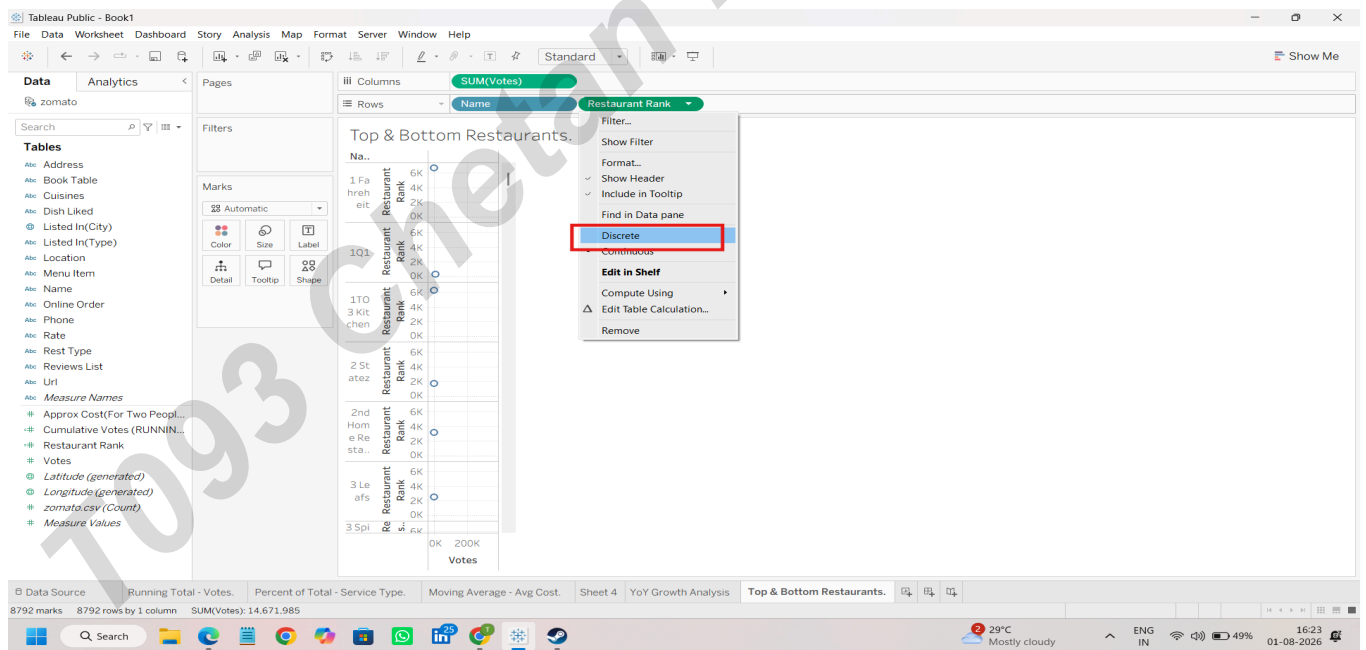
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Step 4:- Convert Restaurant Rank to a Discrete field: Right-click the Restaurant Rank pill on the Rows shelf and select Discrete.

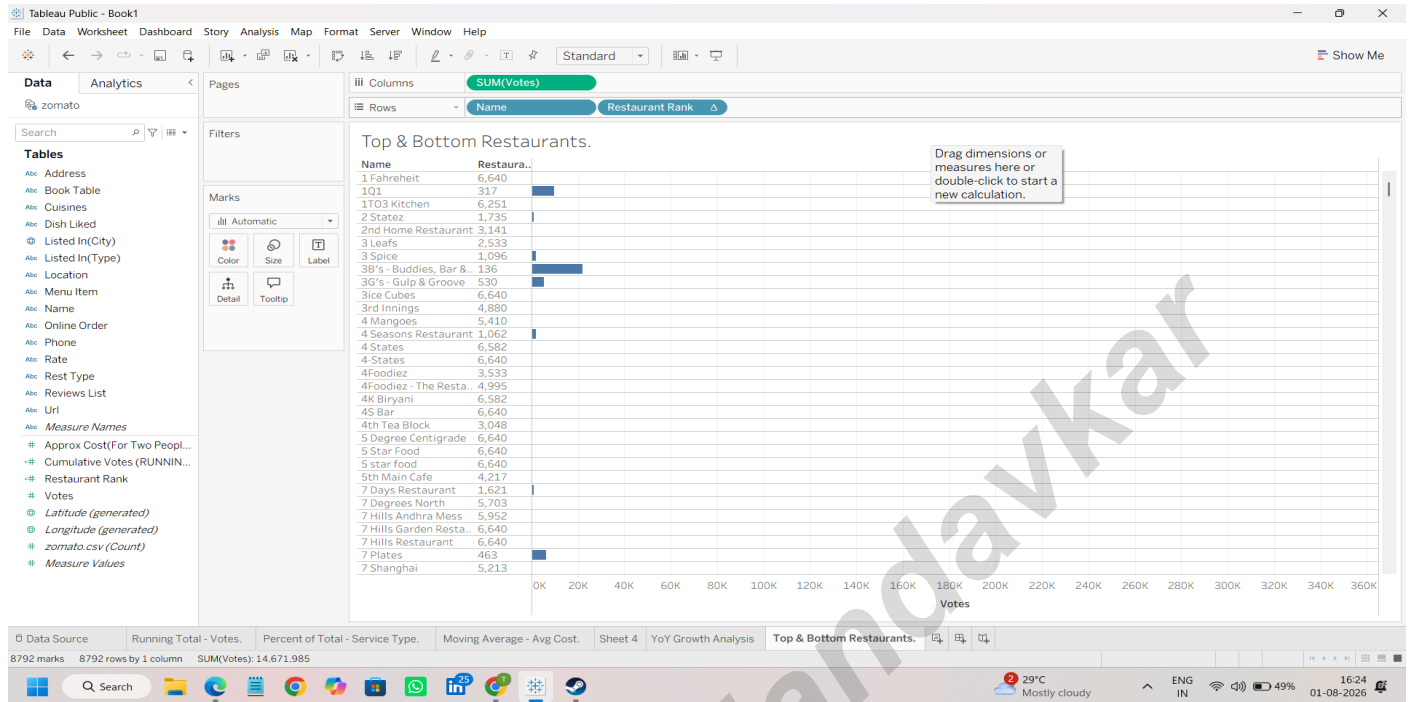


Name:- Chetan Mandavkar

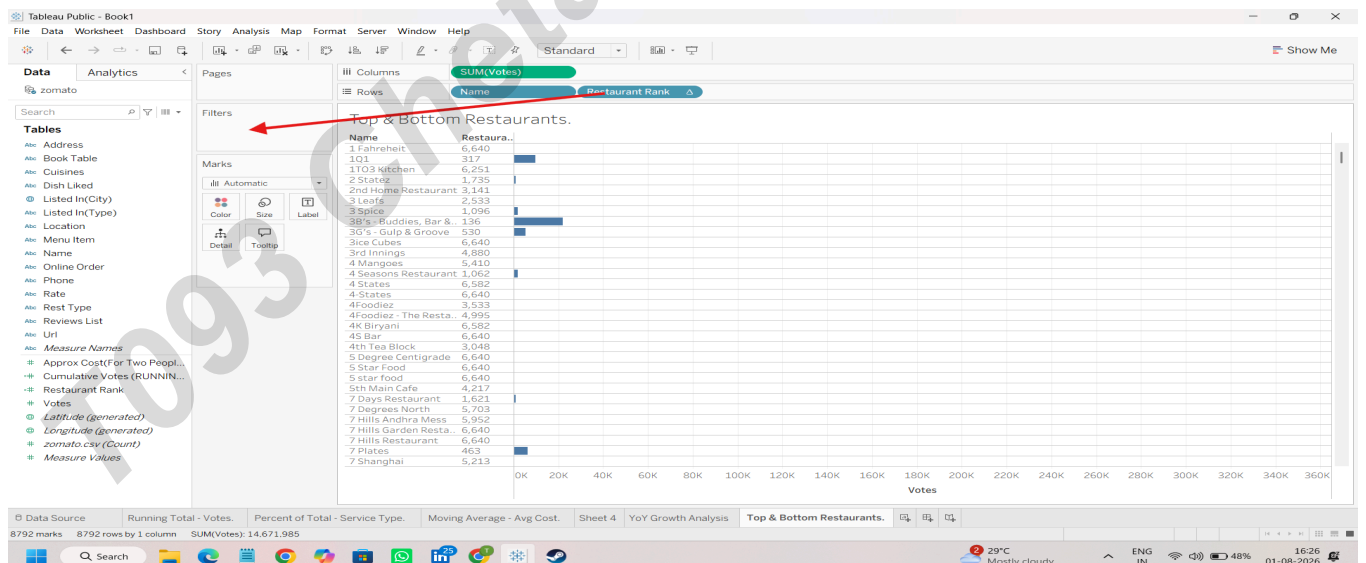
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Step 5:- click and drag the Restaurant Rank pill from the Rows shelf, and drop it onto the Filters card. In the filter dialog box, select a range of ranks (e.g., set the maximum to 10 to show only the Top 10 restaurants). Click OK.

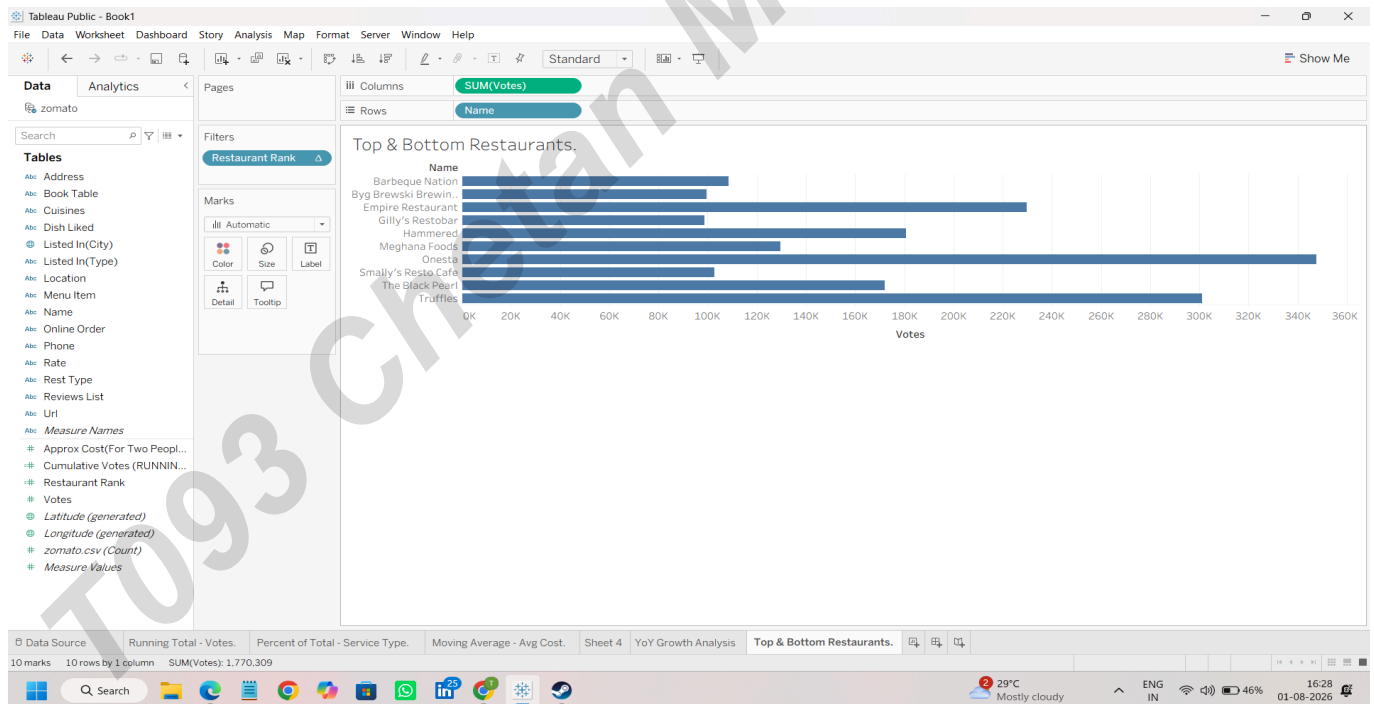
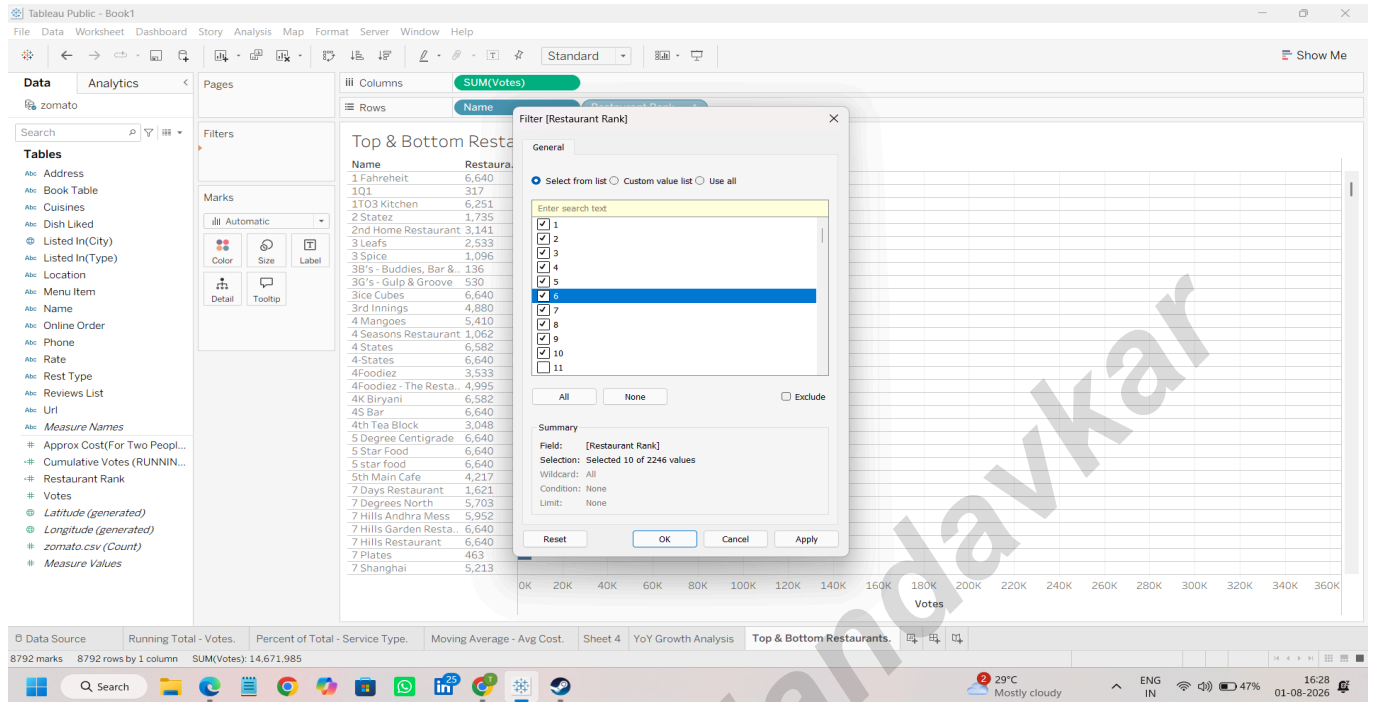


Name:- Chetan Mandavkar

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Name:- Chetan Mandavkar

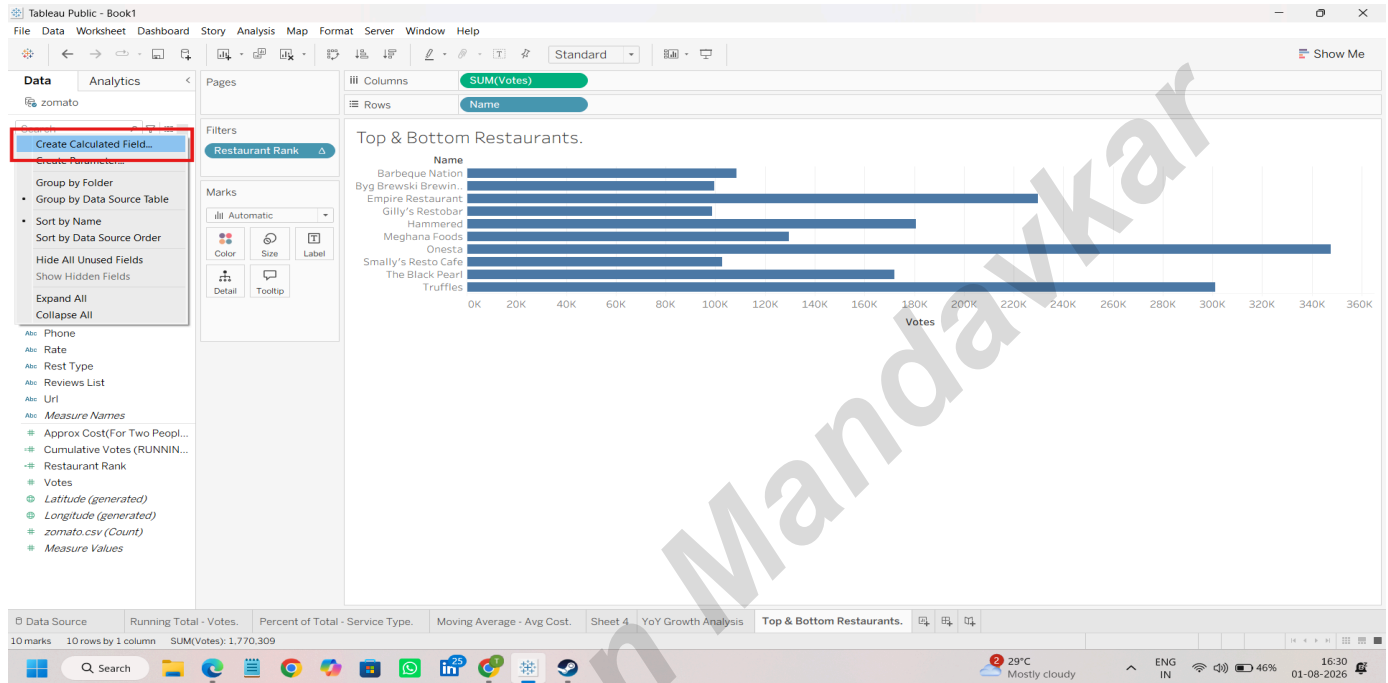
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E] Implement window functions (WINDOW_SUM, WINDOW_AVG)

Step 1:- Click the small drop-down arrow at the top right of the Data Pane (next to the Search bar) and select Create Calculated Field...



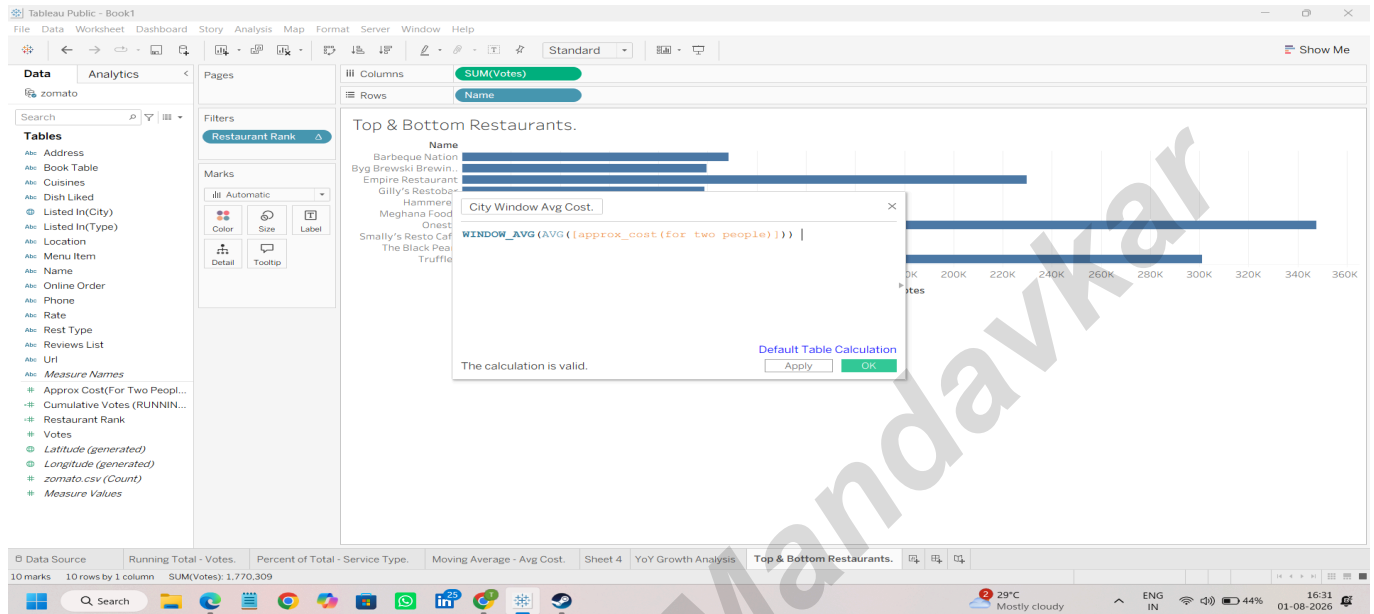
Name:- Chetan Mandavkar

Roll No. T093

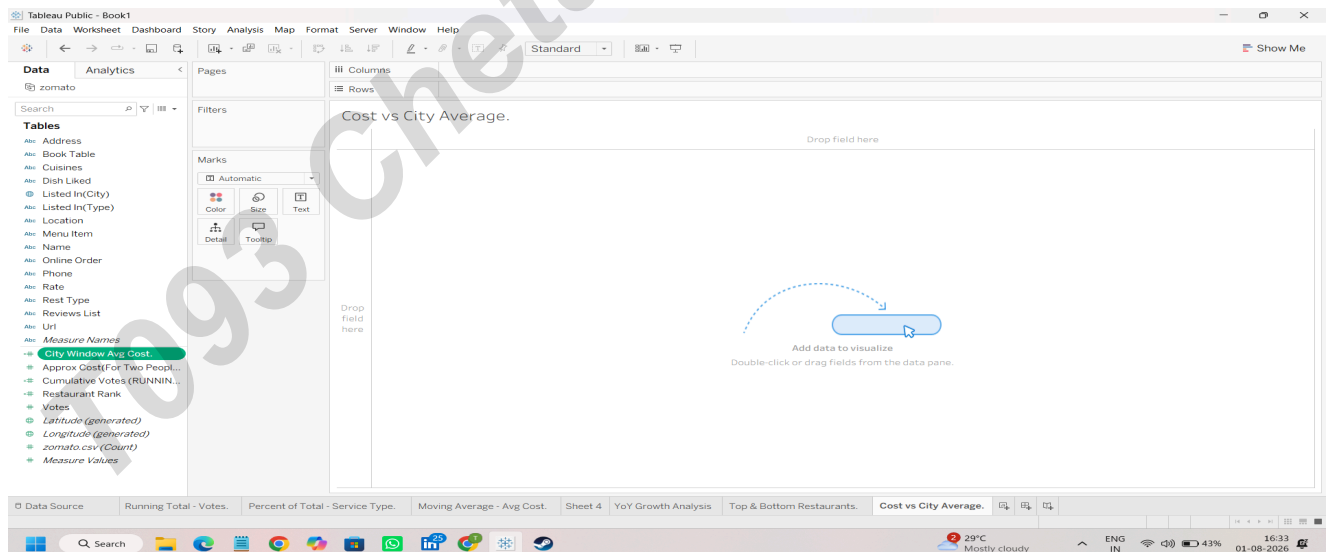
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Step 2:- Name the field: City Window Avg Cost. Input the following window function formula: WINDOW_AVG(AVG([approx_cost(for two people)]))



Step 3:- Open a new worksheet and rename it: Cost vs City Average. Drag listed_in(city) to the Rows shelf. Drag approx_cost(for two people) to the Columns shelf.

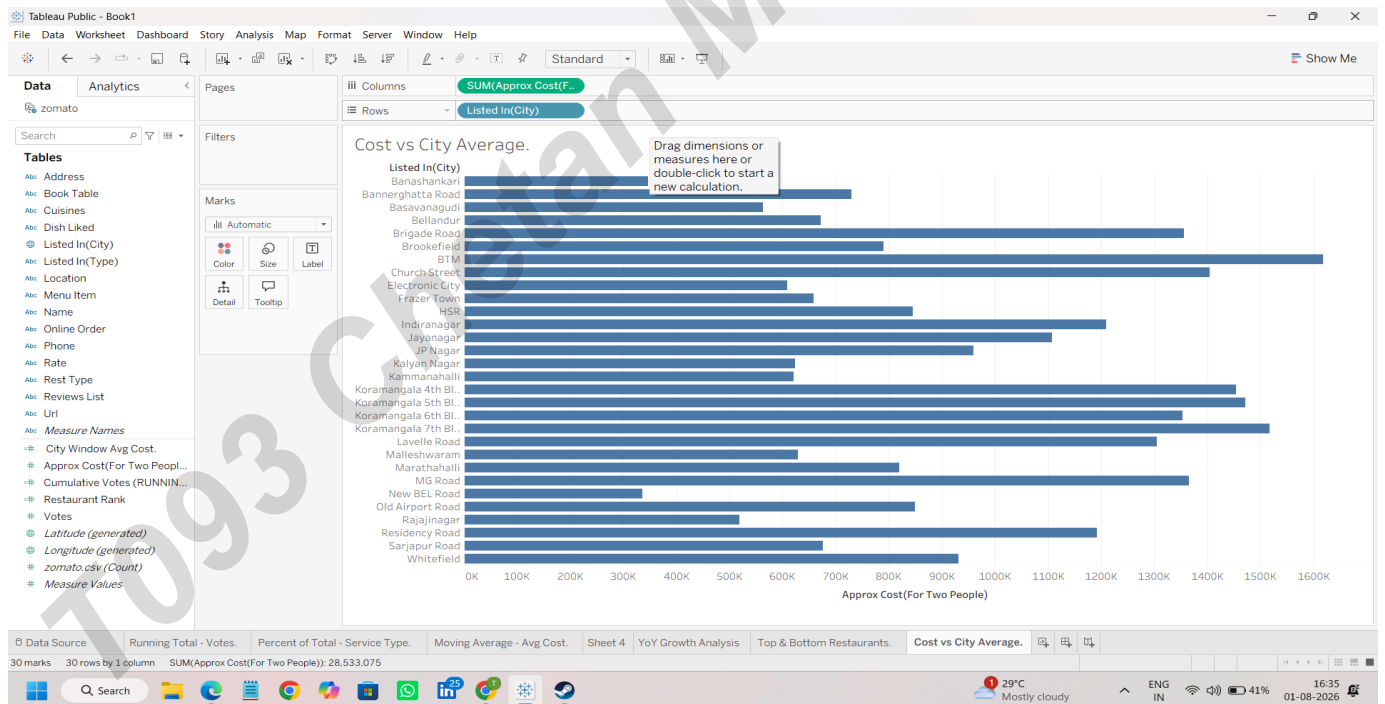
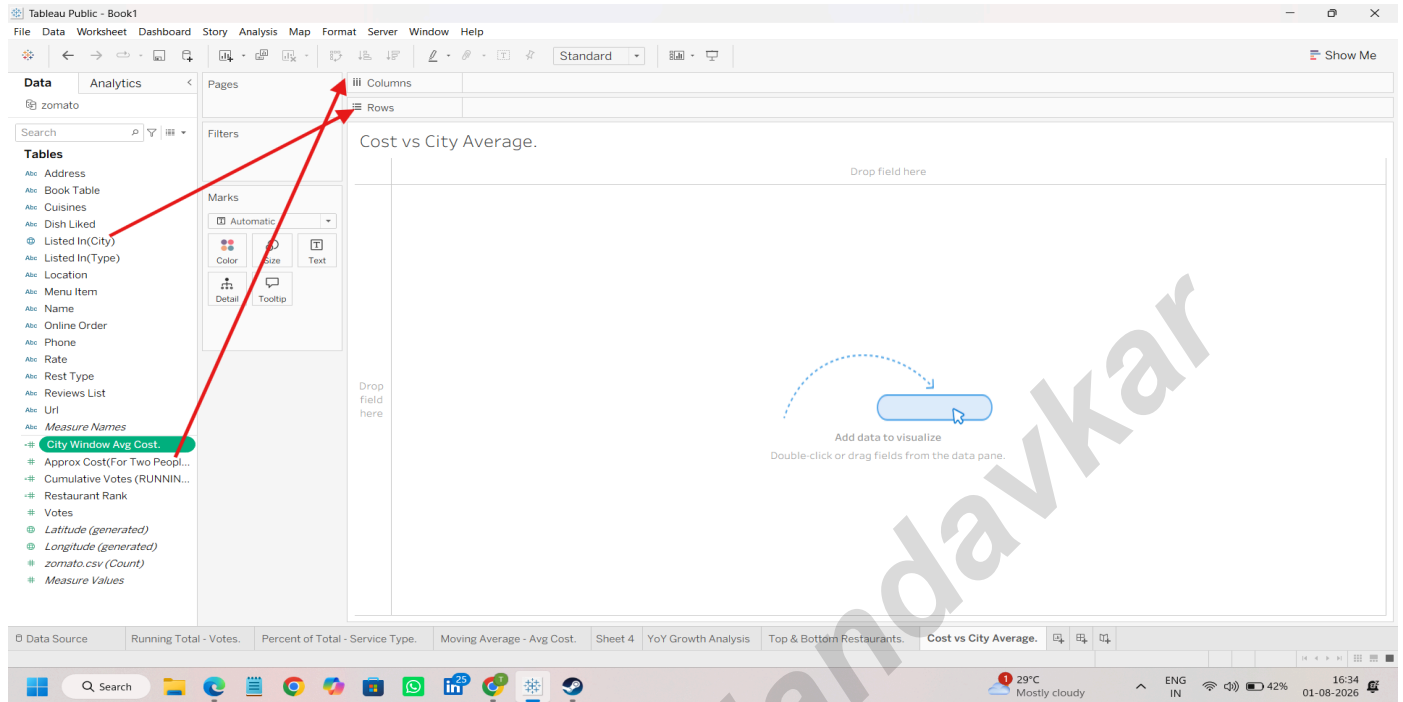


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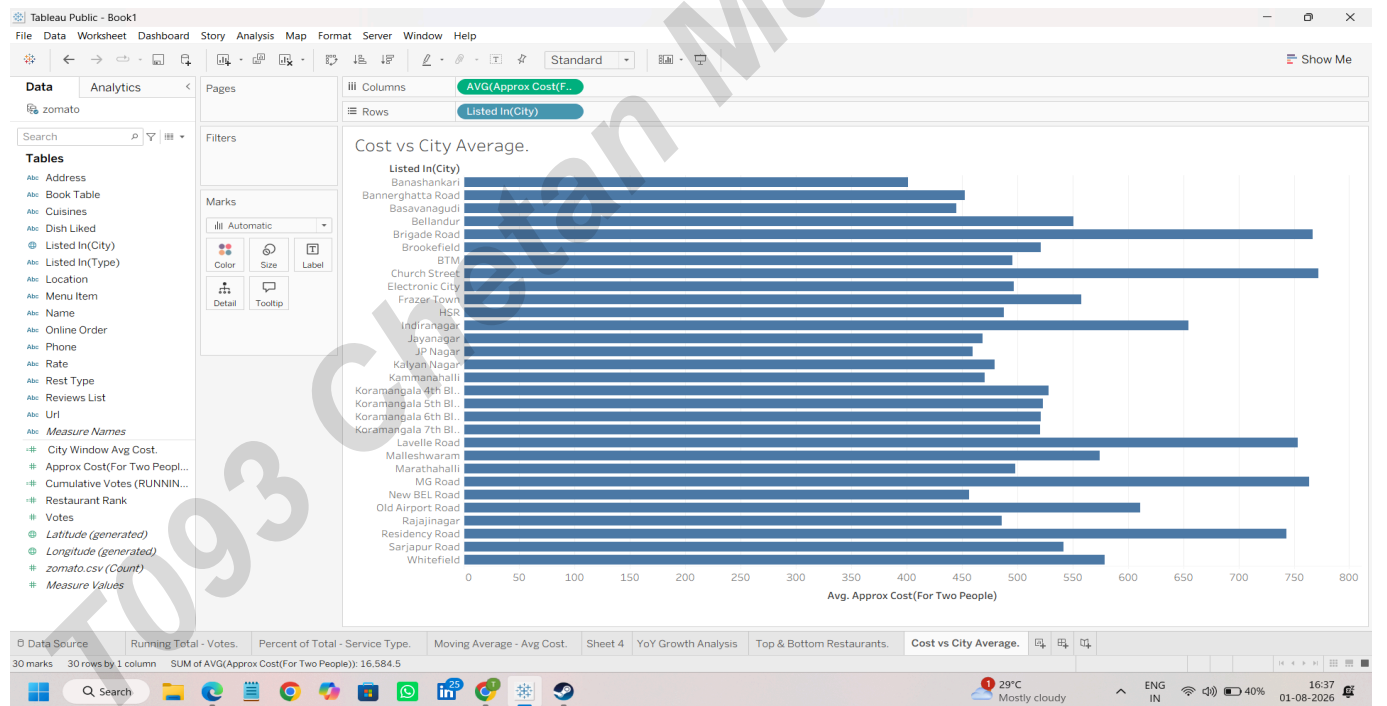
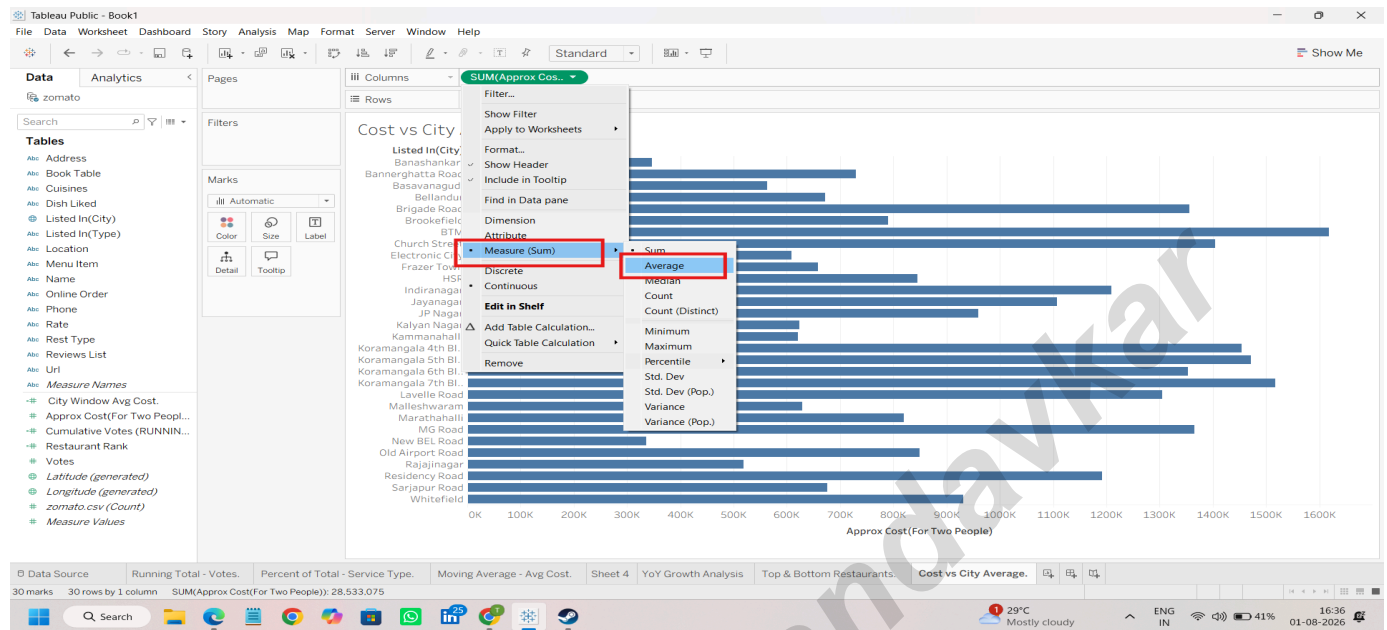
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Step 4:- Change its aggregation to Average



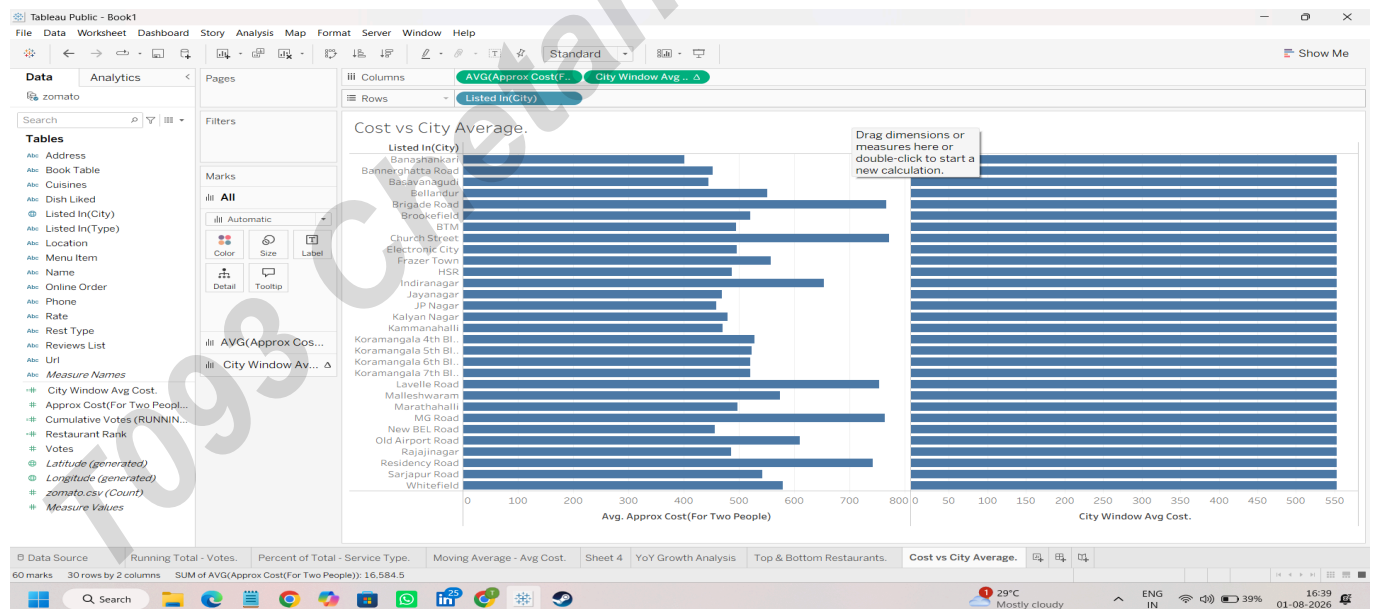
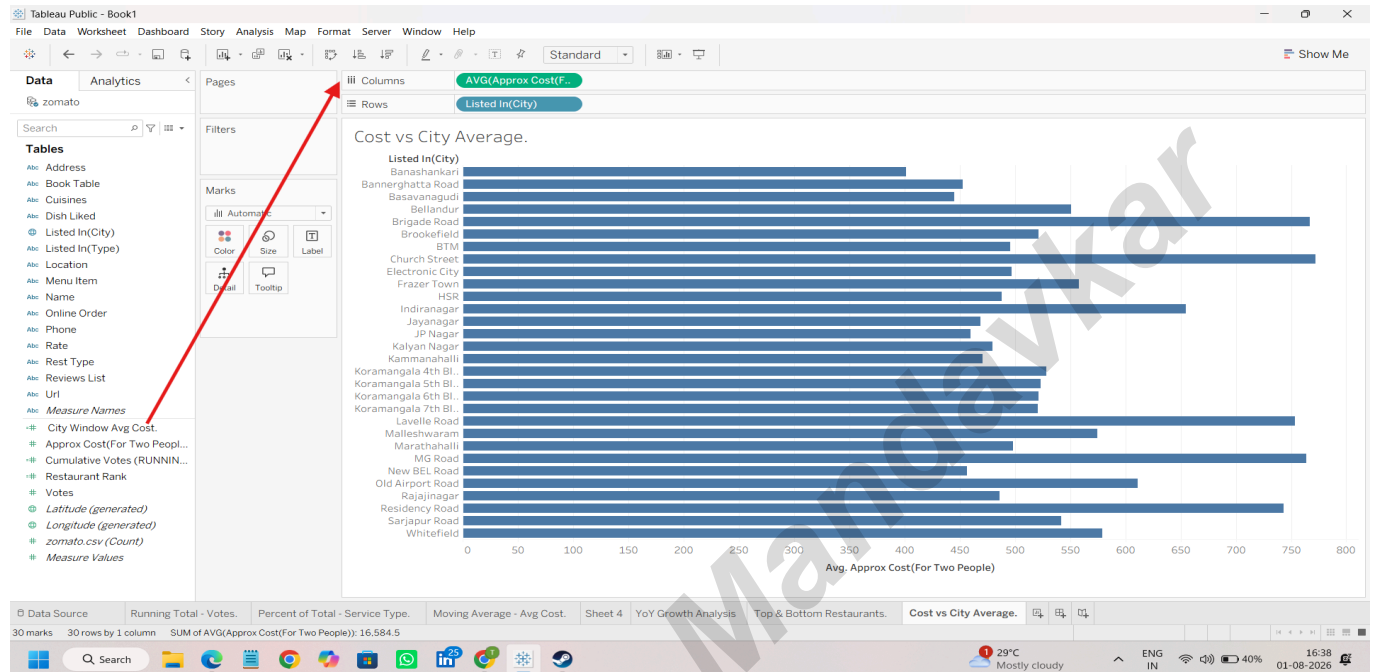
Name:- Chetan Mandavkar

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Step 5:- Drag your newly created City Window Avg Cost calculated field onto the Columns shelf (placing it right beside AVG(approx_cost...)).



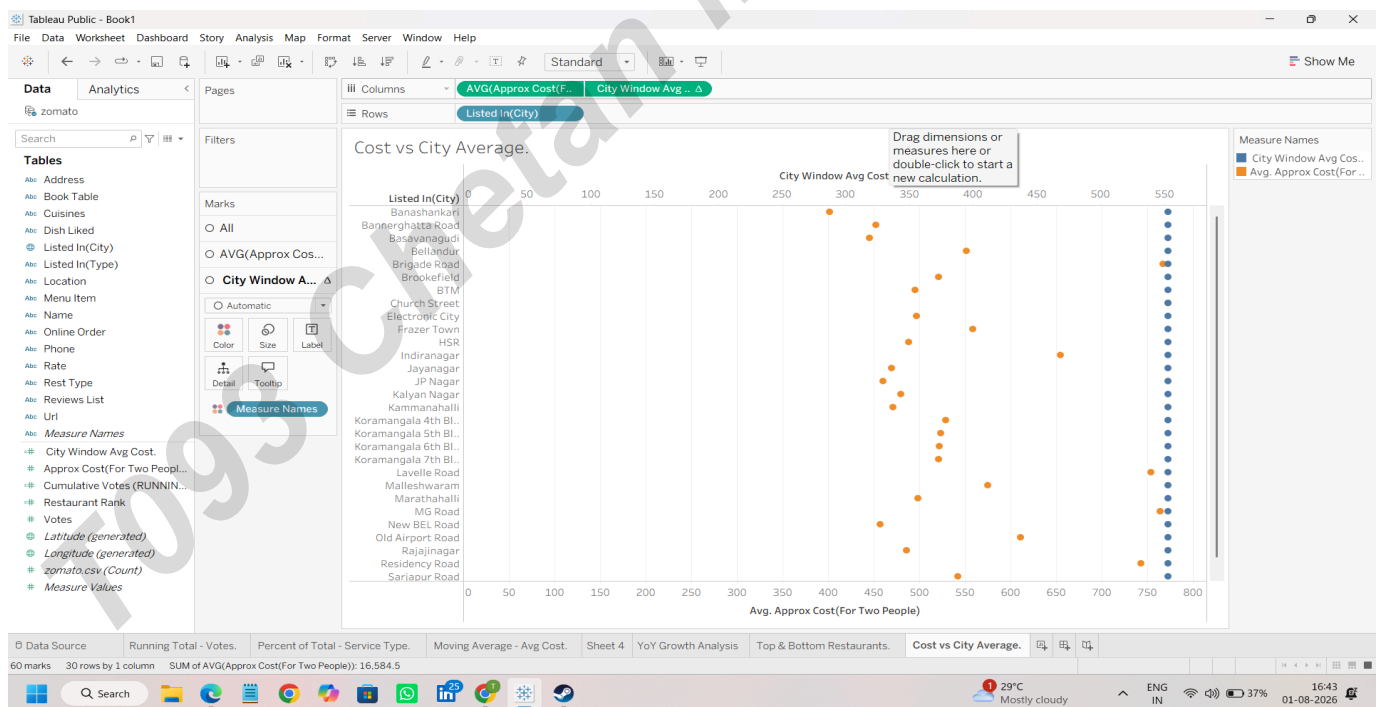
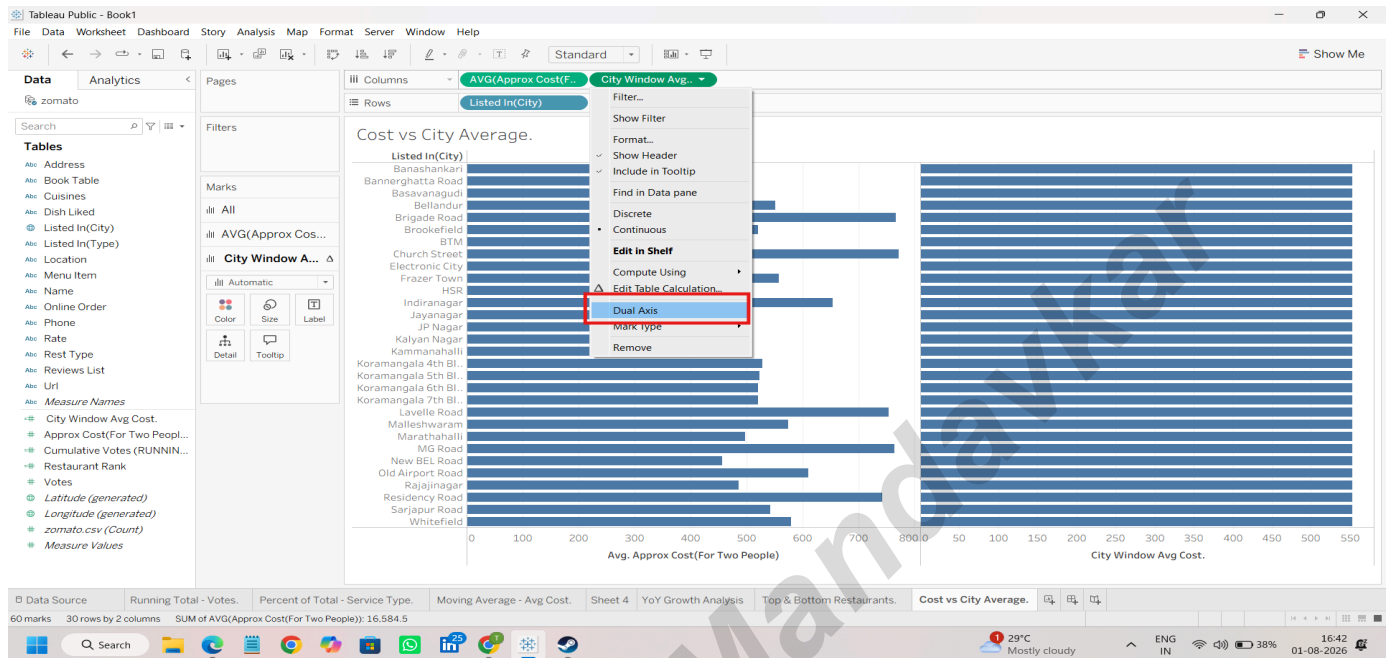
Name:- Chetan Mandavkar

Roll No. T093

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Step 6:- Right-click the City Window Avg Cost pill on the Columns shelf and select Dual Axis.



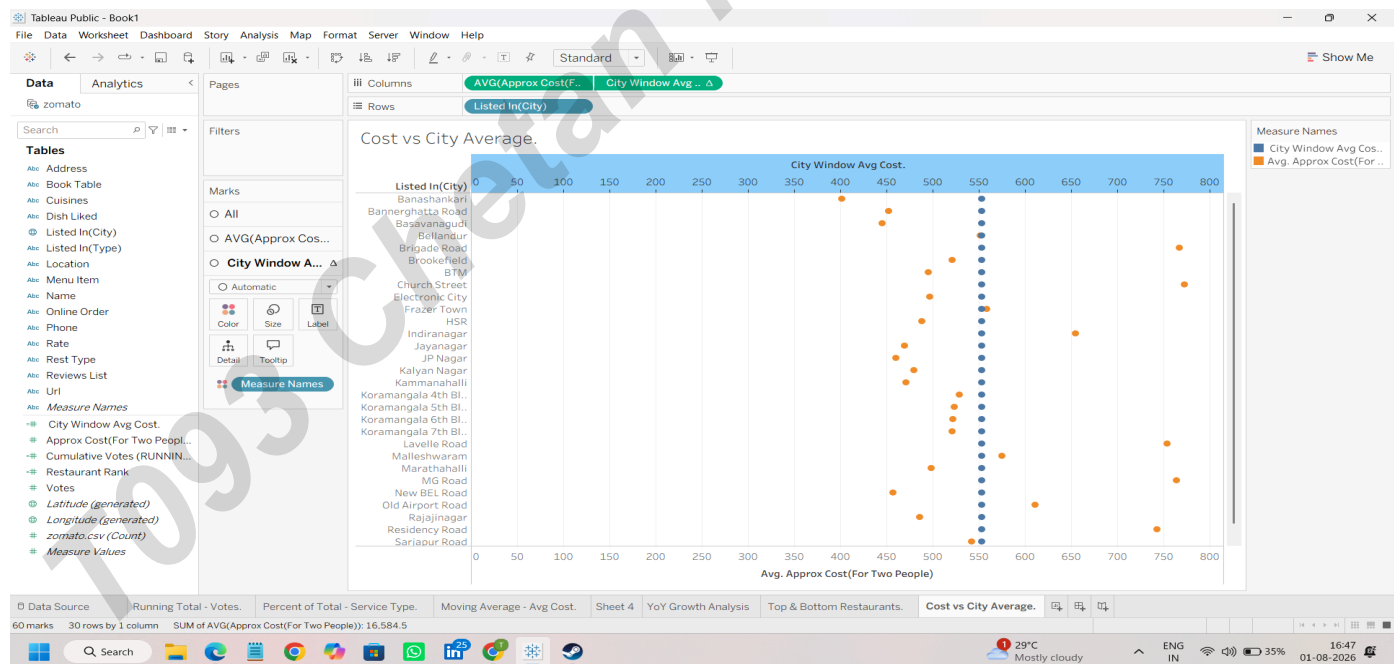
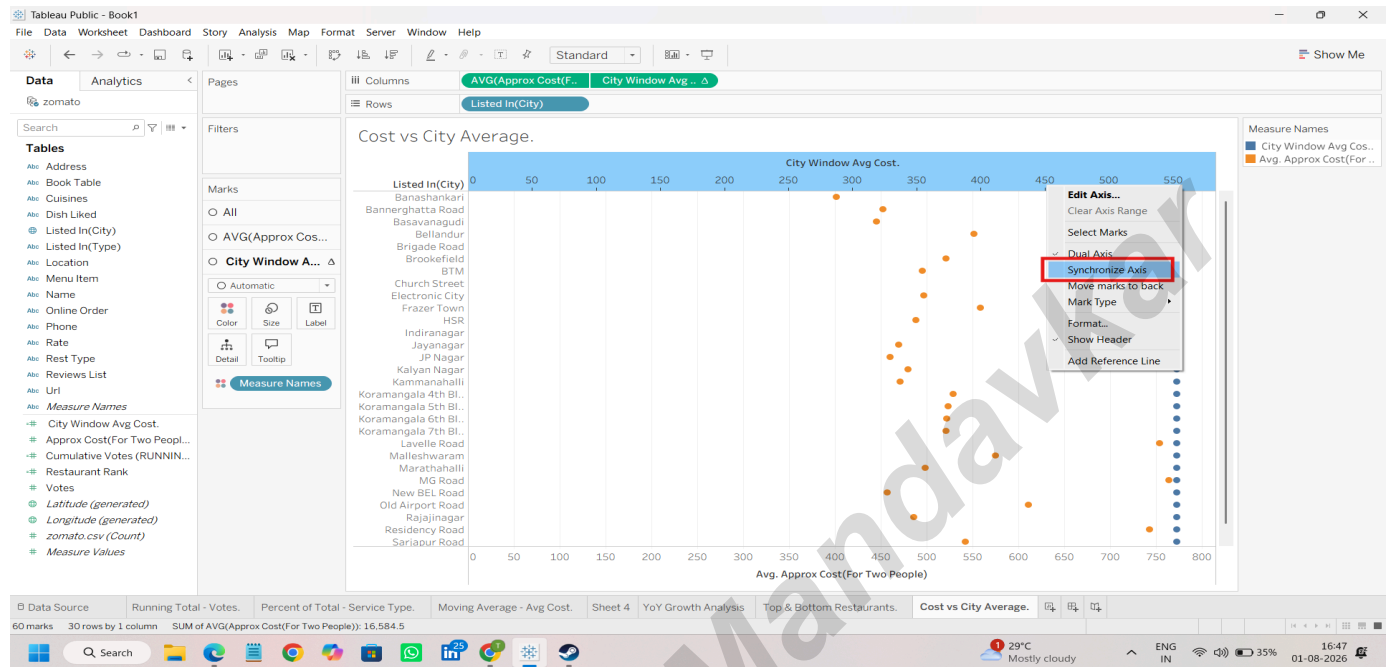
Name:- Chetan Mandavkar

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Step 7:- Right-click the top horizontal axis in the visualization view and select Synchronize Axis.

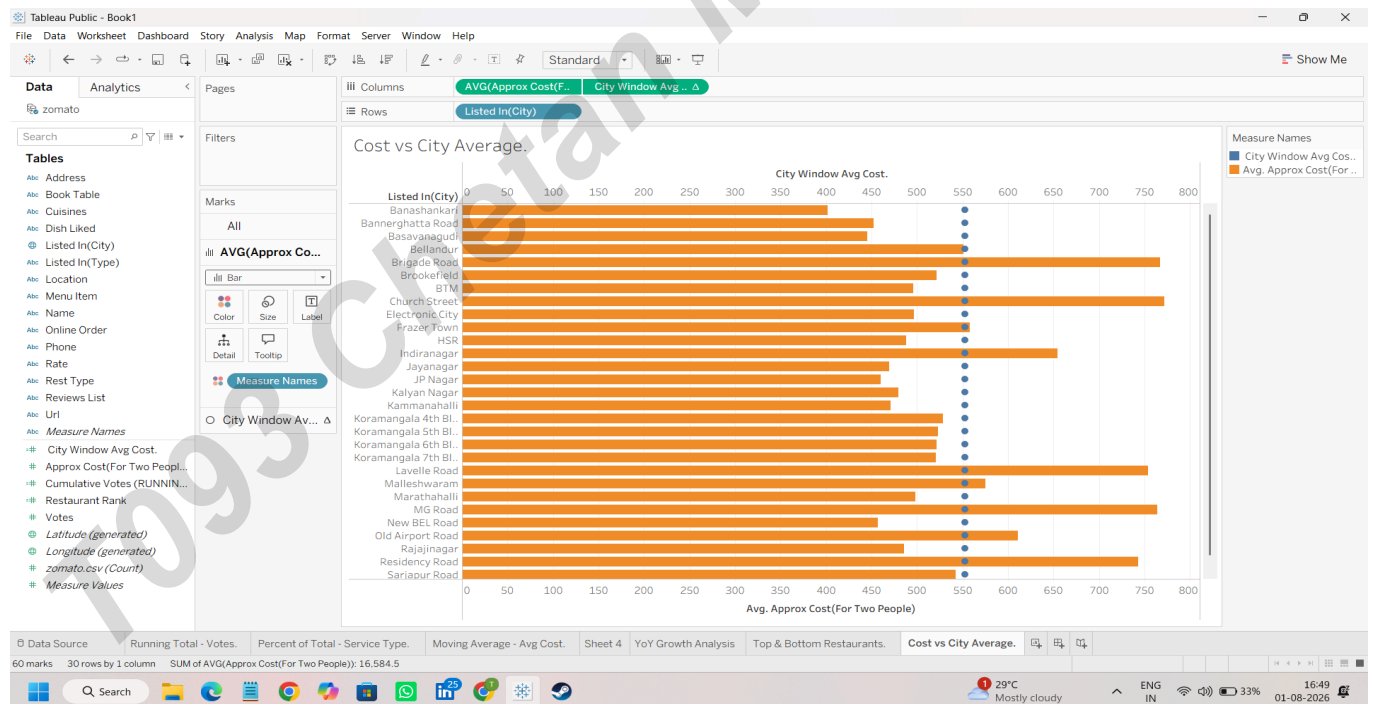
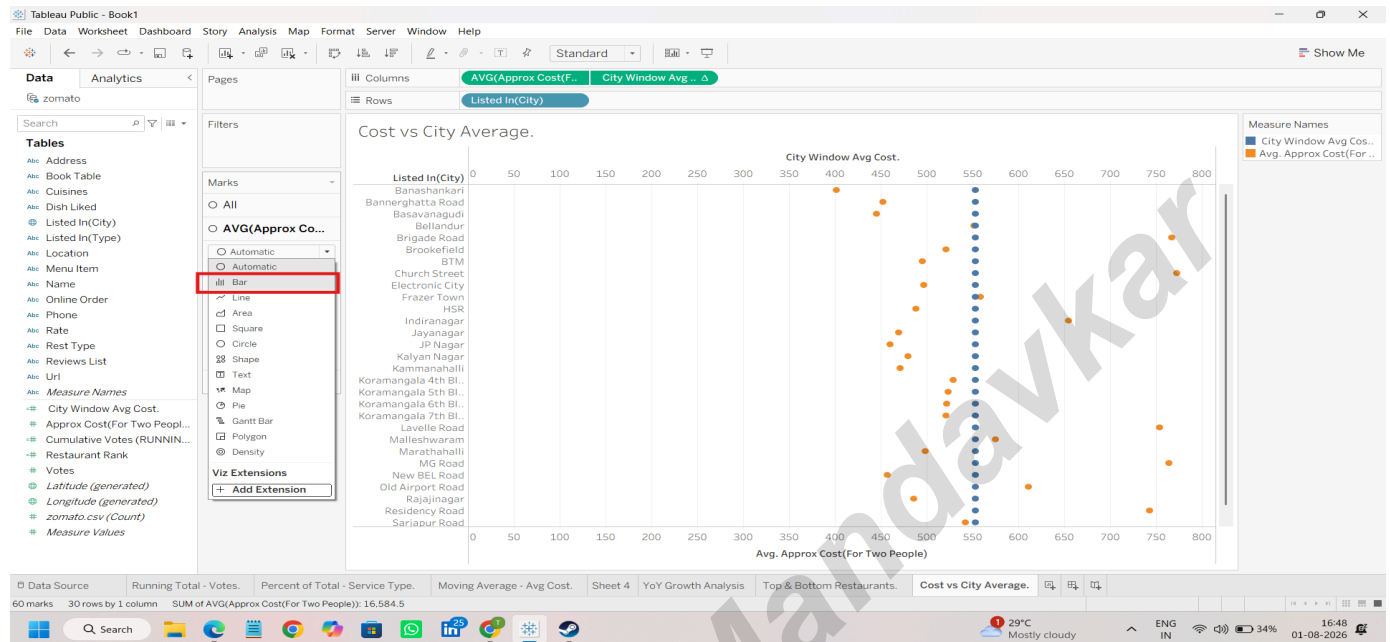


Name:- Chetan Mandavkar

Roll No. T093

Subject:- Data Visualization With PowerBI And Tableau

Step 8:- In the Marks card: Change the mark type for AVG(approx_cost...) to Bar.



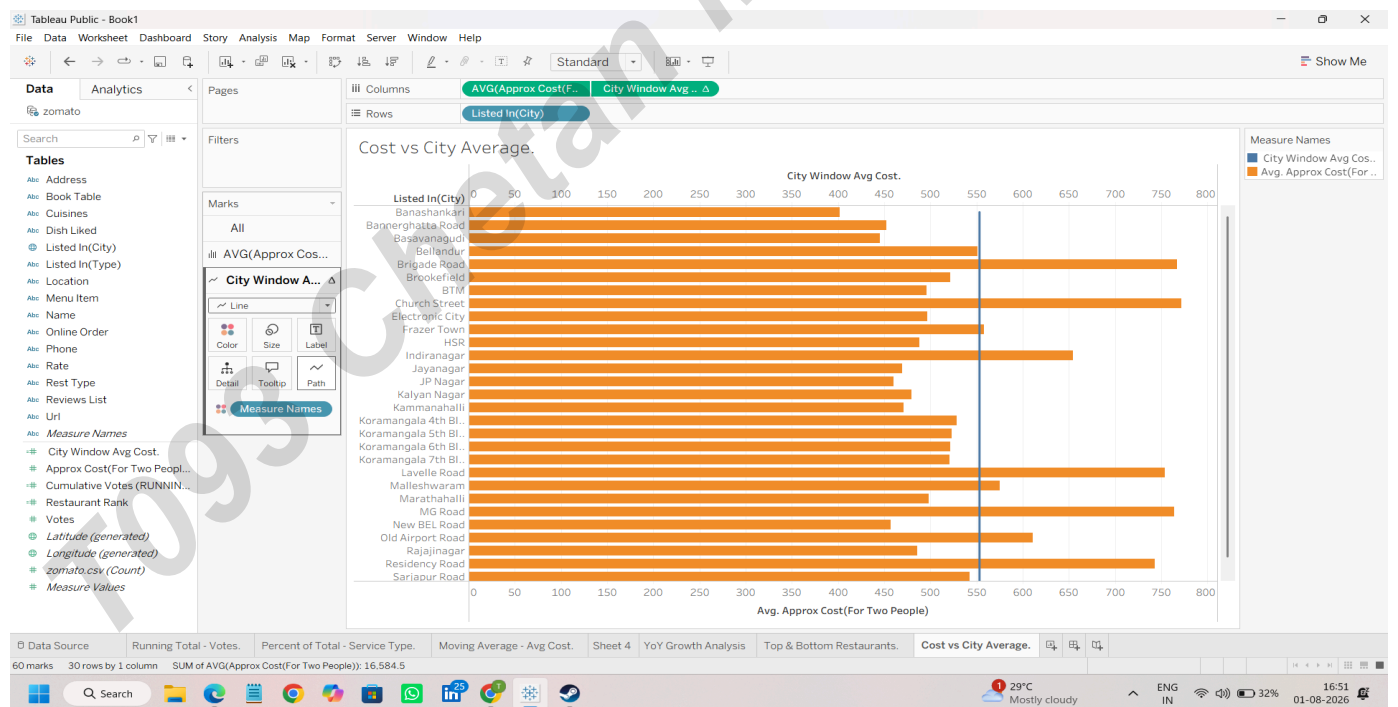
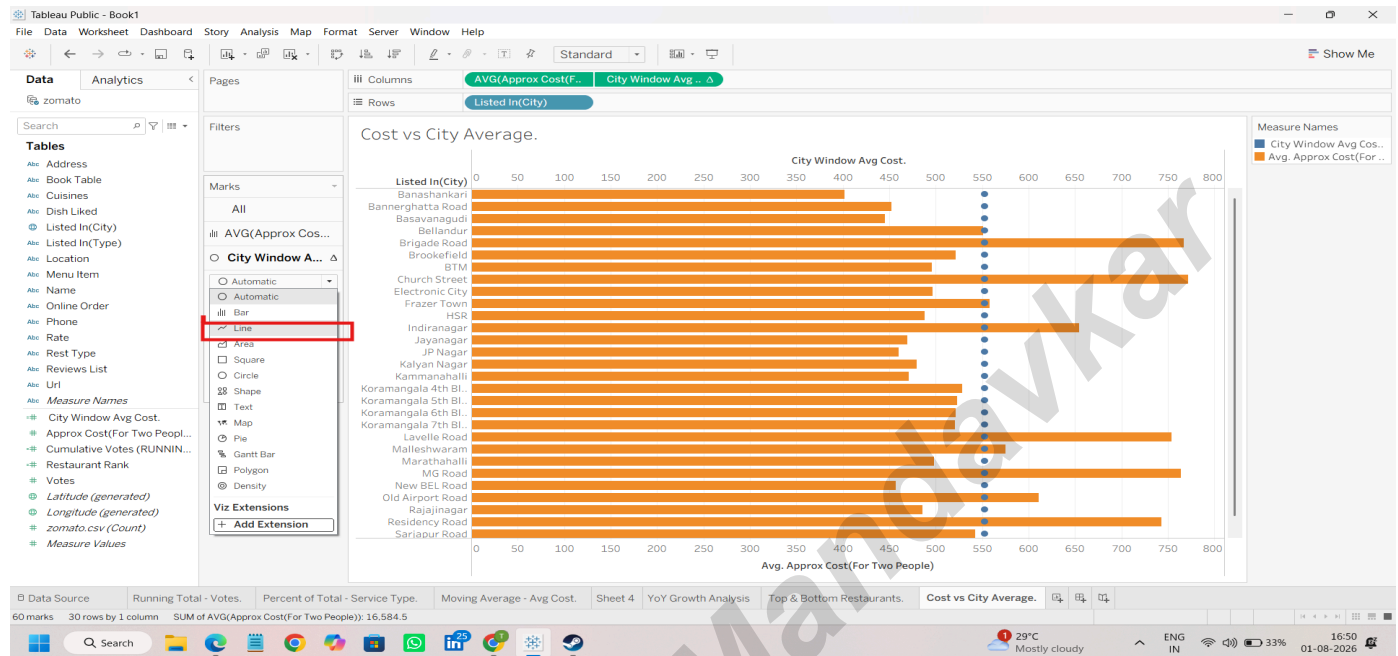
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Step 9:- Change the mark type for City Window Avg Cost to Line (or Gantt Bar).



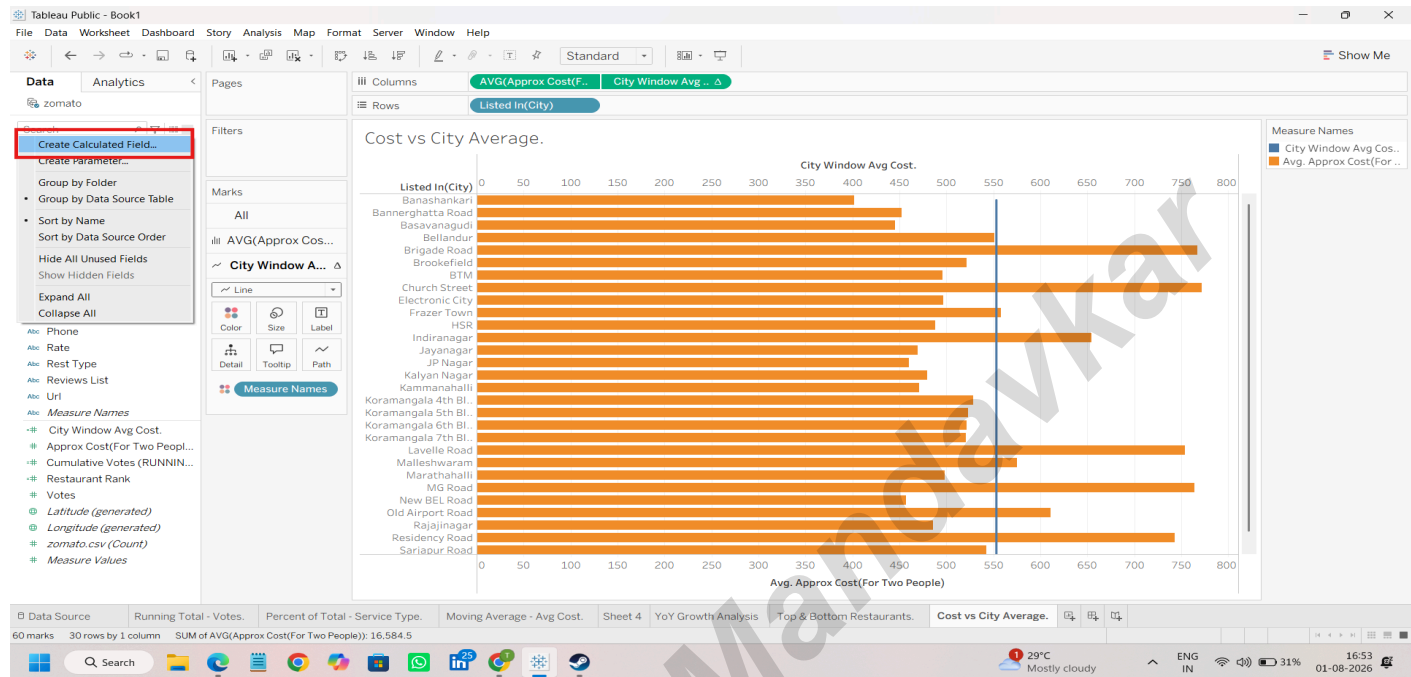
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Step 10:- Click the small drop-down arrow at the top right of the Data Pane and select Create Calculated Field...



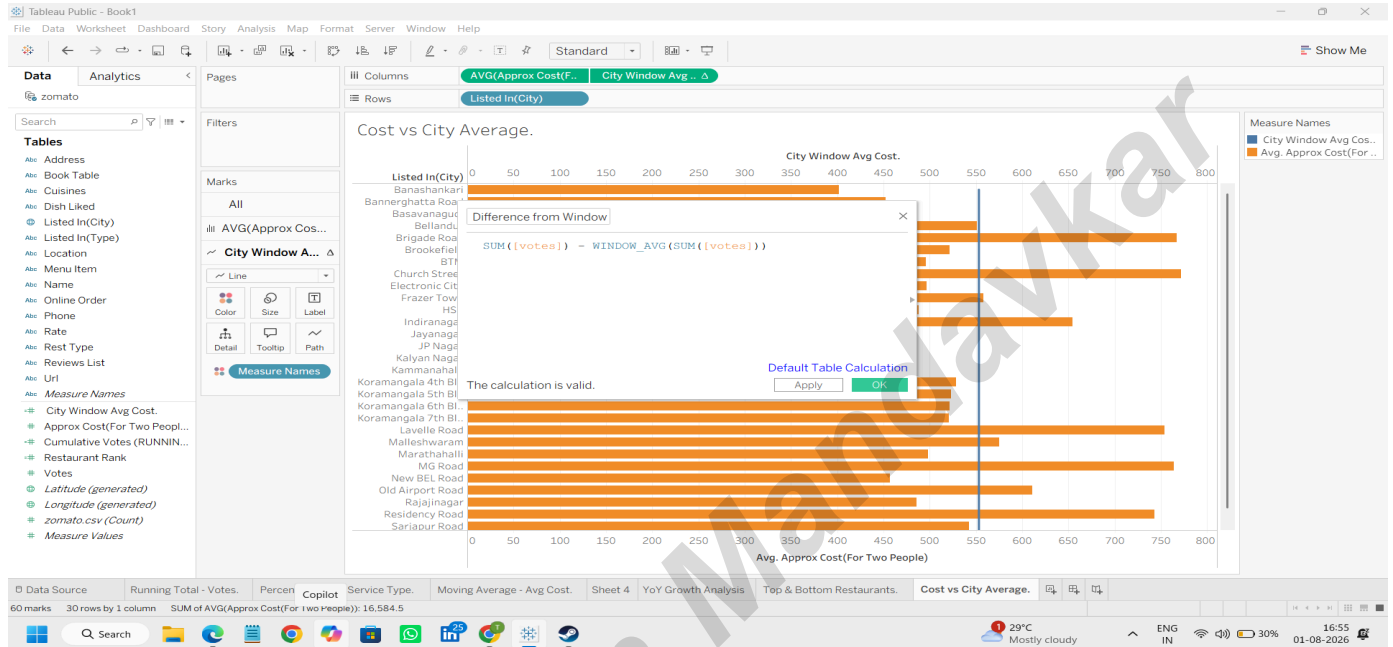
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Step 11:- Name the field: Difference from Window Sum. Input the following expression to compare each city's vote count against the total sum of all votes in the visible window: $SUM([votes]) - WINDOW_AVG(SUM([votes]))$



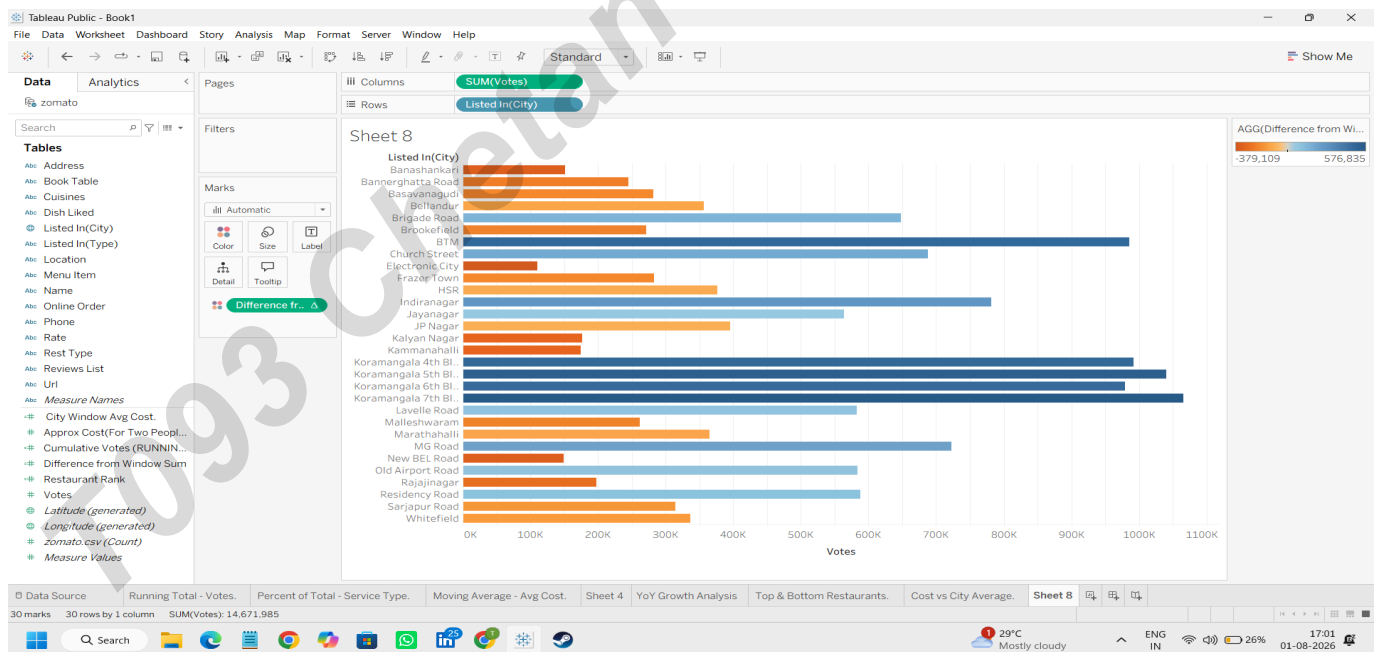
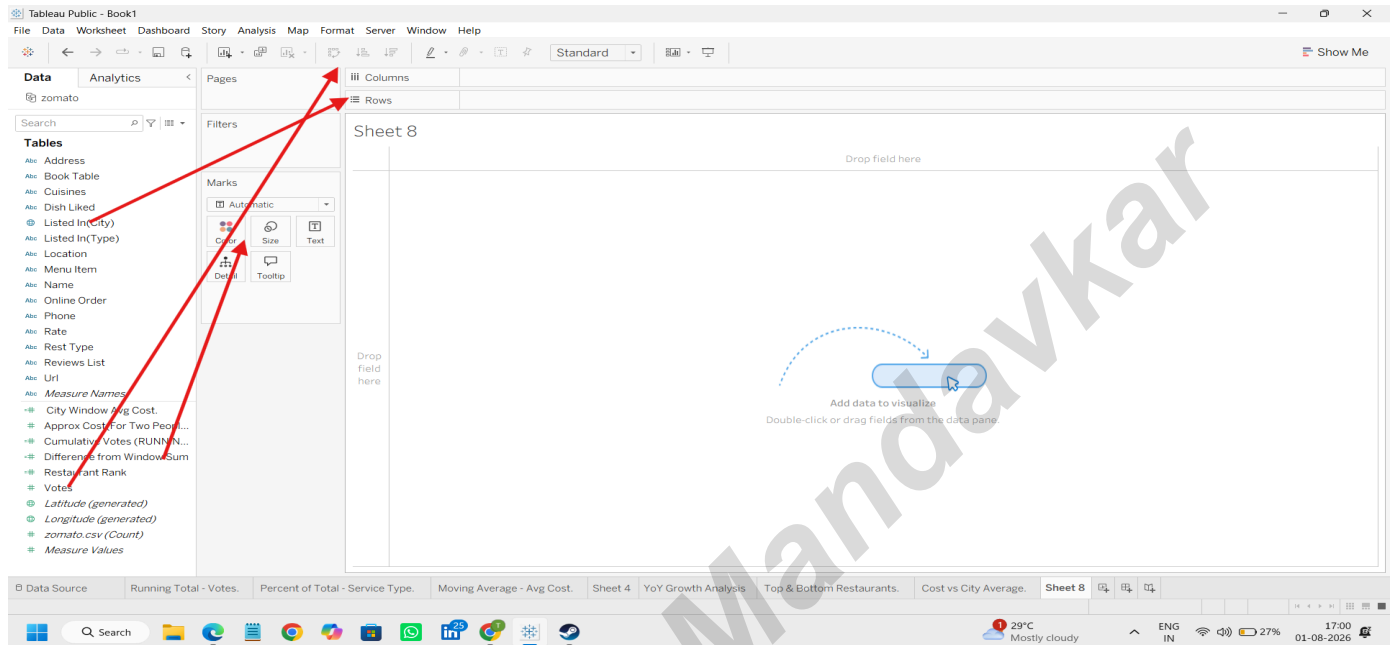
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Step 12:- Open new Sheet Drag listed_in(city) to the Rows shelf and votes to the Columns shelf. Drag your new Difference from Window Sum calculated field onto the Color mark on the Marks card.



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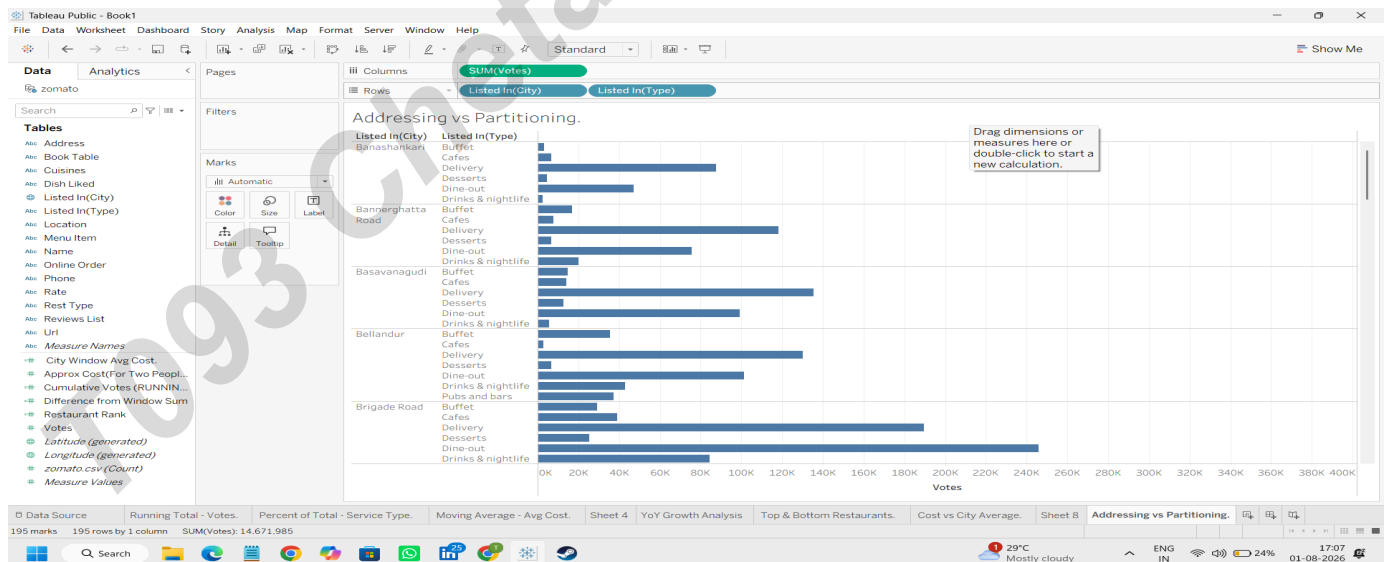
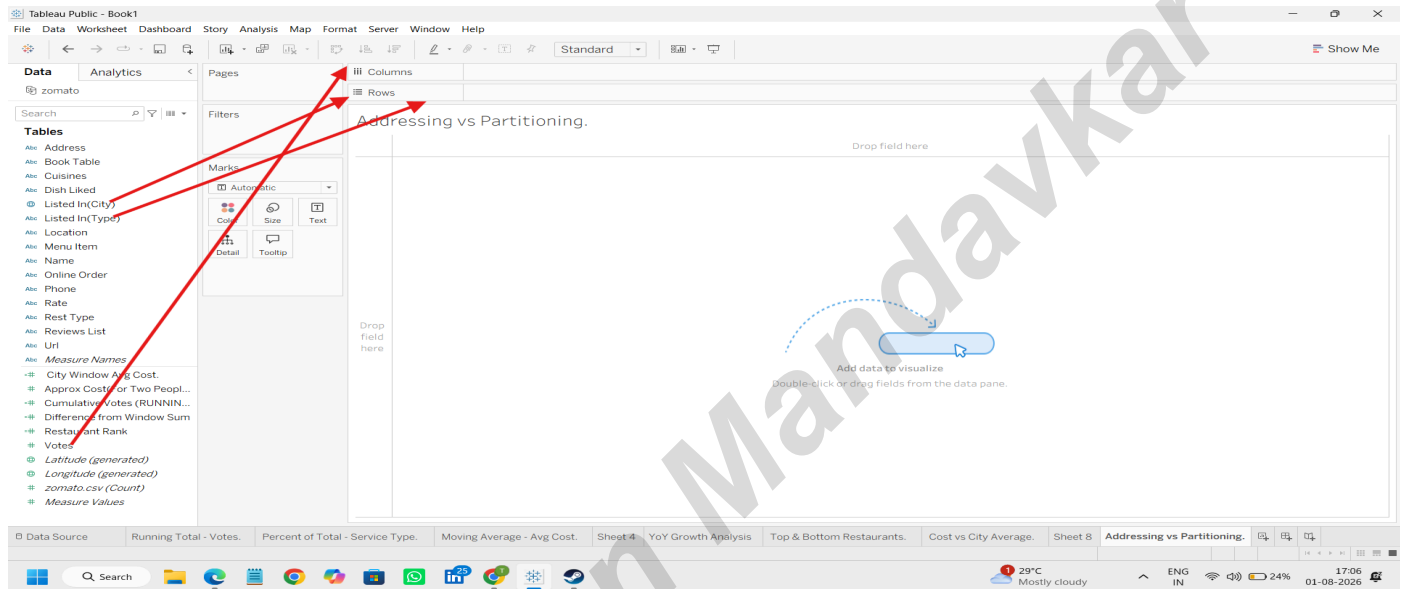
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F] Customize addressing and partitioning

Step 1:- Open a new worksheet and rename it: Addressing vs Partitioning. Drag listed_in(city) to the Rows shelf. Drag listed_in(type) to the Rows shelf, placing it right next to listed_in(city) (creating a two-level nested hierarchy). Drag votes to the Columns shelf as SUM(votes).



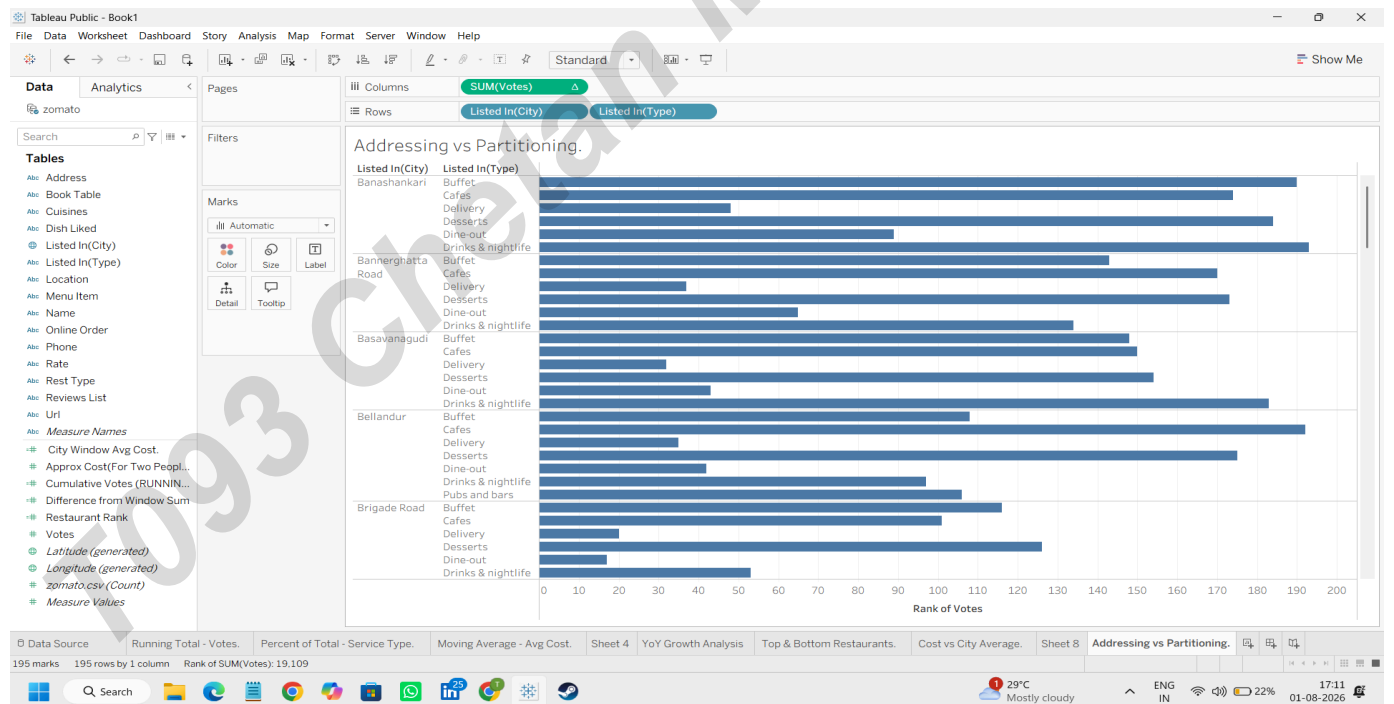
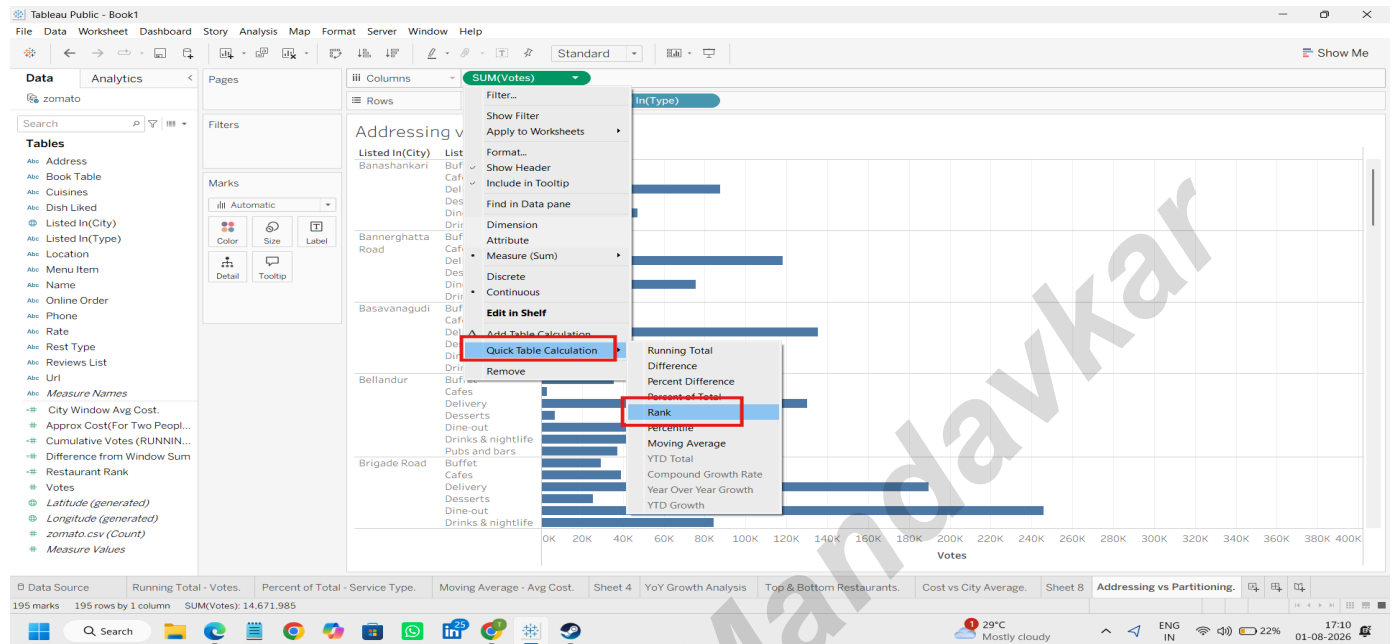
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Step 2:- Right-click the SUM(votes) pill on the Columns shelf, hover over Quick Table Calculation, and select Rank.



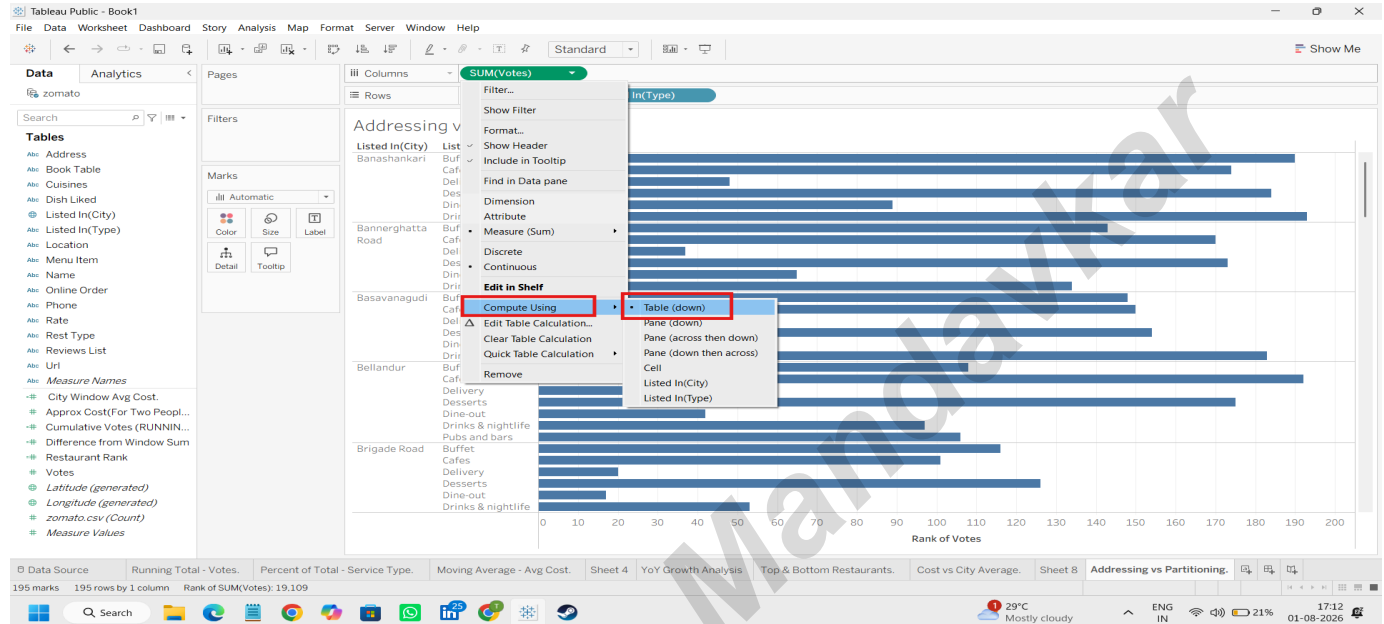
Name:- Chetan Mandavkar

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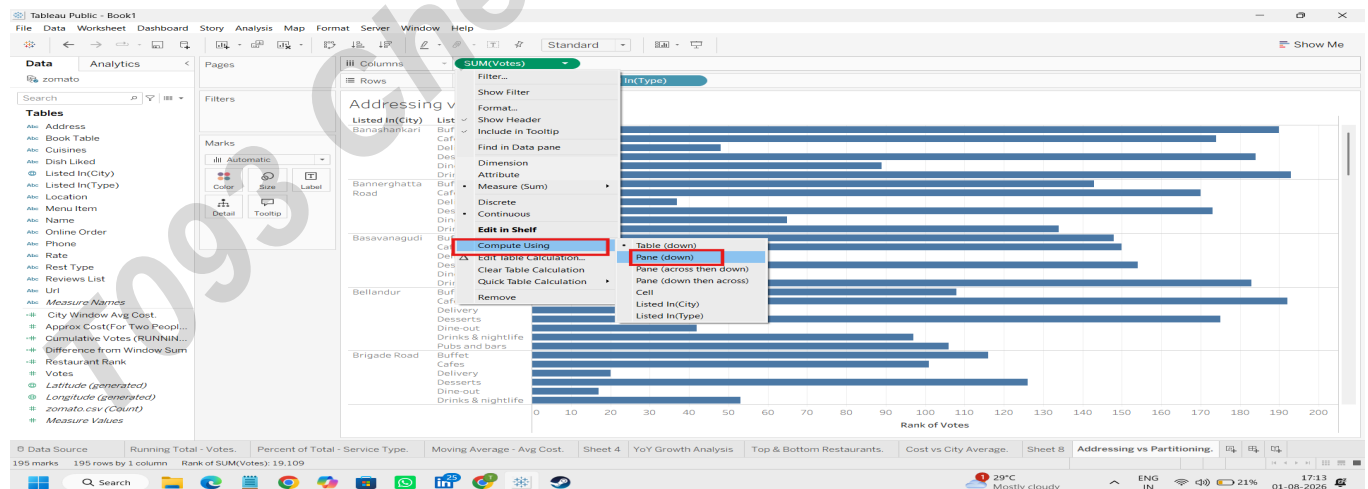
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Step 3:- Right-click the SUM(votes) pill again and hover over Compute Using
Using: o Select Table (down): Tableau addresses the entire table from top to bottom. It ranks every single row relative to the whole sheet (Partition = Entire Table).



Step 4:- Select Pane (down): Tableau partitions the calculation by listed_in(city). The ranking now restarts at 1 for the service types inside each individual city!

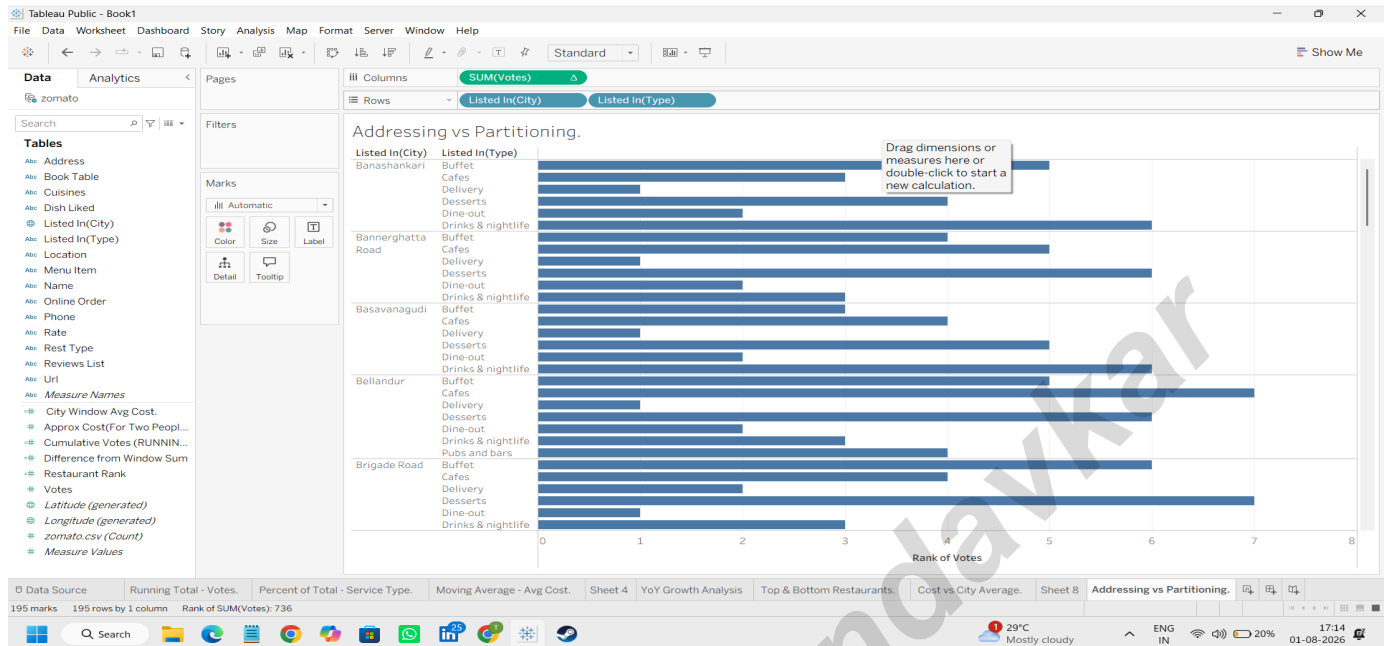


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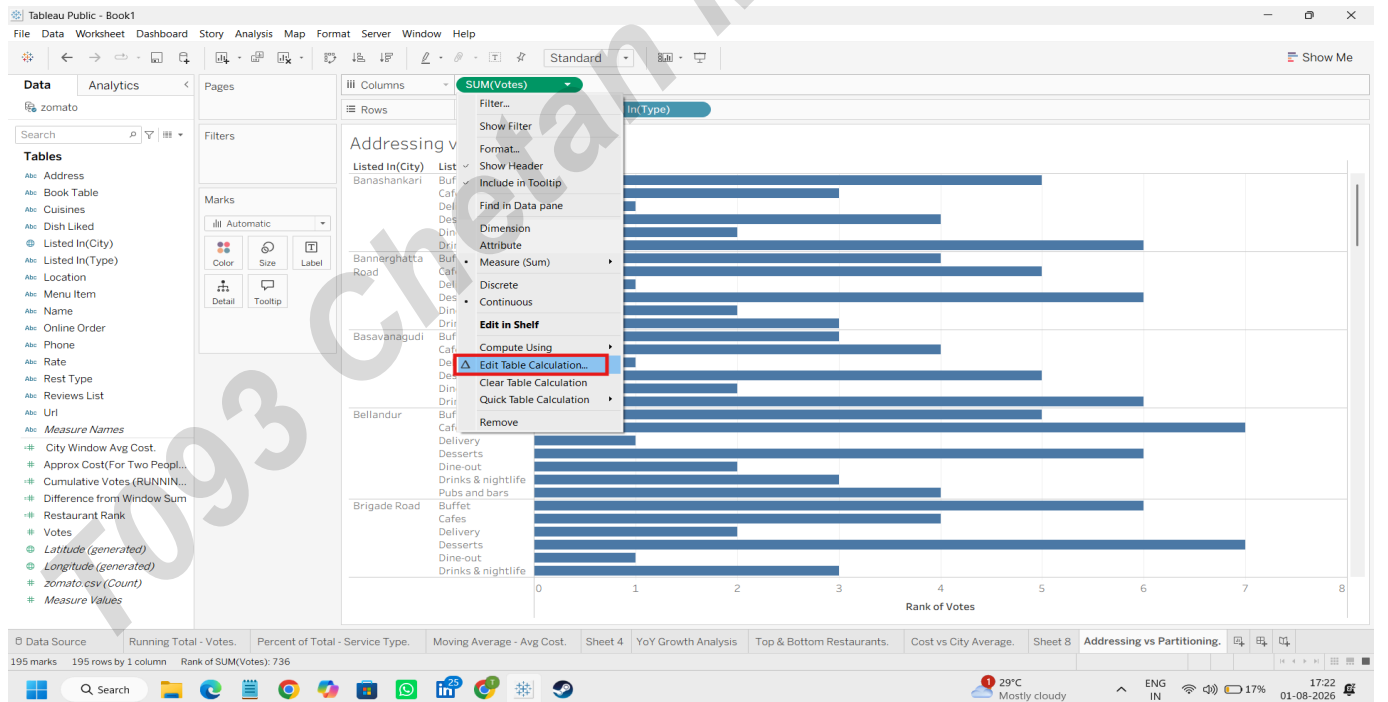
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Step 5:- Right-click the SUM(votes) pill (with the Δ icon) on the Columns shelf and select Edit Table Calculation...



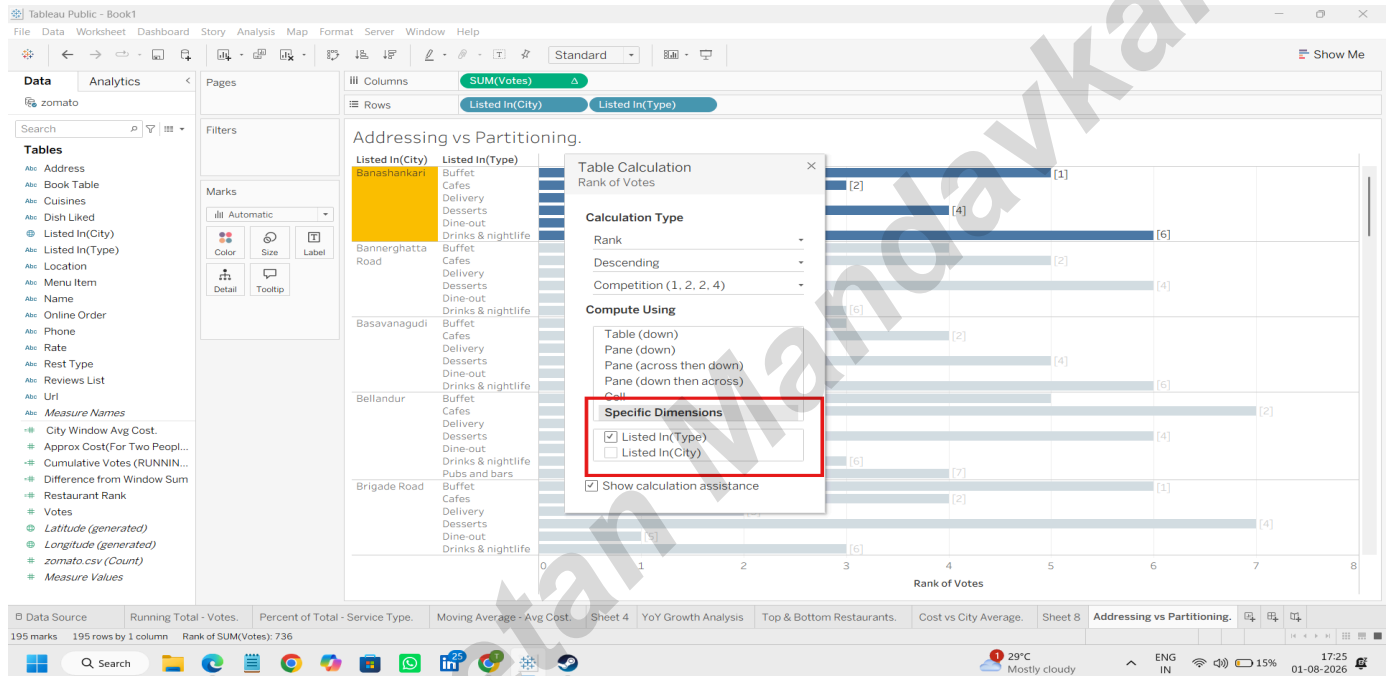
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Step 6:- Under Compute Using, select the radio button for Specific Dimensions. Observe the checked vs. unchecked boxes: Checked boxes = Addressing fields: Tableau computes along these dimensions. Unchecked boxes = Partitioning fields: Tableau restarts the calculation whenever this dimension changes. Uncheck listed_in(city) and leave listed_in(type) checked.



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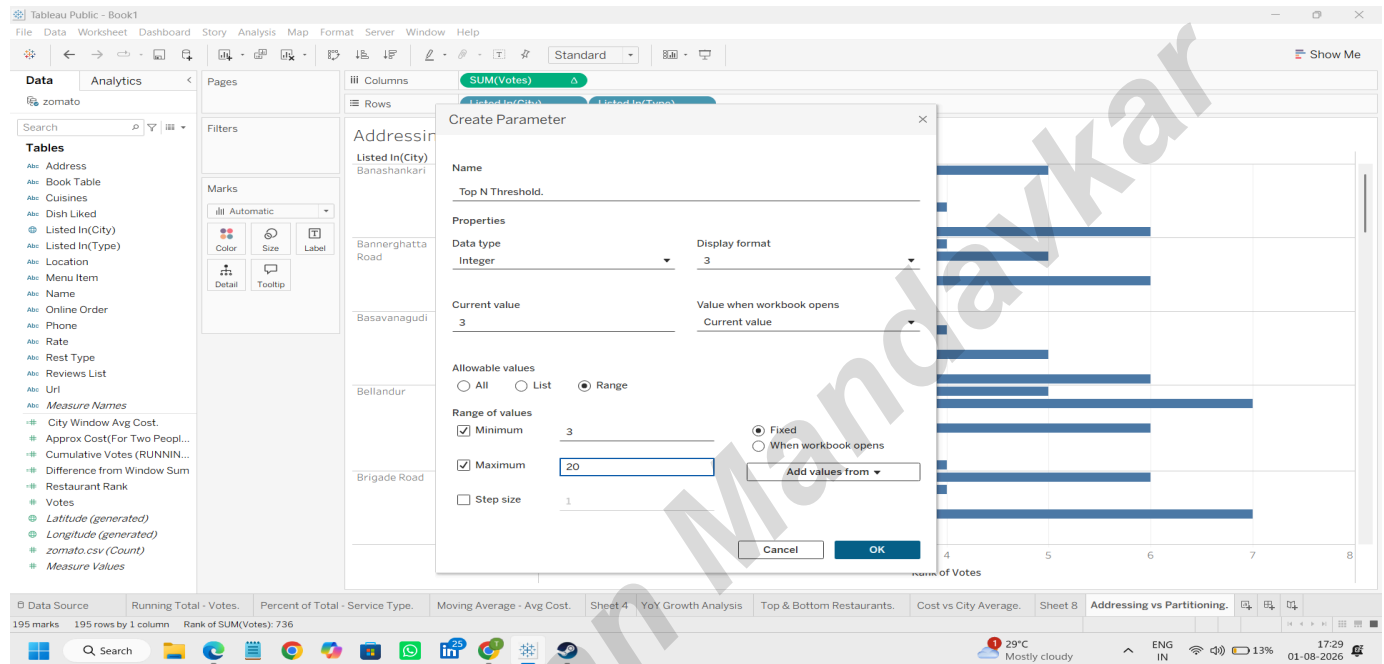
Roll No. T093

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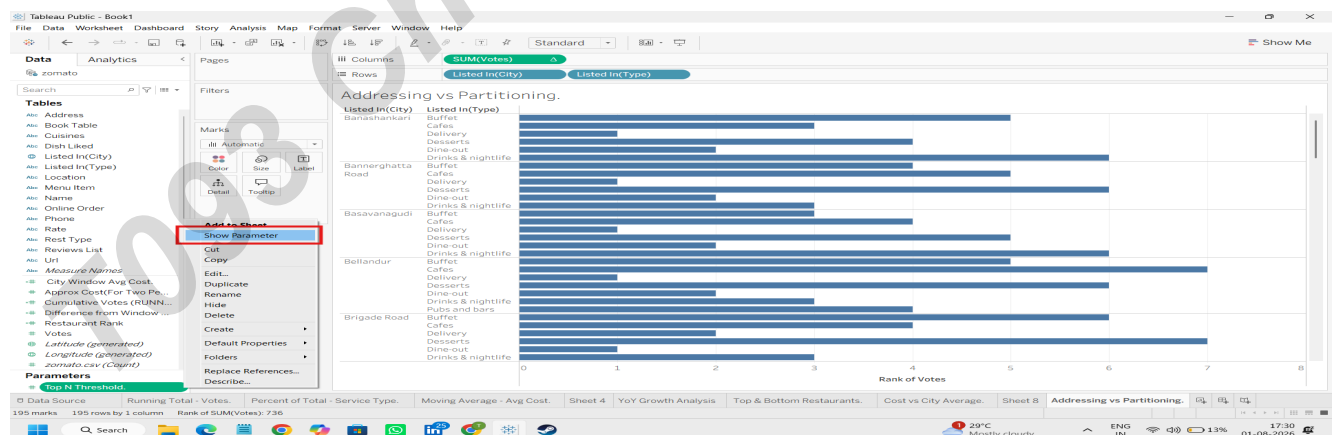
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G] Create dynamic titles and labels using table calculations

Step 1:- Right-click anywhere in the Data Pane and select Create Parameter.. Name the parameter: Top N Threshold. Set Data type to Integer. Set Current value to 5. Under Allowable values, select Range: o Minimum: 3 o Maximum: 20 o Step size: 1 6. Click OK.



Step 2:- Right-click Top N Threshold in the Data Pane and select Show Parameter Control.



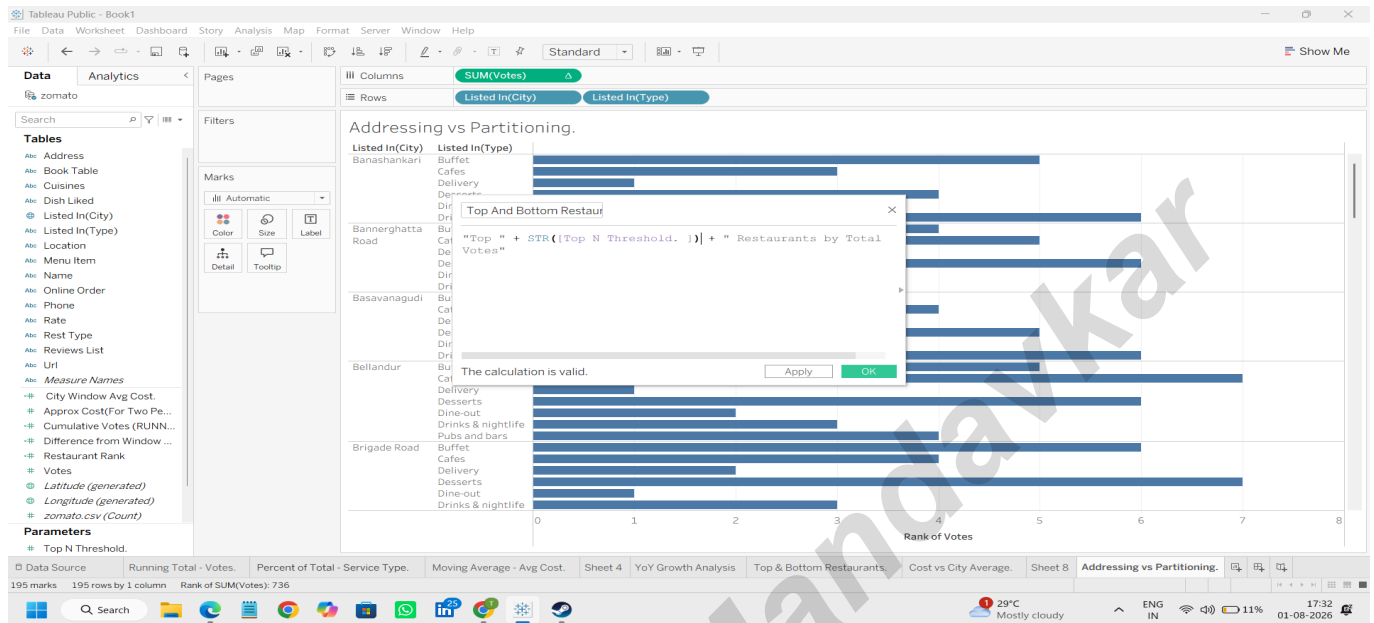
Name:- Chetan Mandavkar

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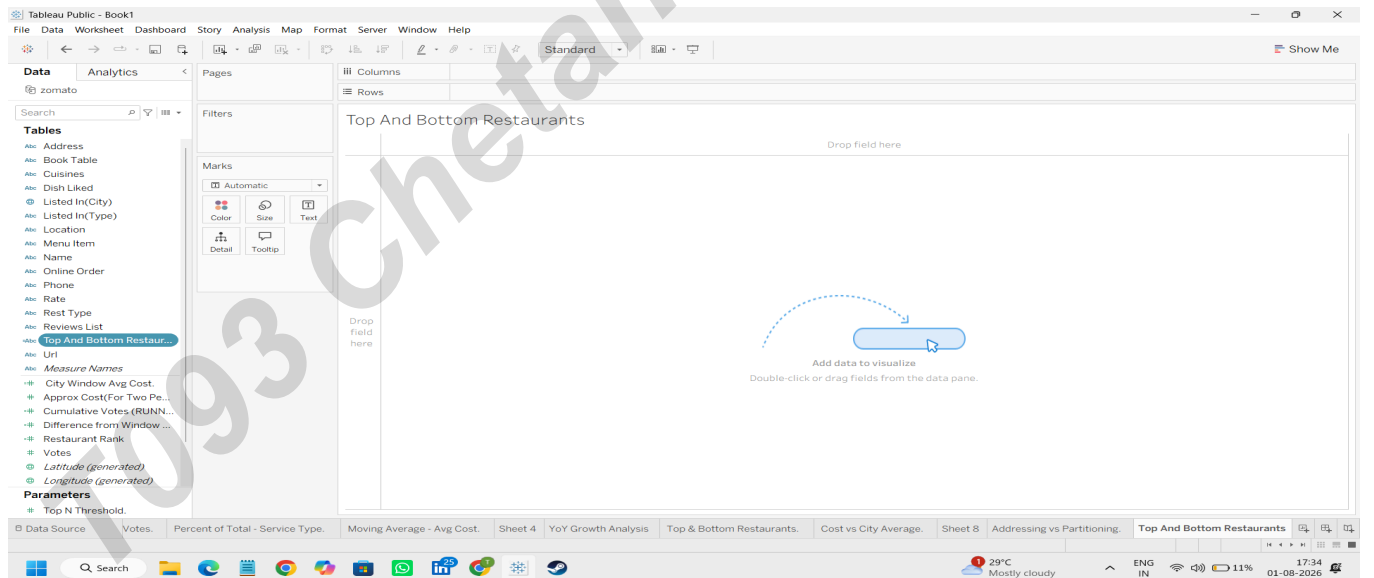
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Step 3:- Create a new Calculated Field named Dynamic Title: Top And Bottom Restaurants



Step 4:- Double-click the Sheet Title at the top of your worksheet (e.g., Top & Bottom Restaurants).



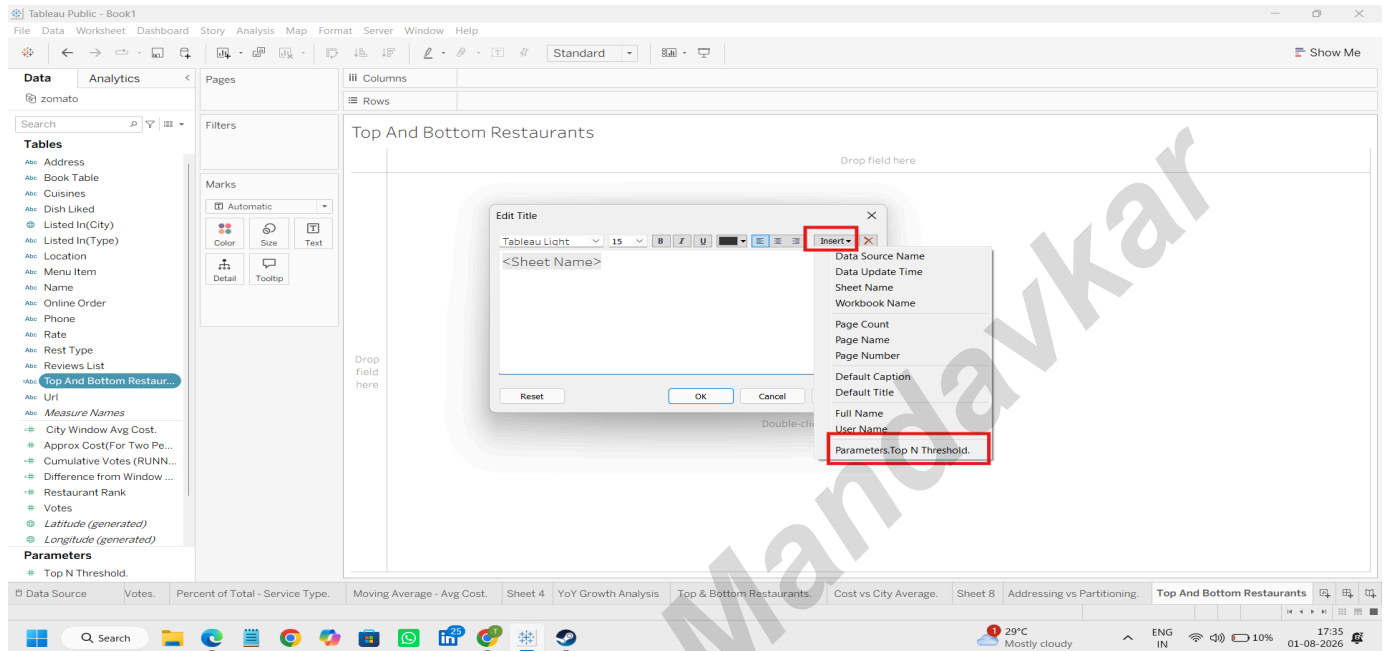
Name:- Chetan Mandavkar

Roll No. T093

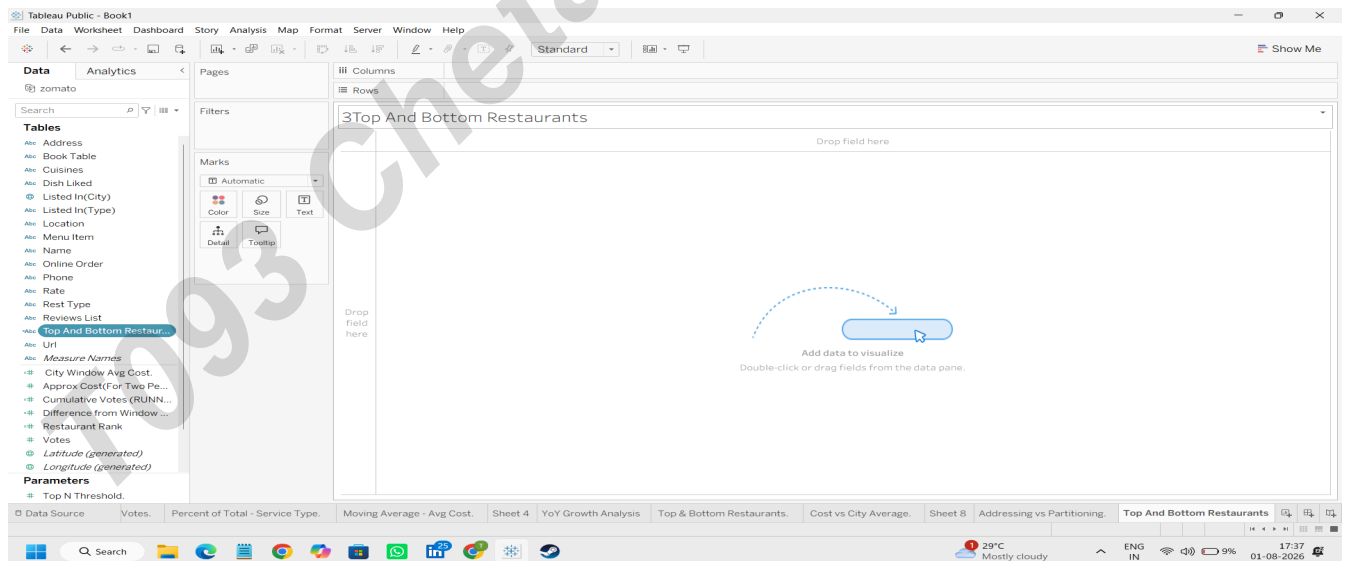
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Step 5:- Clear the existing text. In the Edit Title dialog box, click the Insert button in the top right corner. Select Parameters.Top N Threshold from the list.



Step 6:- Adjust the Top N Threshold parameter slider on your sheet to test how the title updates dynamically!



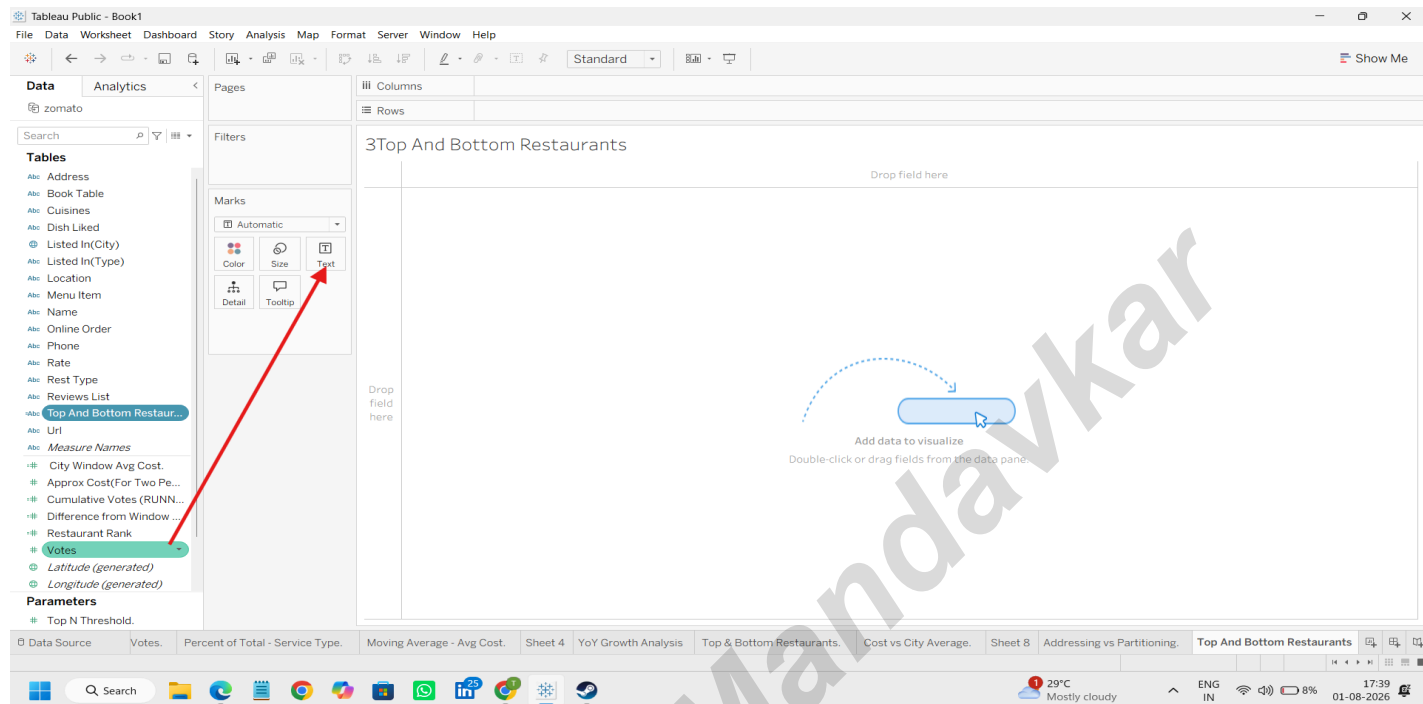
Name:- Chetan Mandavkar

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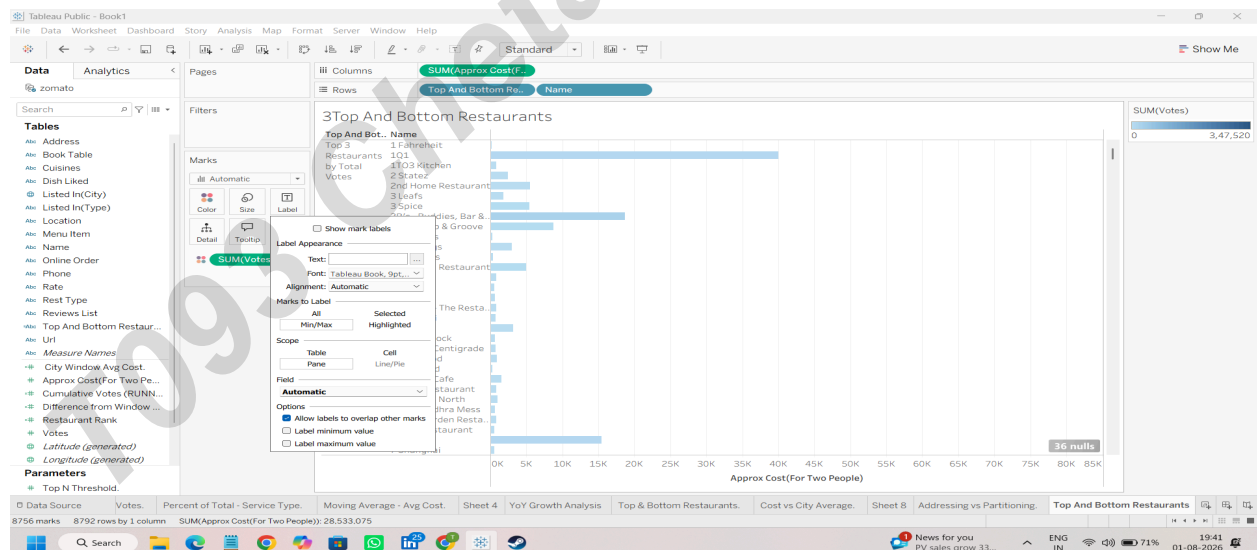
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Step 7:- Drag SUM(votes) to the Label card on the Marks card.



Step 8:- Under Marks to Label, select Min/Max. Under Field, select SUM(votes). o Under mark Options, check Allow labels to overlap other marks

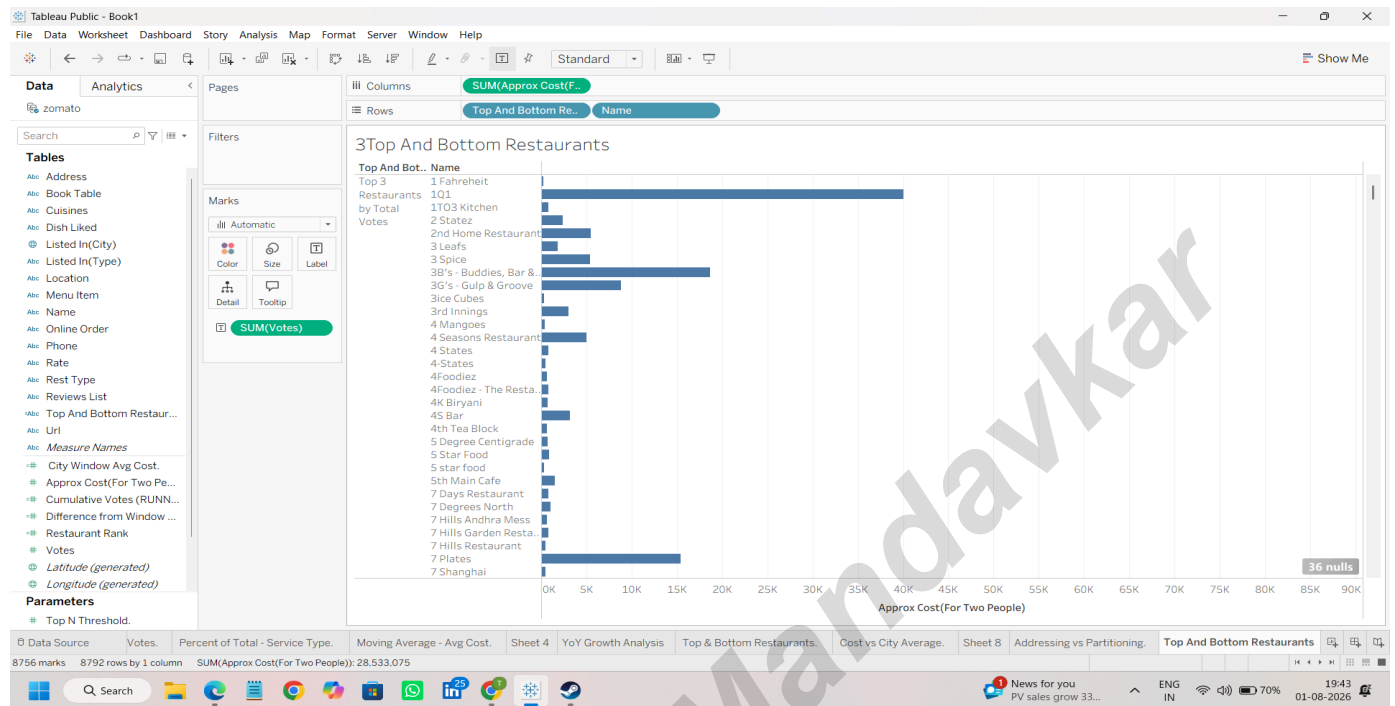


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